Navigating the Chemical Universe of Natural Products: From Structural Diversity to Drug Discovery Applications

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Navigating the Chemical Universe of Natural Products: From Structural Diversity to Drug Discovery Applications

Abstract

This comprehensive review explores the vast chemical space and structural diversity of natural products (NPs) and their critical importance in modern drug discovery. It examines the foundational concepts defining NP chemical space, highlighting how NPs occupy unique regions not explored by synthetic compounds while largely adhering to drug-like properties. The article details cutting-edge methodological approaches for analyzing and navigating this chemical space, including cheminformatics tools, database resources like COCONUT and NPAtlas, and computational screening techniques. Significant challenges in NP research are addressed, such as data curation, stereochemical accuracy, and material supply bottlenecks, along with optimization strategies like diversity-oriented synthesis. Finally, the review provides comparative validation of NP drug-likeness and success rates, demonstrating NPs' proven track record as sources of new pharmacological entities. This resource is designed for researchers, scientists, and drug development professionals seeking to leverage natural product diversity for therapeutic innovation.

Defining Natural Product Chemical Space: The Foundation of Bioactive Diversity

Chemical space is a foundational concept in cheminformatics and modern drug discovery, representing a theoretical multidimensional domain where each point corresponds to a unique chemical structure positioned according to its specific properties [1]. This conceptual framework systematically organizes molecular diversity, enabling researchers to analyze, classify, and visualize relationships within vast compound collections [1]. The chemical universe of possible small organic molecules is astronomically large, with estimates exceeding 10^60 compounds, making exhaustive exploration impractical [1]. Consequently, the field focuses on navigating and mapping biologically relevant regions of this space to identify promising drug-like molecules with desired biological activities [2].

The structural diversity of natural products plays a crucial role in populating chemical space with biologically validated starting points [3]. Natural products exhibit unique chemical diversity complementary to synthetic collections, possessing greater steric complexity and a wider variety of ring systems [3]. This diversity stems from evolutionary pressure and provides privileged scaffolds for interacting with biological targets, making natural products unsurpassed sources of leading structures in drug discovery [3]. As the drug discovery field evolves, artificial intelligence and novel computational approaches are revolutionizing how we explore and exploit chemical space, creating paradigm shifts across research and development platforms [2].

The Fundamentals of Chemical Space

Theoretical Framework and Definitions

Chemical space serves as a systematic tool to organize molecular diversity by postulating that different molecules occupy different regions of a mathematical space where each molecule's position is defined by its properties [1]. While no single unified definition exists, the core concept encompasses all compounds that could potentially exist, with spaces often constricted to specific regions depending on the included compounds and molecular representations [1]. These representations can utilize chemical descriptors (e.g., fingerprints, physicochemical, or quantum properties), biological descriptors (e.g., bioactivity, bioavailability), or clinical descriptors (e.g., side effects) [1].

The exploration of chemical space is fundamentally governed by the structure of small molecule libraries, which serve as essential collections for identifying molecules with desired biological activity [2]. These libraries can be broadly categorized into **diverse libraries** offering broad structural variety and **focused libraries** targeting specific protein families or biological pathways [2]. The generation of these libraries employs various methodologies, including combinatorial chemistry, diversity-oriented synthesis, fragment-based approaches, natural product extraction, and computational generation of virtual libraries [2].

Key Dimensions and Molecular Descriptors

The dimensions of chemical space are defined by molecular descriptors that capture critical structural and physicochemical properties. Key descriptors and filters used to navigate drug-relevant chemical space include:

Table 1: Key Molecular Descriptors and Filters for Navigating Chemical Space

Descriptor/Filter	Description	Application in Drug Discovery
Lipinski's Rule of 5 [2]	Molecular weight <500 Da, CLogP <5, H-bond donors ≤5, H-bond acceptors ≤10	Predicts oral bioavailability of drug-like molecules
Fragment-Based "Rule of 3" [2]	Molecular weight <300 Da, CLogP ≤3, H-bond donors ≤3, H-bond acceptors ≤3	Guides design of fragment libraries for FBDD
ADMET Properties [2]	Absorption, distribution, metabolism, excretion, toxicity	Optimizes pharmacokinetics and reduces toxicity
Synthetic Accessibility Score [2]	Score based on molecular complexity (SAS >6 indicates challenging synthesis)	Assesses synthetic feasibility of designed compounds
Structural Complexity [2]	Chirality, stereocenters, sp2:sp3 hybridization ratios	Measures molecular complexity and synthetic challenge

These descriptors enable the quantitative assessment of chemical space regions most likely to yield successful drug candidates by ensuring appropriate absorption, distribution, metabolism, and excretion characteristics while minimizing toxicity risks [2].

Methodologies for Chemical Space Analysis

Computational Tools and Algorithms

The analysis of chemical space requires specialized computational tools capable of handling millions of compounds efficiently. Recent methodological advances address the steep computational challenges of traditional similarity indices, which scale as $O(N^2)$ when comparing N molecules [1]. Innovative approaches like the iSIM framework bypass this quadratic scaling problem by comparing all molecules simultaneously with O(N) complexity, enabling practical analysis of large libraries [1]. The iSIM Tanimoto value corresponds to the average of all distinct pairwise Tanimoto comparisons, providing a global indicator of library diversity where lower values indicate more diverse collections [1].

Clustering algorithms are equally important for dissecting chemical spaces granularly. The BitBIRCH algorithm draws inspiration from Balanced Iterative Reducing and Clustering using Hierarchies but adapts it for chemical informatics by relying on iSIM to process binary vectors using Tanimoto similarity [1]. This approach uses a tree structure to reduce comparison requirements, making it suitable for large-scale chemical space analysis [1].

Dimensionality Reduction for Chemical Space Visualization

Dimensionality reduction techniques transform high-dimensional molecular descriptor data into human-interpretable 2D or 3D chemical space maps, a process known as "chemography" [4]. These methods enable researchers to visualize complex chemical relationships and identify patterns within compound libraries.

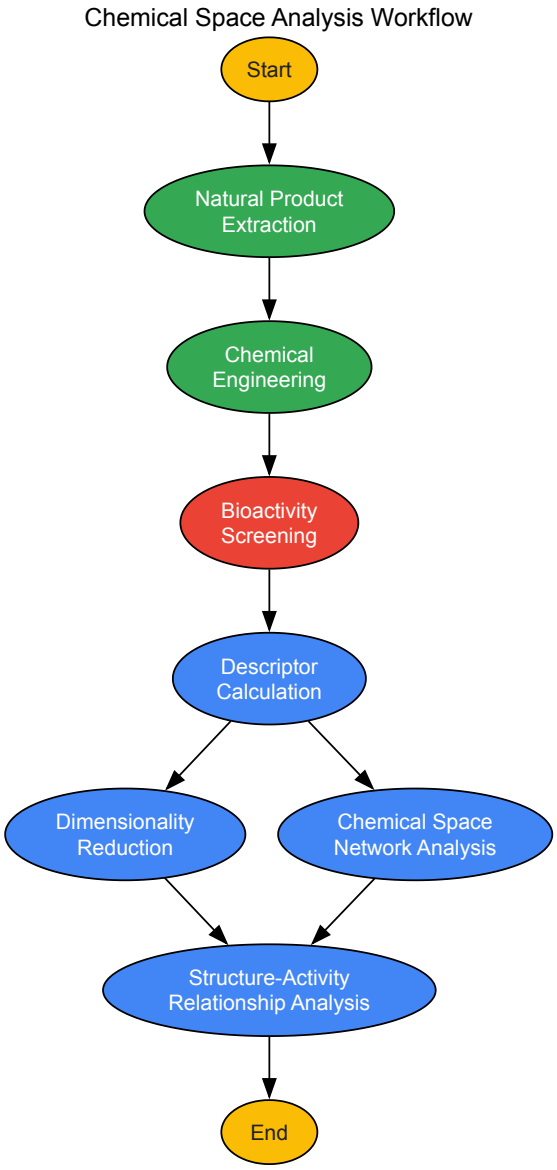
Table 2: Dimensionality Reduction Methods for Chemical Space Visualization

Method	Type	Key Characteristics	Optimal Use Cases
Principal Component Analysis [4]	Linear	Preserves global data structure; computationally efficient	Initial exploration; when linear assumptions hold
t-SNE [4]	Non-linear	Excellent at preserving local neighborhoods; emphasizes clusters	Detailed analysis of specific compound classes
UMAP [4]	Non-linear	Balances local and global structure preservation; faster than t-SNE	General-purpose mapping of diverse compound sets
Generative Topographic Mapping [4]	Non-linear	Generates interpretable property "landscapes"; supports NB-compliant maps	Structure-activity relationship analysis

Benchmarking studies highlight non-linear DR algorithms (t-SNE, UMAP) as best-performing for neighborhood preservation, though PCA remains popular and sometimes more efficient for specific tasks [4]. The choice of method should be guided by suitability for particular analysis objectives rather than seeking a universally superior approach [4].

Experimental Protocol: Chemical Space Analysis of Natural Product Derivatives

The following workflow illustrates a comprehensive approach for analyzing chemical space, particularly relevant to natural products research:



Step 1: Natural Product Extraction and Modification

- Global ethanolic extraction of plant material (e.g., *Ambrosia tenuifolia*) followed by partitioning with solvents of increasing polarity yields subextracts enriched in specific compound classes [3].
- Chemical engineering of extracts employs diversity-oriented synthesis strategies, such as treating sesquiterpene lactone-enriched subextracts with p-toluenesulfonic acid in toluene under reflux conditions to remodel molecular scaffolds [3].
- This complexity-to-diversity approach generates structural complexity through ring distortion reactions including expansions, contractions, fusions, cleavages, and rearrangements [3].

Step 2: Bioactivity Screening

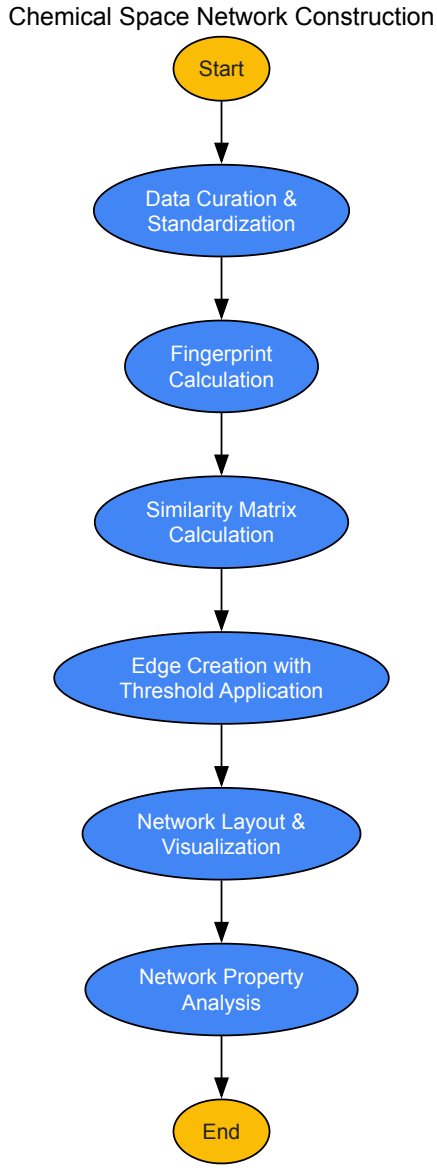
- Bioguided fractionation follows anti-glioblastoma activity in T98G cell cultures to identify bioactive fractions convergently [3].
- Chronobiological experiments evaluate temporal susceptibility to treatments on GBM cell cultures to define optimal administration timing [3].

Step 3: Chemical Space Analysis

- Calculate molecular descriptors including Morgan fingerprints (radius 2, size 1024), MACCS keys, or ChemDist embeddings from graph neural networks [4].
- Apply dimensionality reduction techniques (PCA, t-SNE, UMAP, GTM) with optimized hyperparameters based on neighborhood preservation metrics [4].
- Construct Chemical Space Networks using RDKit and NetworkX, representing compounds as nodes connected by edges based on Tanimoto similarity thresholds [5].
- Analyze network properties including clustering coefficient, degree assortativity, and modularity to identify structural relationships [5].

Chemical Space Networks: A Practical Implementation

Chemical Space Networks provide an alternative representation to coordinate-based visualizations by depicting compounds as nodes connected by edges defined by specific relationships [5]. The following diagram illustrates the CSN construction process:



Implementation Protocol:

- **Data Curation**: Load compound datasets (e.g., from ChEMBL) into Python using Pandas DataFrames. Remove compounds missing bioactivity data, check for salts as disconnected SMILES, and merge duplicate compounds by averaging activity values [5].
- **Fingerprint Calculation**: Generate molecular representations using RDKit. Standard choices include:
  - Morgan fingerprints (circular fingerprints encoding atomic environments)
  - MACCS keys (predefined structural features)
  - Pattern fingerprints (structural key fingerprints) [5]
- **Similarity Calculation**: Compute pairwise Tanimoto similarity values between all compounds. For large datasets, optimize using the iSIM framework to avoid $O(N^2)$ scaling [1].
- **Network Construction with Threshold**: Create a network where nodes represent compounds and edges represent similarity relationships exceeding a defined threshold (typically Tanimoto ≥ 0.7-0.85 for 2D fingerprints) [5].
- **Visualization and Analysis**:
  - Visualize networks using NetworkX or Gephi, optionally replacing circle nodes with 2D structure depictions [5].
  - Color nodes by property values (e.g., bioactivity, physicochemical properties) [5].
  - Calculate network properties including clustering coefficient, degree assortativity, and modularity to identify structural relationships [5].

Research Reagents and Computational Tools

Table 3: Essential Research Reagents and Computational Tools for Chemical Space Analysis

Tool/Resource	Type	Function	Access
RDKit [5]	Cheminformatics Library	Calculates molecular descriptors, fingerprints, and similarity metrics	Open-source
NetworkX [5]	Network Analysis Library	Constructs and analyzes chemical space networks	Open-source
ChEMBL [1]	Chemical Database	Provides curated bioactivity data for drug-like molecules	Public
DrugBank [1]	Pharmaceutical Database	Contains information on drugs and drug targets	Public
PubChem [1]	Chemical Database	Offers extensive compound information and bioassays	Public
iSIM Framework [1]	Computational Algorithm	Enables efficient similarity calculations for large libraries	Algorithm
BitBIRCH [1]	Clustering Algorithm	Groups chemical structures based on molecular similarity	Algorithm
Natural Product Extracts [3]	Experimental Material	Provides biologically validated starting points for diversification	Laboratory preparation

Applications in Natural Products Research

Chemical space analysis provides powerful approaches for exploring and enhancing the structural diversity of natural products. The inherent complementarity between natural and synthetic chemical spaces makes natural products invaluable starting points for drug discovery [3]. Natural products exhibit structural features distinct from synthetic compounds, including greater steric complexity, wider variety of ring systems, and different distribution of heteroatoms [3]. This diversity is biologically prevalidated through evolutionary selection, enhancing the probability of bioactivity [3].

The "complexity to diversity" paradigm has emerged as a particularly powerful approach in natural products research, wherein complex natural products undergo ring distortion reactions to generate new structures that are both complex and diverse from the original natural product and from each other [3]. This process plays a fundamental role in exploring biologically relevant chemical space by taking advantage of nature's biosynthetic machinery strengthened through chemical reactions that form new carbon-carbon bonds and modify molecular scaffolds [3]. Applications include:

- **Diversity-Enhanced Extracts**: Direct chemical modification of natural extracts to yield mixtures containing fragments or elements conducive to bioactivity, modifying functional groups in original natural products or their molecular skeletons [3].
- **Bioguided Fractionation**: Using bioactivity measurements to guide fractionation and purification convergently, enabling discovery of bioactive compounds not previously described in traditional phytochemical studies [3].
- **Chemical Space Navigation**: Mapping natural product derivatives into chemical space visualizations to identify regions of high bioactivity and structure-activity relationships [3].

Quantitative analysis of chemical space diversity reveals that while the number of compounds in public repositories is rapidly increasing, this growth does not automatically translate to increased chemical diversity [1]. Tools like iSIM and BitBIRCH clustering enable researchers to identify which compound additions genuinely expand chemical space versus those that merely populate existing regions [1]. This understanding is crucial for designing natural product-derived libraries that effectively explore uncharted regions of biologically relevant chemical space.

Natural products (NPs) are chemical compounds produced by living organisms—including plants, microorganisms, and marine organisms—that have served as invaluable resources for drug discovery and biomedical research. These molecules are characterized by their **complex chemical structures**, **diverse three-dimensional architectures**, and **precise stereochemistry**, properties that have evolved to fulfill specific biological functions. Within the broader context of chemical space—the theoretical multidimensional space encompassing all possible molecules and compounds—natural products occupy a **distinct and privileged region** known as the biologically relevant chemical space (BioReCS) [6]. This region comprises molecules with biological activity, both beneficial and detrimental, making NPs particularly significant for therapeutic development [6].

The structural complexity of natural products arises from biosynthetic pathways that have been optimized through evolution, resulting in molecules with **intricate scaffolds** and **defined spatial configurations** that often underlie their biological efficacy. Approximately 30% of FDA-approved drugs from 1981 to 2019 originated from natural products or their derivatives, particularly in areas such as anti-infectives and anti-tumor therapies

[7]. This review provides an in-depth technical examination of the unique structural properties of natural products, with a focus on their chemical complexity, three-dimensional architecture, and stereochemical features, framed within the context of chemical space exploration and analysis.

Chemical Complexity and Scaffold Diversity

The chemical complexity of natural products manifests primarily through their intricate molecular scaffolds and diverse functional group decorations. Unlike many synthetic compounds, natural products often feature **highly elaborate carbon skeletons** with multiple stereocenters and complex ring systems. This complexity arises from enzyme-mediated biosynthetic pathways that facilitate chemical transformations often challenging to achieve through conventional synthetic chemistry.

Quantifying Natural Product Complexity

Analysis of microbial natural products in the Natural Products Atlas database (version v2024_09), containing 36,454 compounds, reveals distinct clustering patterns based on structural similarity [8]. When using the Morgan fingerprint method (radius 2) and Dice similarity metric (cutoff = 0.75), the database organizes into 4,148 clusters containing two or more compounds, encompassing 30,094 compounds (82.6% of the database) [8]. The median cluster size is 3, with 1,209 clusters containing at least five members [8]. This distribution demonstrates both the extensive diversity and the presence of structural families within natural product space.

Table 1: Chemical Clustering in Microbial Natural Products

Metric	Value	Description
Total Compounds	36,454	Microbial natural products in NP Atlas [8]
Clusters (≥2 compounds)	4,148	Dice similarity ≥ 0.75 [8]
Clustered Compounds	30,094	82.6% of total database [8]
Median Cluster Size	3	[8]
Large Clusters (≥5 members)	1,209	[8]
Taxonomy-Specific Clusters	1,093	≥95% fungal or bacterial origin [8]

Structural Hotspots in Chemical Space

Certain classes of natural products form particularly dense regions in chemical space, characterized by high structural similarity within the class but significant distinction from other scaffolds. Notable examples include:

- **Microcystins** (cluster 50): 245 compounds with median edge count of 196, indicating high interconnectivity [8]
- **Peptaibols** (cluster 263): Well-known compound class with high structural similarity [8]
- **Anabaenopeptins** (cluster 415): Distinct structural class with characteristic features [8]

These "hotspots" represent regions of chemical space where evolutionary processes have generated structural variants around privileged scaffolds, potentially reflecting optimized interactions with biological targets [8].

Three-Dimensional Architecture and Stereochemistry

The biological functions of natural products depend not only on their two-dimensional structures but crucially on their three-dimensional configurations and stereochemical properties. This section examines the precise spatial arrangements that characterize natural products and the methodologies for their study.

Stereochemical Features of Natural Products

Natural products frequently contain multiple **chiral centers** with defined configurations, a result of stereospecific biosynthetic enzymes. This stereochemical complexity creates a vast configurational space; for example, a natural product with 10 chiral centers theoretically has 1,024 (2^{10}) possible stereoisomers, though biosynthetic pathways typically produce single, defined configurations [9]. This precise stereochemistry is essential for biological activity, as it determines molecular recognition by biological targets.

The challenge of stereochemical characterization is significant: over 20% of known natural products lack complete chiral configuration annotations, and only 1-2% have fully resolved crystal structures [9]. This represents a substantial knowledge gap in natural product research, as incomplete stereochemical assignment hinders accurate understanding of structure-activity relationships.

Experimental and Computational Approaches to 3D Structure Determination

Experimental Structure Validation Protocols

The stereochemical quality of three-dimensional molecular structures is typically validated using standardized computational tools that assess geometrical parameters against established standards:

- **MAXIT Software Analysis**: Evaluates abnormal stereochemical parameters across six categories: close contacts, bond length deviations, bond angle deviations, deviation from planarity, chirality issues, and phosphate bond linkages [10]
- **MolProbity Software Package**: Assesses structural accuracy through Clashscore, which identifies overlapping or too-close atoms in molecular models [10]
- **X3DNA-DSSR**: Verifies base-pairing geometries and handedness of helices in nucleic acid structures [10]
- **Barnaba**: Analyzes base-pairing geometries in RNA structures [10]

Validation of RNA tertiary structures using these tools has revealed that bond angle deviations represent the most common type of geometrical inaccuracy (183 errors across 17 reference structures), followed by close contacts (54 errors across 7 structures) and bond length deviations (32 errors across 5 structures) [10].

Table 2: Stereochemical Quality Assessment of RNA 3D Structures

Stereochemical Parameter	Structures with Errors	Total Errors	Assessment Method
Bond Angle Deviations	17 structures	183	MAXIT [10]
Close Contacts	7 structures	54	MAXIT [10]
Bond Length Deviations	5 structures	32	MAXIT [10]
Phosphate Bond Linkages	7 structures	9	MAXIT [10]
Deviation from Planarity	2 structures	9	MAXIT [10]
Chirality Issues	0 structures	0	MAXIT [10]

NatGen: Deep Learning for 3D Structure Prediction

The NatGen framework represents a significant advancement in computational prediction of natural product structures. This deep learning approach leverages structure augmentation and generative modeling to predict chiral configurations and 3D conformations with remarkable accuracy [9]. The methodology achieves:

- 96.87% accuracy in chiral configuration prediction on benchmark natural product structural datasets [9]
- 100% accuracy in a prospective study involving 17 recently resolved plant-derived natural products [9]
- Average root-mean-square deviation (RMSD) of predicted 3D structures below 1Å—smaller than the radius of a single atom [9]

Using NatGen, researchers have successfully predicted the 3D structures of 684,619 natural products from COCONUT (the largest open natural product repository), significantly expanding the structural landscape of publicly available natural product data [9].

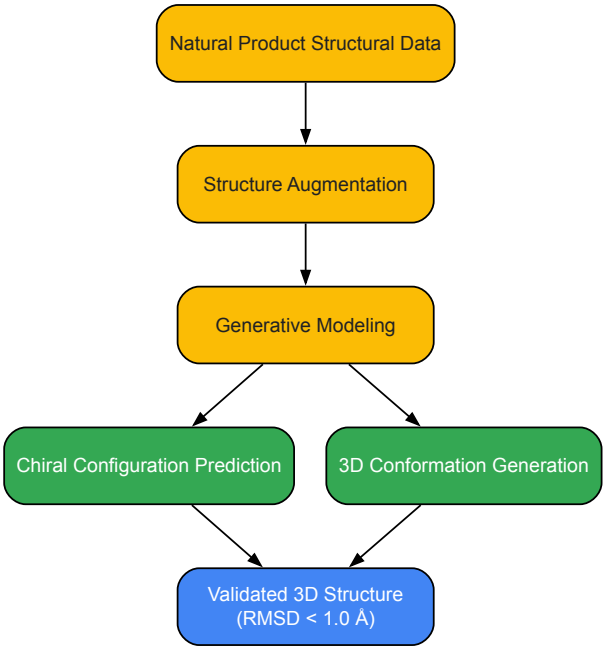


Diagram 1: NatGen 3D Structure Prediction Workflow. This diagram illustrates the deep learning framework for predicting natural product structures, from data input to validated 3D model output.

Research Reagent Solutions for Natural Product Analysis

The experimental study of natural product structures requires specialized reagents and computational tools. The following table details key resources used in contemporary natural products research.

Table 3: Essential Research Reagents and Tools for Natural Product Structural Analysis

Reagent/Resource	Function/Application	Experimental Context
Caco-2 Cell Model	In vitro prediction of intestinal permeability, transport, absorption, and bioavailability of phytochemicals [11]	Bioavailability studies of 84 phytochemicals for drug development [11]
PaDEL-Descriptor & alvaDesc	Computation of molecular descriptors from Isomeric SMILES representations for QSPR modeling [11]	Encoding stereochemistry, chemical structure, and properties into 40 molecular descriptors [11]
Natural Products Atlas	Database of published microbial natural products structures for diversity analysis and chemical space exploration [8]	Analysis of 36,454 microbial compounds using Morgan fingerprints and Dice similarity metrics [8]
COCONUT Database	Largest open natural product repository used for large-scale 3D structure prediction [9]	Source database for predicting 3D structures of 684,619 natural products using NatGen [9]
Factotum System	EPA data management platform for chemical curation, quality assurance, and controlled vocabulary assignment [12]	Curation of chemical use information and product composition data in CPDat [12]

Structural Modification Strategies for Natural Products

Structural modification of natural products represents a crucial approach for optimizing their pharmacological properties while maintaining their biologically privileged scaffolds. This process typically addresses limitations such as unfavorable ADMET properties, low potency, limited specificity, and high toxicity [7].

Computational Approaches to Natural Product Optimization

Artificial intelligence and molecular generative models have emerged as transformative technologies for the rational structural modification of natural products. These approaches can be categorized based on their modification strategies and applicability to different research scenarios:

Target-Interaction-Driven Strategies

When the biological target of a natural product is known, structure-based design approaches leverage protein-ligand interaction data to guide modifications:

- **Fragment Splicing Methods**: Programs including DeepFrag, FREED, DEVELOP, STRIFE, POEM, DrugGPS, TACOGFN, PGMG, FRAME, D3FG, AUTODIFF, and MolEdit3D select fragments from predefined chemical libraries and splice them onto natural product scaffolds, considering target binding requirements [7]
- **Molecular Growth Methods**: Tools such as 3D-MolGNNRL, DiffDec, AutoFragDiff, PMDM, DeepICL, DiffInt, and TargetSA generate molecules directly in the 3D space of target pockets through atom-by-atom or substructure autoregressive generation or global generation based on diffusion models [7]

Molecular Activity-Data-Driven Strategies

For natural products with unknown molecular targets or when optimizing physicochemical properties:

- **Group Modification Models**: Focus on structural modification of characteristic regions (side chains and functional groups) to enhance biological activity and improve properties through "fine-tuning" [7]
- **Scaffold Hopping Approaches**: Reconstruct core scaffolds through linker design and scaffold hopping to optimize the central connected portion of molecules [7]

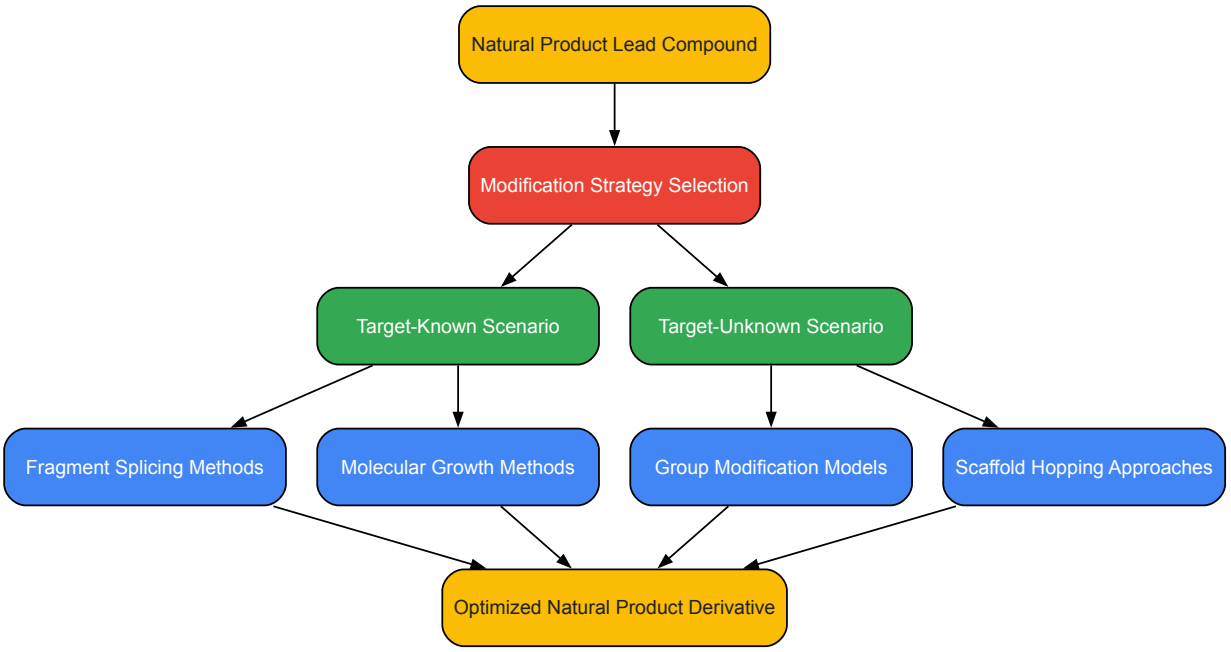


Diagram 2: AI-Driven Structural Modification Workflow. This diagram outlines computational strategies for natural product optimization based on target information availability.

Natural products possess unique structural properties—including complex molecular scaffolds, defined three-dimensional architectures, and precise stereochemistry—that distinguish them in chemical space and underpin their biological activities. Their position within the biologically relevant chemical space (BioReCS) reflects evolutionary optimization for molecular recognition and biological function. Contemporary research approaches, ranging from AI-driven structure prediction to rational modification strategies, continue to reveal the structural sophistication of natural products while providing methodologies to overcome challenges associated with their development as therapeutic agents. The integration of computational structural prediction with experimental validation represents a promising frontier for expanding our understanding and utilization of natural product diversity in drug discovery and chemical biology.

The concept of "chemical space" is a core theoretical construct in cheminformatics, representing a multidimensional space where the position of each molecule is defined by its structural and physicochemical properties [6] [1]. Within this vast universe, two major families of compounds, natural products (NPs) and synthetic compounds (SCs), occupy distinct and characteristic regions. Understanding the differences in how these compounds occupy chemical space is not merely an academic exercise; it is fundamental to guiding drug discovery and the design of novel bioactive molecules [13] [14].

Natural products, forged by billions of years of natural selection, are essential reservoirs for innovative drug discovery, with a significant proportion of approved small-molecule drugs being directly or indirectly derived from them [13] [14]. Conversely, synthetic compounds, born from human ingenuity in the laboratory, offer access to vast regions of chemical space not explored by nature. However, a critical and often overlooked question is the extent to which the structural characteristics of NPs have historically influenced the evolution of SCs [13]. This whitepaper provides an in-depth, time-dependent comparison of the structural variations and chemical space occupation of NPs versus SCs, framing the analysis within the context of a broader thesis on the structural diversity of natural products research. By integrating recent chemoinformatic analyses, we delineate the evolving landscapes of these compound classes, offering a strategic perspective for researchers and drug development professionals aiming to navigate the biologically relevant chemical space (BioReCS) for future discoveries [13] [6].

Methodological Framework for Chemical Space Analysis

A rigorous, time-dependent chemoinformatic analysis requires standardized protocols for data curation, descriptor calculation, and diversity assessment. The following methodologies underpin the key findings discussed in subsequent sections.

Data Curation and Time-Series Grouping

To enable a chronological comparison, large datasets of NPs and SCs must be sorted and grouped. One robust approach involves the following steps:

- **Data Sources**: NPs are typically sourced from curated databases such as the Dictionary of Natural Products (DNP) or the Collection of Open Natural Products (COCONUT). SCs can be compiled from multiple synthetic compound databases (e.g., ChEMBL, PubChem) [13] [15].
- **Temporal Ordering**: Molecules are sorted in early-to-late order based on their CAS Registry Numbers, which serve as a proxy for their discovery or publication date [13].

- **Grouping Strategy**: The sorted molecules are divided into sequential groups, each containing a fixed number of compounds (e.g., 5,000 molecules per group). The remaining molecules after equal grouping are excluded from the analysis to maintain consistency [13].

Calculation of Molecular Descriptors and Fragments

A comprehensive set of descriptors is calculated to characterize the physicochemical and structural properties of the compounds.

- **Physicochemical Properties**: Key descriptors include molecular weight, molecular volume, molecular surface area, number of heavy atoms, number of bonds, and calculated logP (a measure of hydrophobicity) [13].
- **Ring System Analysis**: The number of rings, ring assemblies, aromatic rings, and non-aromatic rings are computed for each molecule. The analysis is extended to include ring sizes (e.g., 4-, 5-, 6-membered) and glycosylation patterns for NPs [13].
- **Molecular Fragmentation**: Molecules are decomposed into fundamental structural units using several methods:
  - **Bemis-Murcko Scaffolds**: These represent the core ring systems and linkers of a molecule, defining its central framework [13].
  - **RECAP Fragments**: The Retrosynthetic Combinatorial Analysis Procedure (RECAP) generates fragments based on chemically sensible and synthetically accessible bonds [13].
  - **Side Chains**: The substituents attached to the core scaffolds are analyzed separately to understand decoration patterns [13].

Assessment of Diversity and Chemical Space

The diversity of compound libraries and their occupation of chemical space are quantified using several advanced algorithms.

- **Intrinsic Similarity (iSIM) Framework**: This tool efficiently quantifies the internal diversity of a library by calculating the average of all distinct pairwise Tanimoto similarities between molecular fingerprints within the set. Its O(N) computational complexity allows it to handle very large libraries. A lower iSIM Tanimoto (iT) value indicates a more diverse collection [1].
- **Complementary Similarity**: This concept identifies central (medoid-like) and outlier molecules within a set. Removing a central molecule significantly decreases the set's diversity, while removing an outlier has little effect. This helps map the internal structure of the chemical space [1].
- **BitBIRCH Clustering**: An efficient clustering algorithm for binary fingerprints, BitBIRCH uses a tree structure to group molecules based on Tanimoto similarity, enabling a "granular" view of the formation and evolution of clusters within chemical space over time [1].
- **Chemical Space Visualization**: Techniques such as Principal Component Analysis (PCA), Tree MAP (TMAP), and SAR Map are employed to project high-dimensional descriptor data into two or three dimensions for visual interpretation of the distribution and overlap of NPs and SCs [13].

Evaluation of Biological Relevance

The biological relevance of a compound or a library can be inferred from its presence in databases annotated with bioactivity data, such as ChEMBL [6]. The premise is that molecules with known biological activities occupy the biologically relevant chemical space (BioReCS). The enrichment of generated fragments or scaffolds in these bioactive databases can serve as a metric for their potential biological relevance [6] [15].

Comparative Analysis of Structural and Physicochemical Properties

Applying the aforementioned methodologies reveals profound and systematic differences between NPs and SCs, which have evolved over time.

Molecular Size and Complexity

Table 1: Time-Dependent Trends in Molecular Size Descriptors

Property	Natural Products (NPs)	Synthetic Compounds (SCs)
Molecular Weight	Consistent increase over time; recently discovered NPs are larger [13].	Variation within a limited range, constrained by synthesis technology and drug-like rules (e.g., Lipinski's Rule of Five) [13].
Heavy Atoms	Number of heavy atoms shows a consistent increase [13].	Number of heavy atoms varies within a constrained range [13].
Molecular Volume/Surface Area	Mean values exhibit a consistent increase [13].	Average values vary within a limited range [13].
Number of Rings	Average number gradually increases over time [13].	Evident rise in the mean number of rings and ring assemblies [13].

Property	Natural Products (NPs)	Synthetic Compounds (SCs)
Aromatic vs. Aliphatic Rings	Most rings are non-aromatic. The number of aromatic rings changes little over time [13].	Distinguished by a greater prevalence of aromatic rings, due to the widespread use of compounds like benzene in synthesis [13].
Ring Assemblies	NPs have more rings but fewer ring assemblies, indicating the presence of bigger fused rings (e.g., bridged rings, spiral rings) [13].	The increase in ring assemblies suggests more complex, linked ring systems [13].
Glycosylation	Glycosylation ratios and the mean number of sugar rings in each glycoside increase gradually over time [13].	Not a common feature in typical SC libraries.

The data clearly indicate that NPs are generally larger and more complex than SCs, a trend that has become more pronounced over time. This is attributed to technological advancements in the isolation and characterization of larger NPs. SCs, in contrast, have their size bounded by synthetic feasibility and the historical influence of "drug-like" rules [13].

Ring Systems and Scaffolds

The analysis of ring systems provides deep insight into the structural foundations of these compound classes.

- **NPs Prefer Complex, Saturated Systems**: NPs contain a higher proportion of non-aromatic rings and larger, fused ring systems (as evidenced by fewer ring assemblies). This points to a structural landscape rich in aliphatic and stereochemically complex scaffolds [13] [13].
- **SCs Favor Aromatic and Simple Rings**: SCs are characterized by a greater involvement of aromatic rings, particularly stable five- and six-membered rings. A notable trend is the sharp increase in four-membered rings in SCs after 2009, likely driven by a desire to enhance pharmacokinetic properties [13].

Fragmentation and Functional Groups

Deconstructing molecules into their core scaffolds and side chains reveals divergent evolutionary paths.

Table 2: Key Differences in Molecular Fragments and Substituents

Fragment Component	Natural Products (NPs)	Synthetic Compounds (SCs)
Core Scaffolds	Larger, more diverse, and more complex ring systems [13] [13]. Contain more aliphatic rings and fewer heteroatoms (except oxygen) [13].	Contain more heteroatoms (N, S) and phenyl rings. Scaffolds are generally less complex and more synthetically accessible [13] [16].
Side Chains/Substituents	Have more oxygen atoms, stereocenters, and very few heteroatoms other than oxygen. Exhibit higher structural complexity [13] [14].	Rich in nitrogen and sulfur atoms, halogens, and aromatic rings. Generally have lower structural complexity [13] [14].
Functional Groups	Feature more oxygen atoms, ethylene-derived groups, and unsaturated systems [13] [17].	Feature more nitrogen atoms. Functional groups are generally chemically easily accessible [13] [17].

The fragment analysis underscores that NPs and SCs are built from fundamentally different chemical "vocabularies." NPs are rich in oxygen-based functionality and complex, saturated architectures, while SCs are rich in nitrogen, aromatics, and halogens, reflecting the synthetic chemist's toolkit [13] [17] [14].

Occupation and Evolution of Chemical Space

The distinct structural features of NPs and SCs translate into unique patterns of chemical space occupation.

Diversity and Uniqueness

- **Structural Diversity**: NPs consistently cover a more diverse chemical space than SCs. The iSIM framework analysis of library releases shows that merely increasing the number of SCs does not automatically translate to increased chemical diversity [1]. The intrinsic diversity of NP libraries often remains higher.
- **Uniqueness of Space**: The chemical space of NPs has become less concentrated and more unique over time. Recently discovered NPs increasingly occupy regions of chemical space that are sparsely populated by SCs [13]. Conversely, the chemical space of SCs, while broad, is more densely concentrated and homogeneous in comparison.

Temporal Evolution of Chemical Space

A time-dependent analysis reveals divergent evolutionary trajectories:

- **NPs**: The NP chemical space has been expanding outward, with newer compounds becoming larger, more complex, and more hydrophobic. This expansion is driven by the discovery of structurally novel compounds that push the boundaries of known chemical space [13].
- **SCs**: The evolution of SCs has been heavily influenced by the pursuit of "drug-likeness." While SCs have undergone a continuous shift in physicochemical properties, these changes are constrained within a defined range governed by rules like Lipinski's Rule of Five [13]. Although SCs possess a broader range of synthetic pathways and the potential for immense structural diversity, their evolution has not fully mirrored the direction of NPs. There has been a noted decline in the biological relevance of SCs over time, suggesting that sheer numbers do not guarantee coverage of BioReCS [13] [1].

Biologically Relevant Chemical Space (BioReCS)

The biologically relevant chemical space is the subset of the chemical universe containing molecules with biological activity [6]. NPs, by virtue of their co-evolution with biological targets, are inherently enriched in BioReCS. It is estimated that 68% of approved small-molecule drugs between 1981 and 2019 were directly or indirectly derived from NPs [13]. This highlights that the NP chemical subspace is a privileged region for identifying bioactive leads. While SC libraries can be enormous, their coverage of BioReCS is often less efficient, a phenomenon that contributes to high attrition rates in high-throughput screening campaigns focused purely on synthetic libraries [13] [14].

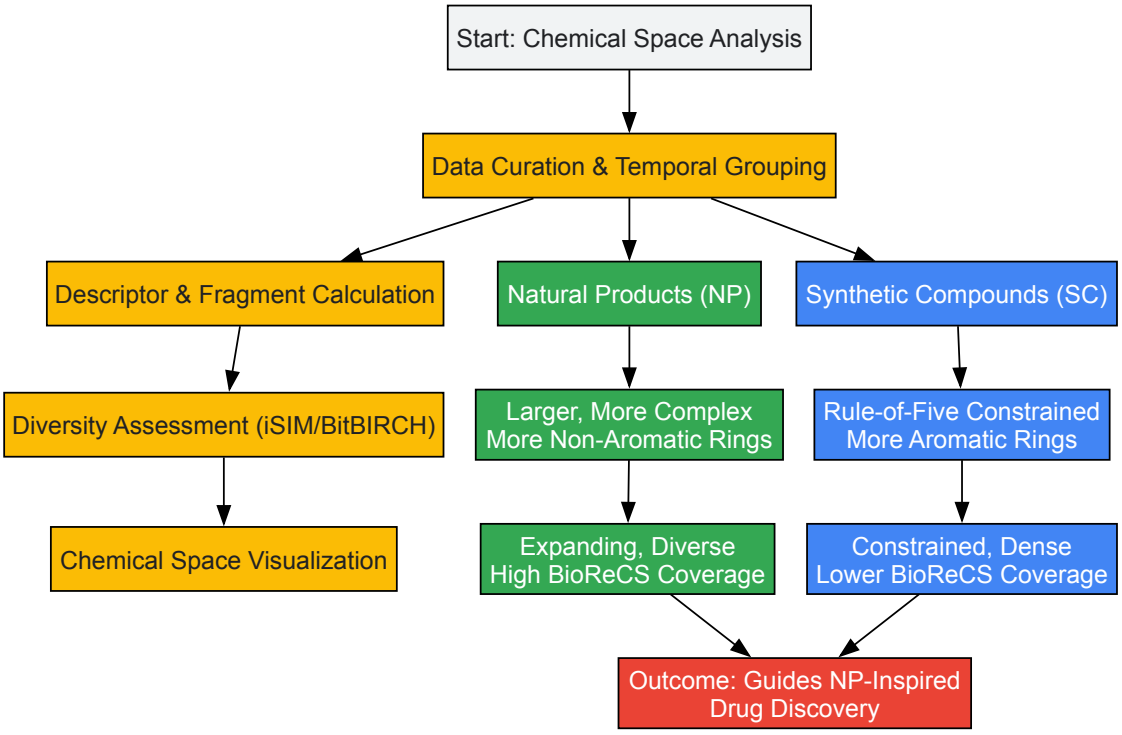


Figure 1: Workflow for comparative analysis of chemical space occupation.

The Scientist's Toolkit: Essential Research Reagents and Materials

Table 3: Key Research Reagents and Computational Tools for Chemical Space Analysis

Item/Tool	Function/Brief Explanation	Relevance to NPs vs. SCs
Dictionary of Natural Products (DNP)	A curated database of natural product structures used as a primary source for NP data [13].	Essential for obtaining standardized and reliable NP structures for comparative analysis.
ChEMBL / PubChem	Large, public databases of bioactive molecules and synthetic compounds with extensive bioactivity annotations [6] [1].	Serve as primary sources for SC data and bioactivity data to define BioReCS.
RDKit / CACTVS Toolkit	Open-source and commercial cheminformatics toolkits for calculating molecular descriptors, generating fingerprints, and processing chemical structures [13] [16].	Fundamental for descriptor calculation, structure standardization, and fragment generation.
iSIM Framework	A computational tool for efficiently calculating the intrinsic similarity (diversity) of large compound libraries with O(N) complexity [1].	Crucial for quantifying and comparing the internal diversity of massive NP and SC libraries.

Item/Tool	Function/Brief Explanation	Relevance to NPs vs. SCs
BitBIRCH Algorithm	An efficient clustering algorithm for binary fingerprints that enables the grouping of millions of molecules [1].	Allows for a granular analysis of the cluster formation and evolution in NP and SC chemical spaces over time.
RECAP Fragmentation	A method to generate molecular fragments based on chemically sensible retrosynthetic rules [13].	Used to decompose NPs and SCs into comparable, chemically meaningful fragments for diversity analysis.
Bemis-Murcko Scaffolds	An algorithm to extract the core ring system and linker of a molecule, ignoring side chains [13].	Enables comparison of the central scaffolds of NPs and SCs, highlighting differences in core complexity.
LHASA Transform Rules	A set of rules originally developed for retrosynthetic analysis, encoding robust chemical reactions [16].	Used to generate synthetically accessible virtual inventories (SAVI) and define the synthetic feasibility of chemical space regions.

Discussion and Future Perspectives

The comprehensive, time-dependent analysis confirms that natural products and synthetic compounds occupy distinct and evolving regions of the chemical universe. NPs continue to be a source of unprecedented structural complexity and diversity, densely populating the biologically relevant chemical space due to their evolutionary history. SCs, while vast in number and accessible through synthesis, have not fully evolved toward the structural profiles of NPs, remaining constrained by drug-like paradigms and synthetic practicality [13].

This divergence presents both a challenge and an opportunity for drug discovery. The challenge lies in the inefficient coverage of BioReCS by conventional SC libraries. The opportunity, however, is to leverage the unique attributes of NPs to guide the design of next-generation synthetic libraries. Strategies such as designing pseudo-natural products (pseudo-NPs) by combining NP fragments in novel arrangements represent a human-driven branch of chemical evolution that inherits the biological relevance of NPs while exploring new chemical territory [13] [18]. Furthermore, the application of generative artificial intelligence (AI) models for the structural modification of NPs is a transformative approach. These models can be trained to generate novel, NP-inspired structures that optimize desired properties while maintaining synthetic accessibility, effectively bridging the gap between the NP and SC chemical subspaces [18].

In conclusion, the future of productive drug discovery lies not in choosing between natural products and synthetic compounds, but in intelligently integrating them. By understanding their complementary strengths—the biological relevance and complexity of NPs, and the vast synthetic accessibility and tunability of SCs—researchers can more effectively navigate and populate the chemical universe to discover the life-saving therapeutics of tomorrow.

The concept of "chemical space" provides a powerful framework for understanding the relationship between molecular structure and biological activity in drug discovery. This multidimensional space, where molecular properties define coordinates and relationships between compounds, contains a specialized region known as the Biologically Relevant Chemical Space (BioReCS), which encompasses molecules with biological activity—both beneficial and detrimental [6]. Natural products (NPs) represent a uniquely privileged region within BioReCS, having evolved over millions of years to interact with biological systems through **evolutionary refinement** [19]. Unlike synthetic compound libraries designed primarily around synthetic accessibility and compliance with simplified rules, NPs originate from **biological necessity**, functioning as defense chemicals, signaling agents, and ecological mediators fine-tuned for optimal interactions with living systems [19].

The drug-likeness paradigm has historically been dominated by rule-based approaches such as Lipinski's Rule of Five, which established molecular guidelines for oral bioavailability. However, NPs consistently challenge these conventions, demonstrating that **molecular complexity** and **structural diversity** can confer superior biological targeting despite deviations from traditional drug-like properties [19]. This apparent paradox—how NPs balance structural complexity with favorable pharmacological properties—stems from their unique position within chemical space and their evolutionary optimization for biological interfaces, offering valuable lessons for modern drug discovery.

Structural and Physicochemical Characteristics of Natural Products

Defining Features in Chemical Space

Natural products occupy a distinct region of chemical space characterized by structural properties that differ significantly from those of synthetic compounds (SCs). When analyzed through chemoinformatic approaches, NPs exhibit several distinguishing characteristics that contribute to their biological relevance and drug-likeness:

Table 1: Key Structural Differences Between Natural Products and Synthetic Compounds

Characteristic	Natural Products	Synthetic Compounds	Biological Implication
Molecular complexity	Higher proportions of sp^3-hybridized carbon atoms, increased stereochemical complexity [19]	More planar structures, lower sp^3 character	Enhanced 3D shape complementarity with biological targets
Oxygen content	Higher oxygenation [19]	Lower oxygen content	Improved hydrogen bonding capacity
Nitrogen content	Lower nitrogen content [19]	Higher nitrogen content	Different target recognition patterns
Aromatic rings	Fewer aromatic rings [13]	Predominance of aromatic rings, especially benzene derivatives	Reduced planarity, better solubility
Ring systems	Larger fused rings (bridged rings, spiral rings) [13]	More five- and six-membered rings	Structural rigidity and defined 3D geometry
Molecular weight	Generally larger [13]	Constrained by synthetic and drug-like rules	Broader interaction surface with targets

Property Distributions and Drug-Likeness

Analysis of property distributions reveals that NPs often fall outside the traditional "drug-like" space defined by synthetic compounds yet demonstrate favorable bioavailability. This apparent contradiction can be explained by their **structural biosynthesis** and **evolutionary optimization**. While synthetic compounds are typically designed with strict adherence to rules such as Lipinski's Rule of Five, NPs frequently violate these guidelines yet maintain excellent pharmacological profiles [19]. For instance, many NP-based drugs exhibit exceptional oral bioavailability despite non-compliance with traditional rules, as evidenced by the increasing average molecular weight of newly approved oral medications [19].

Time-dependent analyses of NPs and SCs reveal divergent evolutionary trajectories in chemical space. NPs have progressively become larger, more complex, and more hydrophobic over time, exhibiting **increased structural diversity** and **uniqueness** [13]. Conversely, SCs have undergone continuous shifts in physicochemical properties constrained within a defined range governed by drug-like constraints, resulting in a decline in biological relevance despite broader synthetic diversity [13].

The Chemical Space Perspective: Why Natural Products Succeed

Privileged Positioning in BioReCS

The biologically relevant chemical space (BioReCS) comprises molecules with biological activity, both beneficial and detrimental [6]. Natural products occupy a privileged position within BioReCS due to their evolutionary history and biological origins. Several factors contribute to this advantageous positioning:

- **Evolutionary Optimization**: NPs have evolved through natural selection to interact specifically with biological macromolecules, leading to **inherent bio-compatibility** and **target affinity** [19]. This evolutionary refinement provides NPs with mechanisms of action that exploit biological vulnerabilities, particularly in pathogens and cancer cells [19].
- **Structural Complementarity**: The elevated molecular complexity of NPs, including higher sp^3 character and increased stereochemical complexity, enables superior **three-dimensional binding** with protein targets compared to the more planar structures typical of synthetic libraries [19].
- **Polypharmacology Potential**: The structural complexity of NPs facilitates simultaneous interactions with multiple biological targets, making them particularly valuable for treating complex diseases through **network modulation** rather than single-target inhibition [20].

Multi-Target Engagement and Network Pharmacology

Complex diseases often involve intricate molecular networks that are difficult to modulate with single-target drugs. Natural products offer inherent advantages in this context due to their **multi-target capabilities** [20]. The "single target, single disease" model has shown limitations in clinical practice, often resulting in insufficient therapeutic effects, adverse side effects, and drug resistance [20]. NPs naturally address these challenges through their ability to simultaneously regulate multiple targets within disease network systems, affecting overall physiological balance and potentially improving efficacy while reducing toxicity and resistance [20].

This multi-target engagement represents a fundamental aspect of the drug-likeness paradigm for NPs, where **balanced polypharmacology** compensates for deviations from traditional drug-like rules. Rather than maximizing affinity for a single target, NPs typically exhibit moderate

affinity for multiple targets, creating a more holistic therapeutic effect particularly valuable for complex, multifactorial diseases [20].

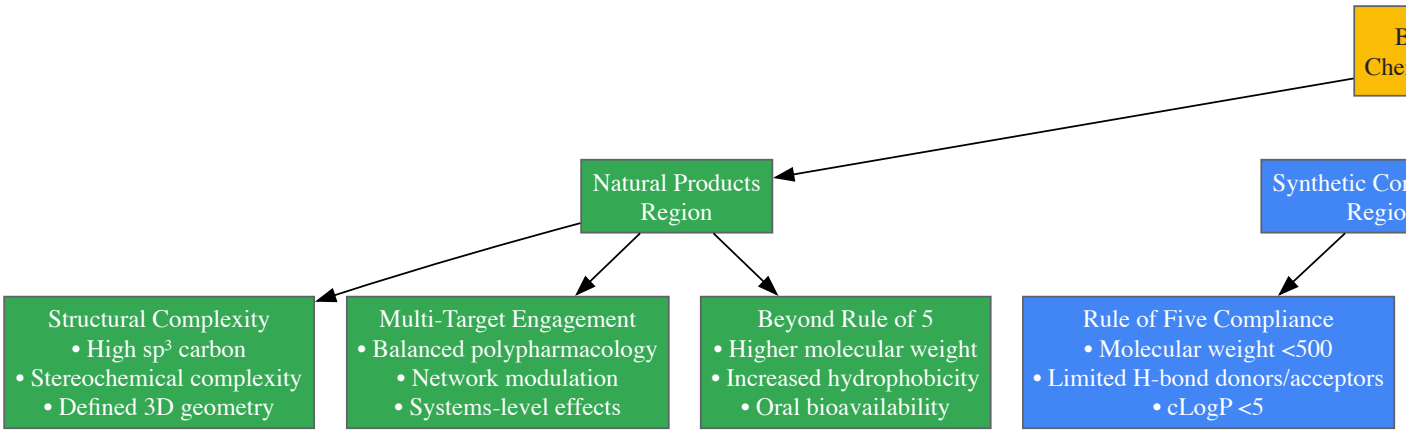


Figure 1: Relationship between natural products, synthetic compounds, and specialized regions within the biologically relevant chemical space (BioReCS)

Experimental Approaches for Natural Product Drug Discovery

Advanced Screening and Characterization Technologies

Modern NP drug discovery employs sophisticated technologies that address historical limitations while leveraging the unique advantages of natural product chemistry:

Table 2: Key Methodologies in Natural Product Drug Discovery

Methodology	Technical Approach	Application in NP Drug Discovery
Liquid Chromatography-Mass Spectrometry (LC-MS)	High-resolution separation coupled with mass detection	Metabolome profiling, chemical feature identification, and dereplication [21]
High-Throughput Screening (HTS)	Automated screening of compound libraries against biological targets	Identification of bioactive NPs from large collections [20] [19]
Genome Mining	Bioinformatics analysis of biosynthetic gene clusters (BGCs)	Prediction of NP diversity and prioritization of strains for chemical investigation [19]
Artificial Intelligence (AI)	Machine learning and generative models for pattern recognition	NP structural modification, property prediction, and activity optimization [18]
Quantitative In Vitro to In Vivo Extrapolation (QIVIVE)	Mathematical modeling to translate in vitro activity to in vivo doses	Prediction of human pharmacokinetics and efficacious doses [22]

Quantitative Library Design and Metabolomics

Rational design of NP screening libraries has been revolutionized by quantitative approaches that maximize chemical diversity while minimizing redundancy. The integration of **genetic barcoding** with **metabolomics profiling** enables researchers to build NP libraries with predetermined levels of chemical coverage [21]. This approach allows for the identification of overlooked pockets of chemical diversity within taxa, refocusing collection strategies toward underexplored regions of chemical space.

Studies on fungal isolates have demonstrated that a surprisingly modest number of isolates (195 isolates) can capture nearly 99% of chemical features within a dataset [21]. However, the observation that 17.9% of chemical features appeared in single isolates suggests that fungi continuously explore nature's metabolic landscape, presenting both challenges and opportunities for library design [21]. This quantitative framework enables evidence-based decisions about sampling depth and resource allocation in NP discovery programs.

Computational and Modeling Approaches

AI and Generative Models for Natural Product Optimization

Artificial intelligence has emerged as a transformative technology for NP-based drug discovery, particularly in the structural modification of natural products for improved drug-likeness. AI-driven approaches address several key challenges in NP optimization:

- **Target-Guided Optimization**: When target structures are known, generative models can propose structural modifications that maintain or enhance binding affinity while improving pharmacokinetic properties [18].
- **Phenotype-Based Optimization**: For systems where molecular targets remain unidentified, AI models can learn structure-activity relationships from phenotypic screening data to guide optimization [18].
- **Chemical Space Navigation**: AI algorithms can efficiently explore the chemical space around promising NP scaffolds, identifying structural variations that balance complexity with favorable properties [18].

These approaches leverage the privileged positioning of NPs in BioReCS while addressing potential limitations such as poor solubility or metabolic instability through targeted structural modifications.

Read-Across and RASAR Modeling

The Read-Across Structure-Activity Relationship (RASAR) methodology represents an innovative approach that combines elements of quantitative structure-activity relationship (QSAR) modeling with similarity-based read-across predictions [23]. This hybrid approach is particularly valuable for NP research where experimental data may be limited. RASAR modeling incorporates **similarity-based descriptors** and **error-based descriptors** into a machine learning framework, using similarity-based information from close structural neighbors to predict properties of query compounds [23].

This approach has demonstrated superior predictive performance compared to conventional QSAR models, particularly for complex endpoints like hepatotoxicity [23]. For NP research, RASAR modeling offers a powerful tool for predicting ADMET properties while navigating the complex chemical space occupied by natural products.

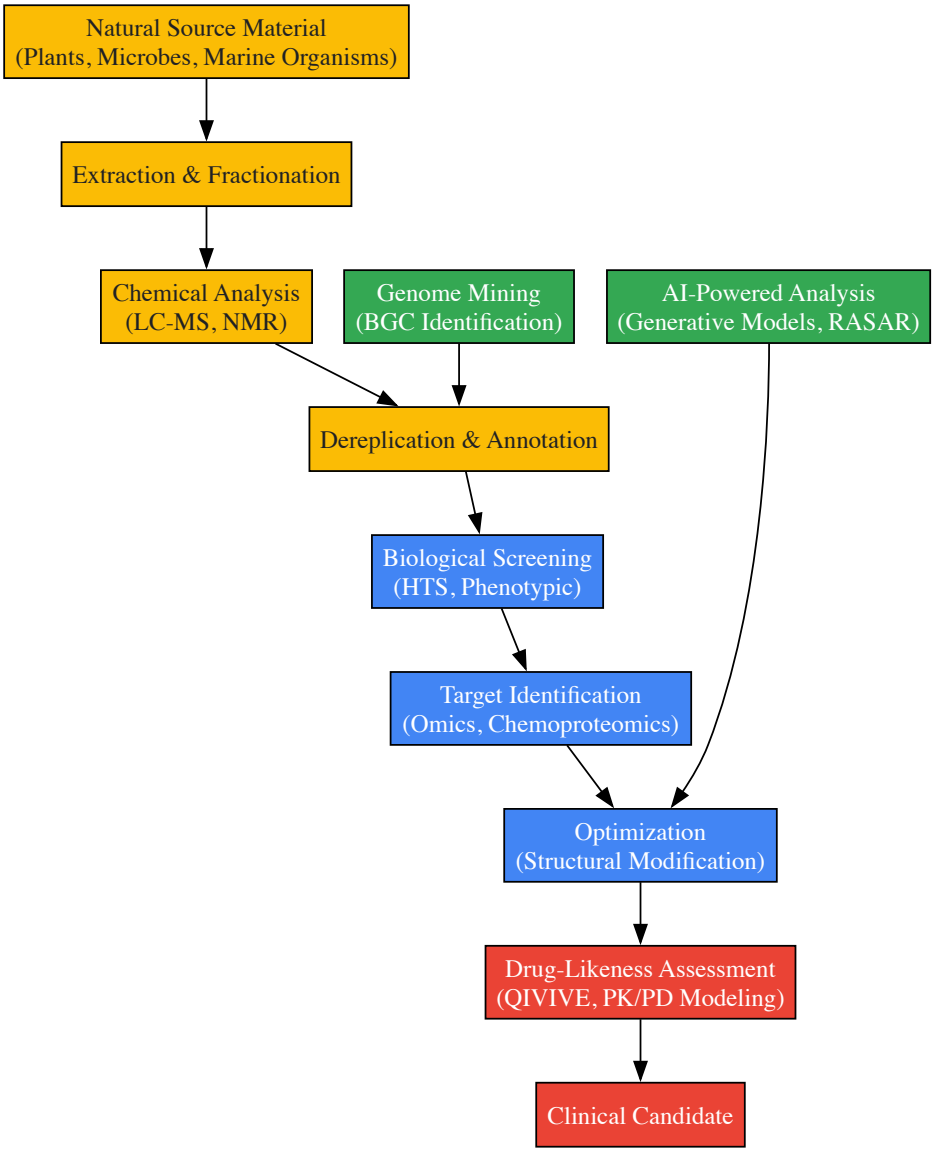


Figure 2: Integrated workflow for natural product drug discovery combining traditional and modern technologies

The Scientist's Toolkit: Essential Research Reagents and Technologies

Table 3: Key Research Reagent Solutions for Natural Product Drug Discovery

Research Tool	Function	Application Context
ChEMBL Database	Public database of bioactive molecules with drug-like properties [6]	Comparison of NP properties with known bioactive compounds
PubChem Database	Public repository of chemical substances and their biological activities [6]	Chemical space analysis and activity prediction
Global Natural Products Social Molecular Networking (GNPS)	Mass spectrometry data sharing and annotation platform [14]	NP identification and dereplication
AntiSMASH	Bioinformatics platform for identifying biosynthetic gene clusters [19]	Genome mining for novel NP discovery
InertDB	Database of curated inactive compounds [6]	Definition of non-biologically relevant chemical space
Induced Pluripotent Stem Cells (iPSCs)	Patient-derived cell models for phenotypic screening [19]	Biologically relevant screening for NP activity
CRISPR-Cas Systems	Gene editing technology for target validation [19]	Mechanism of action studies for NPs

Natural products present a compelling paradox within the drug-likeness paradigm: they frequently violate conventional rules of drug-likeness while demonstrating exceptional pharmacological success. This apparent contradiction resolves when viewed through the lens of chemical space and evolutionary optimization. NPs occupy a privileged region within the biologically relevant chemical space, shaped by millions of years of evolutionary refinement for biological interfaces [19].

The structural complexity of NPs—including higher sp^3 character, increased stereochemical complexity, and diverse ring systems—provides superior three-dimensional complementarity with biological targets compared to synthetic compounds [19] [13]. This structural advantage, combined with inherent multi-target engagement capabilities, positions NPs as ideal starting points for addressing complex diseases requiring systems-level modulation [20].

Modern approaches to NP-based drug discovery leverage advanced technologies including genomics, AI, and sophisticated modeling to navigate the complex relationship between molecular structure and biological activity. By understanding and respecting the unique positioning of NPs within chemical space, researchers can more effectively harness nature's chemical ingenuity for therapeutic innovation. The future of NP-inspired drug discovery lies in integrated approaches that combine traditional knowledge with modern technologies, guided by a deeper understanding of the fundamental principles that govern bio-relevance in chemical space.

Natural products (NPs) represent an invaluable resource for drug discovery and development, with an estimated 80% of all clinically used antibiotics originating from natural compounds [24]. The exploration of natural product **chemical space**—the multi-dimensional descriptor-based domain encompassing all possible natural compounds—is crucial for identifying novel bioactive molecules. However, the experimental characterization of natural products remains resource-intensive, with only approximately 400,000 fully characterized natural products known to date [24]. This limitation has accelerated the development of comprehensive databases that catalog known natural products and enable in-silico exploration of their structural diversity.

Within the context of natural products research, **structural diversity** refers to the variety of molecular scaffolds, functional groups, and physicochemical properties represented across different natural product collections. Understanding this diversity is essential for drug development professionals seeking to identify novel therapeutic candidates or explore structure-activity relationships. This technical guide provides an in-depth analysis of three major natural product databases—COCONUT, NPAtlas, and SuperNatural II—as essential exploratory tools for mapping the chemical space of natural products, complete with experimental methodologies and visualization frameworks for researchers in the field.

Comparative Analysis of Major Natural Product Databases

The landscape of natural product databases has expanded significantly, with resources ranging from comprehensive global collections to specialized focused databases. The table below provides a quantitative comparison of the three primary databases examined in this guide:

Table 1: Technical Comparison of Major Natural Product Databases

Database	Content Size	Update Frequency	Accessibility	License	Unique Features
COCONUT	>400,000 compounds [24]	Regular updates (2024 version available) [25]	Freely accessible without restrictions [25]	Creative Commons CC0 [25]	Integrated NP-likeness scoring, structural diversity analysis, user submissions [25] [26]
NPAtlas	32,552 microbial compounds [27]	2021 version (most recent) [27]	Open access [27]	Not specified	Specialized in microbial natural products, taxonomic origins, chemical ontology terms [27]
SuperNatural II	~326,000 molecules [28]	2022 update (version 3.0) [28]	Freely available without registration [28]	Not specified	Predicted toxicity, vendor information, mechanism of action, taste prediction [28]

Each database offers distinct advantages for researchers exploring chemical space. **COCONUT** provides the most extensive collection with comprehensive annotation, while **NPAtlas** offers specialized coverage of microbial natural products, and **SuperNatural II** includes valuable predictive data on toxicity and biological activity. The selection of an appropriate database depends on the specific research objectives, whether focused on broad chemical space exploration or targeted investigation of particular natural product classes.

Database-Specific Architectures and Functional Capabilities

COCONUT (COlleCtion of Open Natural prodUcTs)

COCONUT represents one of the most comprehensive open natural product databases, launched in 2021 and regularly updated since [25] [29]. The database architecture incorporates not only chemical structures but also extensive metadata including names and synonyms, source organisms and specific plant parts, geographical collection data, and literature references [26]. The platform enables multiple search modalities including textual queries, exact structure matching, substructure search, and chemical similarity analysis, providing researchers with flexible tools for chemical space exploration [26].

A key innovation in COCONUT 2.0 is the implementation of community curation features and user data submissions, enhancing the database's comprehensiveness and accuracy through collaborative scientific effort [26]. The database also provides specialized analytical tools including fragment analysis using the Ertl algorithm for functional group identification, scaffold generation for core structure analysis, and calculated properties including NP-likeness scores, synthetic accessibility scores, and QED drug-likeness metrics [25]. These features enable researchers to quantitatively assess the position of compounds within the broader chemical space of natural products.

NPAtlas (The Natural Products Atlas)

NPAtlas specializes in microbially-derived natural products, providing detailed coverage of bacterial and fungal metabolites [27]. The database incorporates full taxonomic descriptions of source microorganisms and employs dual chemical ontology systems (NP Classifier and ClassyFire) for standardized structural classification [27]. This specialized focus makes NPAtlas particularly valuable for researchers investigating microbial chemical ecology or seeking novel antibiotics and other bioactive compounds from microbial sources.

The NPAtlas platform features a comprehensive application programming interface (API) that supports programmatic access and integration with other bioinformatics resources [27]. The database is explicitly developed following FAIR principles (Findable, Accessible, Interoperable, and Reusable), ensuring optimal utility for computational natural products research [27]. Integration with complementary resources including the Minimum Information About a Biosynthetic Gene Cluster (MIBiG) repository and the Global Natural Products Social Molecular Networking (GNPS) platform enables multi-omics approaches to natural product discovery [27].

SuperNatural II

SuperNatural II provides extensive coverage of natural products and natural product-based derivatives, with particular emphasis on drug discovery applications [28]. The database includes predictive data on biological activity, including target-specific predictions for antiviral, antibacterial, antimalarial, and anticancer applications, as well as central nervous system (CNS) target potential [28]. This functional annotation makes it particularly valuable for drug development professionals seeking to identify lead compounds for specific therapeutic areas.

A distinctive feature of SuperNatural II is its incorporation of taste prediction using VirtualTaste models, expanding its utility beyond pharmaceutical applications to include food and sensory sciences [28]. The database also includes vendor information for many compounds, facilitating the acquisition of physical samples for experimental validation [28]. The platform supports multiple search strategies including template-based similarity searching, compound name queries, substructure searches, and physical property filters, enabling efficient navigation of its extensive chemical space [28].

Experimental Protocols for Database Utilization

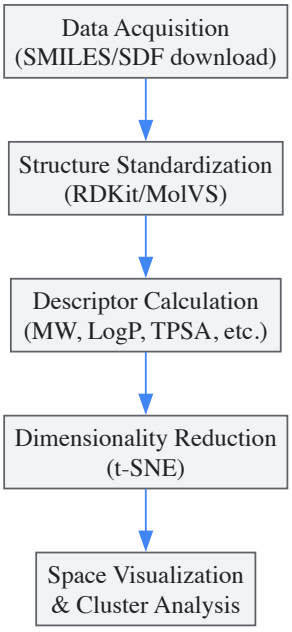
Protocol 1: Chemical Space Mapping Using Molecular Descriptors

Purpose: To quantitatively map and visualize the chemical space covered by natural product databases to assess structural diversity and identify regions of interest for targeted screening.

Methodology:

- **Data Acquisition**: Download canonical SMILES representations of natural products from target databases. For COCONUT, bulk download is available in SDF, CSV, or database dump formats [26].
- **Structure Standardization**: Process structures using standardization tools such as RDKit or MolVS to ensure consistent representation, including neutralization, desalting, and canonical tautomer generation [30].
- **Descriptor Calculation**: Compute key molecular descriptors using cheminformatics toolkits (e.g., RDKit):
  - Molecular weight (MW)
  - Partition coefficient (Wildman-Crippen LogP)
  - Hydrogen bond acceptors/donors (HBA/HBD)
  - Topological polar surface area (TPSA)
  - Fraction of carbon sp^3 atoms ($FCSP^3$)
  - Number of rotatable bonds [24]
- **Chemical Space Visualization**: Apply dimensionality reduction techniques such as t-distributed stochastic neighbor embedding (t-SNE) to project high-dimensional descriptor data into two-dimensional space for visualization [24].
- **Cluster Analysis**: Identify dense regions and outliers in the chemical space mapping to guide targeted exploration efforts.

Figure 1: Workflow for Chemical Space Mapping and Analysis



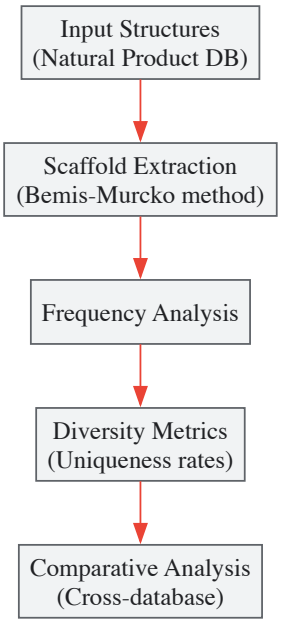
Protocol 2: Scaffold Analysis for Structural Diversity Assessment

Purpose: To identify and quantify molecular scaffolds within natural product databases to assess structural diversity and scaffold representation across different biological sources.

Methodology:

- **Scaffold Extraction**: Generate molecular scaffolds according to the Bemis-Murcko method, which involves systematic removal of side chains to reveal core structural frameworks [30].
- **Scaffold Classification**: Categorize scaffolds based on structural properties (e.g., ring systems, connectivity) using tools such as RDKit or specialized scaffold generators.
- **Frequency Analysis**: Quantify scaffold occurrence to identify prevalent structural motifs within and across databases.
- **Diversity Metrics**: Calculate scaffold diversity metrics including scaffold uniqueness rates and scaffold-hitting compounds to quantify structural diversity [25].
- **Comparative Analysis**: Compare scaffold distributions across different natural product databases or between natural products and synthetic compound libraries.

Figure 2: Scaffold Diversity Analysis Workflow



Protocol 3: Virtual Screening for Natural Product Discovery

Purpose: To employ natural product databases for in-silico screening against biological targets to identify potential lead compounds with desired bioactivity.

Methodology:

- **Library Preparation**: Curate and filter natural product databases based on drug-likeness criteria (e.g., Lipinski's Rule of Five, synthetic accessibility scores) [25].
- **Target Preparation**: Obtain three-dimensional structures of biological targets (e.g., from Protein Data Bank) and prepare for docking (add hydrogens, assign charges).
- **Molecular Docking**: Perform high-throughput docking of natural product libraries against biological targets using software such as AutoDock Vina or Schrödinger Glide.
- **Score Interpretation**: Analyze docking scores and binding interactions to identify promising natural product hits.
- **Hit Prioritization**: Prioritize hits based on multiple criteria including docking scores, chemical novelty, structural diversity, and synthetic accessibility [25].

Advanced Applications in Chemical Space Exploration

Expanding Natural Product Chemical Space with AI-Generated Libraries

Recent advances in deep generative modeling have enabled significant expansion of explorable natural product chemical space. As demonstrated by a 2023 study, recurrent neural networks (RNNs) trained on known natural products from COCONUT can generate over 67 million natural product-like structures—a 165-fold expansion beyond known natural products [24]. This approach dramatically increases the accessible chemical space for virtual screening campaigns while maintaining natural product-like characteristics.

The generated structures exhibit NP Score distributions closely resembling known natural products (Kullback-Leibler divergence of 0.064 nats), confirming their natural product-like character [24]. Furthermore, t-SNE visualization of molecular descriptor space shows that these AI-generated libraries cover significantly expanded physicochemical space while maintaining dense coverage in regions occupied by known natural products, suggesting their utility for exploring novel yet biologically relevant chemical space [24].

Cross-Database Chemical Space Integration

Integrating multiple natural product databases enables more comprehensive mapping of natural product chemical space. The Latin American Natural Product Database (LaNAPDB) initiative represents one such effort, combining regional databases from multiple countries including Argentina (NaturAr), Brazil (NuBBEDB, SistematX, UEFS), Panama (CIFPMA), Peru (PeruNPDB), and Mexico (BIOFACQUIM, UNIIQUIM) [30]. Similar integration approaches can be applied to the major global databases discussed in this guide, creating a unified chemical space map for natural products.

Essential Research Reagent Solutions

The following table details key computational tools and resources essential for natural product database research and chemical space exploration:

Table 2: Essential Research Reagent Solutions for Natural Products Chemical Space Analysis

Tool/Resource	Function	Application in Natural Products Research
RDKit	Open-source cheminformatics toolkit [30]	Structure standardization, molecular descriptor calculation, scaffold analysis [24]
NPClassifier	Deep learning-based natural product classification [24]	Biosynthetic pathway annotation and structural classification [24]
NP Score	Natural product-likeness quantification [24]	Bayesian measure of similarity to natural product structural space [24]
MolVS	Molecule validation and standardization [30]	Structure normalization for comparative analysis [30]
ChEMBL Curation Pipeline	Chemical structure curation [24]	Structure sanitization and standardization following FDA/IUPAC guidelines [24]

COCONUT, NPAtlas, and SuperNatural II provide complementary platforms for exploring the chemical space and structural diversity of natural products. COCONUT offers the most extensive collection with advanced analytical capabilities, NPAtlas provides specialized coverage of microbial metabolites, and SuperNatural II incorporates valuable predictive data for drug discovery applications. Together, these databases enable comprehensive mapping of natural product chemical space, facilitating the identification of novel bioactive compounds and expanding our understanding of nature's structural diversity. The experimental protocols and analytical frameworks presented in this guide provide researchers with standardized methodologies for leveraging these resources in natural product discovery and development efforts.

Tools and Techniques for Mapping and Exploiting NP Chemical Diversity

The exploration of chemical space is a fundamental objective in natural product-based drug discovery. Chemical space can be conceptualized as a multidimensional universe where molecular properties define coordinates and relationships between compounds [6]. Within this vast universe, the biologically relevant chemical space (BioReCS) comprises molecules with biological activity, a region where natural products (NPs) have historically played a paramount role [31] [6]. It has been estimated that from the drugs approved between 1981 and 2019, 3.8% are unaltered natural products, while 18.9% are natural product derivatives [31].

Natural products present unique cheminformatic challenges due to their distinct structural characteristics. Compared to typical synthetic, drug-like compounds, NPs often exhibit a wider range of molecular weight, contain multiple stereocenters, and possess a higher fraction of sp^3-hybridized carbons [32]. This structural complexity, while contributing to their biological potency and selectivity, makes the encoding of natural products via traditional molecular representations particularly challenging [32] [33]. This technical guide comprehensively explores the key cheminformatics methodologies—molecular descriptors, fingerprints, and scaffold analysis—employed to navigate and characterize the complex chemical space of natural products, thereby facilitating modern drug discovery efforts.

Molecular Descriptors and Fingerprints for Natural Products

Defining the Chemical Space of Natural Products

Molecular descriptors are numerical representations derived from molecular structure that serve as the fundamental metrics for quantifying chemical space [34]. Profiling natural product datasets typically involves a core set of physicochemical properties that capture size, polarity, and flexibility: **Molecular Weight (MW)**, the **octanol/water partition coefficient (SlogP)**, **Topological Polar Surface Area (TPSA)**, counts of **Hydrogen Bond Donors (HBD)** and **Acceptors (HBA)**, and the number of **Rotatable Bonds (RB)** [31]. These descriptors help contextualize NPs within the broader drug-like chemical space and are essential for understanding their absorption, distribution, metabolism, excretion, and toxicity (ADME/Tox) profiles [31].

Molecular Fingerprints: Encoding Structure for Computation

Molecular fingerprints convert molecular structures into fixed-length vectors, enabling computational processing and similarity assessment [32]. They are crucial for quantitative structure-activity relationship (QSAR) modeling and virtual screening [32]. The performance of different fingerprinting algorithms varies significantly when applied to natural products due to their unique structural motifs [32] [33].

Table 1: Major Categories of Molecular Fingerprints and Their Application to Natural Products

Fingerprint Category	Description	Examples	Performance on NPs
Path-Based	Generates features by analyzing paths through the molecular graph [32].	Depth First Search (DFS), Atom Pairs (AP) [32]	Variable performance; depends on structural class [32].
Circular	Dynamically constructs fragments from the molecular graph by iteratively adding neighbor information [32].	Extended Connectivity Fingerprints (ECFP), Functional Class Fingerprints (FCFP) [32]	ECFP is the de-facto standard for drug-like compounds but can be matched or outperformed by other fingerprints for NPs [32] [33].
Substructure-Based	Each bit encodes the presence/absence of a predefined structural moiety [32].	MACCS, PUBCHEM [32]	Limited by the predefined list of substructures, which may not cover NP-specific motifs [32].
Pharmacophore-Based	Encodes atoms based on their pharmacophoric features (e.g., hydrogen bond donor/acceptor) rather than pure structure [32].	Pharmacophore Pairs (PH2), Triplets (PH3) [32]	Can capture functional similarities beyond scaffold structure, useful for scaffold hopping [34].
String-Based	Operates on the SMILES string representation of the compound [32].	LINGO, MinHashed (MHFP), MinHashed Atom Pairs (MAP4) [32]	MAP4 shows promise for capturing NP complexity and performs well in bioactivity prediction [32].

A comprehensive benchmark study evaluating 20 different fingerprints on over 100,000 unique natural products revealed that while ECFPs are a common choice, other fingerprints can match or surpass their performance for specific tasks like bioactivity prediction [32] [33]. This highlights the importance of testing multiple fingerprint algorithms for optimal performance on NP-focused projects [32].

Scaffold Analysis in Natural Product Research

The Informacophore: A Data-Driven Pharmacophore

In classical medicinal chemistry, the **pharmacophore** represents the spatial arrangement of chemical features essential for target recognition. A modern extension is the **"informacophore,"** which integrates traditional knowledge with data-driven insights from computed molecular descriptors, fingerprints, and machine-learned representations [35]. The informacophore defines the minimal chemical structure and its essential features for biological activity, acting as a "skeleton key" for triggering biological responses [35]. This concept is particularly powerful for analyzing ultra-large datasets of lead compounds, helping to reduce biased intuitive decisions and accelerate discovery [35].

Privileged Scaffolds and Scaffold Hopping

Privileged scaffolds are molecular frameworks that consistently produce biologically active compounds across multiple targets and indications [36]. Identifying these scaffolds within natural products provides a valuable source of novel leads that can circumvent existing patents [36]. **Scaffold hopping**—the process of identifying structurally distinct compounds that share the same biological activity—is a critical strategy for innovating from these privileged scaffolds [34].

Traditional reductionist descriptors can struggle with scaffold hopping from NPs due to their structural complexity. Holistic molecular representations like **WHALES (Weighted Holestone Atom Localization and Entity Shape)** descriptors have been developed to overcome this. WHALES simultaneously encode information on geometric interatomic distances, molecular shape, and atomic partial charge distributions, facilitating the identification of synthetic mimetics of complex natural products [34]. A prospective application using WHALES to find synthetic modulators of cannabinoid receptors from phytocannabinoid queries achieved a 35% experimental success rate, with several identified scaffolds being novel compared to existing databases [34].

Experimental Protocols and Workflows

Protocol 1: Benchmarking Molecular Fingerprints for Bioactivity Prediction

This protocol is adapted from a large-scale study evaluating fingerprint performance on natural products [32].

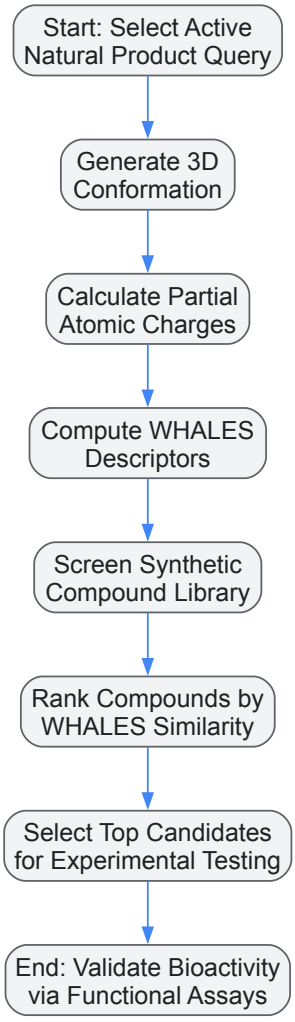
- **Dataset Curation**: Collect a standardized set of natural products. Public databases like COCONUT (COlleCtion of Open Natural prodUcTs) and CMNPD (Comprehensive Marine Natural Products Database) are suitable sources [32]. Preprocess structures using a curation package (e.g., from ChEMBL) to remove salts, neutralize charges, and ensure consistent stereochemistry.
- **Fingerprint Calculation**: Compute a diverse set of molecular fingerprints (e.g., ECFP, FCFP, MAP4, MACCS, Atom Pairs) for all compounds in the dataset using cheminformatics toolkits like RDKit [32].

- **Bioactivity Data Preparation**: Assemble classification datasets by selecting NPs annotated with a specific property (e.g., "antibacterial") as the positive class and a random sample of unannotated NPs as the negative class, enforcing a minimum dataset size (e.g., 1,000 compounds) [32].
- **Model Training and Validation**: Train a machine learning classifier (e.g., Random Forest) for each bioactivity task and fingerprint type. Validate model performance using robust cross-validation techniques.
- **Performance Analysis**: Compare fingerprints based on their average predictive performance (e.g., AUC-ROC) across all bioactivity datasets to determine the optimal encoding for the specific NP chemical space under investigation [32].

Protocol 2: Scaffold Hopping Using Holistic Descriptors

This protocol outlines the workflow for using WHALES descriptors to identify synthetic mimetics of a natural product, as proven in a prospective case study [34].

- **Query Selection and Preparation**: Select a biologically active natural product as the query. Generate a low-energy 3D conformation using a force field (e.g., MMFF94) and compute partial atomic charges (e.g., Gasteiger-Marsili) [34].
- **WHALES Descriptor Calculation**:
  - **Step 1**: For each non-hydrogen atom *j* in the query, calculate an atom-centered weighted covariance matrix (Sw(j)) using the 3D coordinates and the absolute values of the partial charges of all other atoms [34].
  - **Step 2**: From Sw(j), compute the Atom-Centered Mahalanobis (ACM) distance from atom *j* to every other atom *i* [34].
  - **Step 3**: From the ACM matrix, calculate three atomic indices for each atom: Remoteness (Rem, row-average), Isolation degree (Isol, column-minimum), and their ratio (IR) [34].
  - **Step 4**: Convert these variable-length atomic indices into a fixed-length vector by recording their deciles, minimum, and maximum values, resulting in the final 33-dimensional WHALES descriptor [34].
- **Database Screening**: Compute WHALES descriptors for a large library of commercially available synthetic compounds.
- **Similarity Searching and Compound Selection**: Calculate the molecular similarity (e.g., using Euclidean distance) between the query NP WHALES descriptor and all database compounds. Select the top-ranked compounds for experimental testing [34].



Scaffold Hopping with WHALES Descriptors. This workflow visualizes the protocol for identifying synthetic mimetics of a natural product using holistic molecular similarity [34].

Table 2: Key Resources for Cheminformatics Analysis of Natural Products

Resource Name	Type	Function and Application	Access
COCONUT [32] [31]	Database	A comprehensive, open-source collection of over 400,000 unique natural products for chemical space analysis and model training.	Public
CMNPD [32]	Database	The Comprehensive Marine Natural Products Database, used for constructing bioactivity classification datasets.	Public
RDKit [32]	Software	Open-source cheminformatics toolkit used for structure standardization, fingerprint calculation, and descriptor computation.	Public
Python Mordred Library [37]	Software Library	A tool for calculating a comprehensive set of 1,826+ 2D and 3D molecular descriptors for detailed property profiling.	Public
WHALES Descriptors [34]	Algorithm	A holistic molecular representation that integrates shape and pharmacophore features for scaffold hopping from NPs.	Implementation required
MAP4 Fingerprint [32]	Molecular Fingerprint	A string-based, MinHashed fingerprint that shows strong performance in encoding the complex structures of natural products.	Public
Enamine / OTAVA "Make-on-Demand" Libraries [35]	Virtual Compound Library	Ultra-large libraries (billions of compounds) for virtual screening to find novel, synthetically accessible hits.	Commercial

The strategic application of cheminformatics approaches is indispensable for unlocking the therapeutic potential embedded within the chemical space of natural products. As this guide has detailed, the choice of molecular descriptors and fingerprints is critical, with benchmarking studies confirming that no single method is universally superior for all NP-related tasks. The emerging concepts of informacophores and holistic molecular representations, coupled with robust experimental protocols for fingerprint evaluation and scaffold hopping, provide a powerful framework for modern research. By leveraging these tools and the vast, yet underexplored, structural diversity of natural products, researchers can continue to efficiently navigate the biologically relevant chemical space and accelerate the discovery of novel therapeutic agents.

The concept of "chemical space" provides a powerful framework for understanding the structural diversity and biological relevance of natural products (NPs). Chemical space can be defined as a multidimensional domain where each dimension represents a specific molecular property or descriptor, and each compound occupies a distinct coordinate based on its structural characteristics [38]. For researchers investigating natural products, effectively navigating this space is crucial for identifying novel bioactive compounds, understanding structure-activity relationships, and guiding drug discovery efforts. Natural products possess exceptional structural diversity that distinguishes them from synthetic compounds, making them invaluable for exploring biologically relevant regions of chemical space [39] [40]. This diversity stems from their evolutionary selection and optimization for interactions with biological macromolecules, essentially pre-validating them for biological relevance [41] [38].

The structural diversity of natural products manifests at multiple levels, from atomic composition to three-dimensional architecture. Compared to synthetic compounds found in typical medicinal chemistry collections, natural products exhibit higher structural complexity, greater molecular rigidity, fewer aromatic rings, more stereogenic centers, and higher sp3 carbon fractions [40] [38]. These characteristics enable natural products to occupy unique regions of chemical space that often remain unexplored by synthetic compounds. Studies comparing natural products from the Dictionary of Natural Products with bioactive medicinal chemistry compounds from WOMBAT have revealed significant differences in their distribution within chemical space, with natural products covering regions that lack representation in synthetic compound libraries [38]. This makes them exceptional starting points for exploring new biologically relevant chemical space and discovering novel bioactive compounds with unique mechanisms of action.

Fundamental Tools for Chemical Space Navigation

ChemGPS-NP: Global Positioning in Chemical Space

ChemGPS-NP (Chemical Global Positioning System for Natural Products) is a PCA-based tool specifically designed to handle the extensive chemical diversity encountered in natural products research [41]. Unlike earlier chemical navigation tools focused on the more restricted drug-like chemical space, ChemGPS-NP was explicitly tuned for the structurally complex domain of natural products. The system operates by mapping compounds into an eight-dimensional principal component space where each dimension captures fundamental physicochemical properties including **size, shape, polarizability, lipophilicity, polarity, flexibility, rigidity, and hydrogen bond capacity** [38].

The core innovation of ChemGPS-NP lies in its fixed reference system, which allows for consistent positioning of new compounds through PCA score prediction [41] [38]. This approach enables meaningful comparisons across different compound sets and tracking of structural changes over time.

When projected into the ChemGPS-NP space, natural products and synthetic medicinal compounds display distinct distribution patterns. Natural products tend to occupy regions characterized by greater structural rigidity (negative PC4 direction) and lower aromaticity (negative PC2 direction), while synthetic medicinal compounds often exhibit greater flexibility and aromatic character [38]. These differences highlight the unique structural properties of natural products and their ability to access regions of chemical space that are sparsely populated by synthetic compounds.

Table 1: Key Dimensions and Interpretations in ChemGPS-NP

Principal Component	Property Direction (+)	Property Direction (-)
PC1	Increasing molecular size	Decreasing molecular size
PC2	Higher aromaticity	Lower aromaticity
PC3	Increased lipophilicity	Increased polarity
PC4	Greater flexibility	Greater rigidity

Scaffold Hunter: Visual Analytics for Chemical Data

Scaffold Hunter is a comprehensive visual analytics framework specifically designed for drug discovery applications, with particular strength in analyzing natural products [42]. The tool combines techniques from data mining and information visualization to enable interactive exploration of high-dimensional chemical data. At its core, Scaffold Hunter employs a **scaffold tree algorithm** that computes a hierarchical classification for chemical compounds based on their common core structures [42].

The scaffold tree construction process follows a systematic approach: First, each compound is associated with its unique scaffold obtained by removing terminal side chains while preserving double bonds directly attached to rings. Then, each scaffold is progressively pruned through deterministic rules that remove individual rings while preserving the most characteristic core structure. This process continues until a single-ring scaffold remains [42]. The resulting hierarchy enables researchers to visualize chemical space from multiple perspectives, identifying both populated and unexplored regions that may represent opportunities for novel compound discovery.

Scaffold Hunter incorporates multiple interconnected visualization techniques, including:

- **Scaffold Tree View**: Hierarchical representation of molecular scaffolds
- **Tree Map View**: Space-filling alternative to the scaffold tree
- **Molecule Cloud View**: Compact representation of compound sets by common scaffolds
- **Heat Map View**: Matrix visualization combining property values with hierarchical clustering
- **Plot View**: For general data visualization and analysis [42]

This multi-view approach enables researchers to maintain context while focusing on specific aspects of their data, facilitating the identification of meaningful patterns in complex natural product datasets.

Dimensionality Reduction Techniques for Chemical Space Visualization

Core Mathematical Approaches

Dimensionality reduction (DR) techniques are essential for transforming high-dimensional chemical descriptor data into human-interpretable 2D or 3D visualizations, a process known as "chemography" [4]. These methods address the fundamental challenge of representing chemical structures described by dozens or hundreds of molecular descriptors in a visually comprehensible format. The main DR techniques applied in chemical space analysis include:

Principal Component Analysis (PCA) is a linear dimensionality reduction method that identifies orthogonal axes of maximum variance in the original high-dimensional data [4]. PCA is computationally efficient, deterministic (producing the same result for a given dataset), and easily interpretable as the principal components often correlate with specific physicochemical properties. However, its linear nature limits its ability to capture complex nonlinear relationships in chemical data [43].

t-Distributed Stochastic Neighbor Embedding (t-SNE) is a nonlinear technique that focuses on preserving local neighborhood structures rather than global data geometry [4]. It converts high-dimensional Euclidean distances between points into conditional probabilities representing similarities, then constructs a probability distribution over pairs of objects in the low-dimensional mapping that minimizes the Kullback-Leibler divergence between the two distributions. t-SNE excels at revealing local cluster structures but can be computationally intensive for very large datasets [4].

Uniform Manifold Approximation and Projection (UMAP) is a relatively recent nonlinear dimensionality reduction technique based on Riemannian geometry and algebraic topology [4]. UMAP assumes the data is uniformly distributed on a Riemannian manifold and seeks to learn the

manifold structure before projecting it into lower dimensions. It typically provides better preservation of global data structure compared to t-SNE while maintaining computational efficiency, making it increasingly popular for chemical space visualization [4] [43].

Generative Topographic Mapping (GTM) is a probabilistic alternative to self-organizing maps that models the high-dimensional data as arising from a latent space through a nonlinear transformation [4]. GTM creates a generative model of the data that can be used for both visualization and clustering, providing a principled probabilistic framework for chemical space analysis.

Performance Comparison and Evaluation

Recent comprehensive studies have evaluated these dimensionality reduction techniques specifically in the context of chemical space visualization using subsets from the ChEMBL database [4]. The performance was assessed based on neighborhood preservation metrics and visual interpretability, with key findings summarized in the table below:

Table 2: Performance Comparison of Dimensionality Reduction Techniques for Chemical Space Visualization

Method	Neighborhood Preservation	Global Structure	Local Structure	Computational Efficiency	Key Strengths
PCA	Moderate	Strong	Moderate	High	Interpretability, linear relationships, computational efficiency
t-SNE	High (local)	Weak	Strong	Moderate	Cluster separation, local structure preservation
UMAP	High	Strong	Strong	Moderate	Balance of local and global structure, clear clustering
GTM	High	Strong	Strong	Moderate	Probabilistic framework, property landscapes

The evaluation utilized multiple metrics to quantify neighborhood preservation, including:

- **PNNk**: Percentage of preserved nearest neighbors
- **QNN**: Co-k-nearest neighbor size
- **AUC(QNN)**: Area under the QNN curve
- **LCMC**: Local continuity meta criterion
- **Trustworthiness**: Measure of false neighbors in the projection
- **Continuity**: Measure of missing neighbors in the projection [4]

Nonlinear methods generally outperformed PCA in neighborhood preservation, with UMAP and GTM providing the best balance between local and global structure preservation. However, PCA remains valuable when interpretability and identification of linear structure-property relationships are prioritized [4] [43].

Experimental Protocols and Workflows

Chemical Space Analysis Protocol Using Dimensionality Reduction

A standardized workflow for chemical space analysis using dimensionality reduction techniques ensures consistent and reproducible results. The following protocol outlines the key steps:

Step 1: Data Collection and Curation

- Select natural product datasets from sources such as Dictionary of Natural Products, ChEMBL, or in-house compound libraries
- Apply standardization procedures to normalize molecular representations
- Remove duplicates and compounds with structural errors
- Annotate compounds with relevant biological activity data when available [4] [38]

Step 2: Molecular Descriptor Calculation

- Calculate molecular descriptors capturing structural and physicochemical properties
- Common descriptor types include:
  - **Morgan Fingerprints**: Circular fingerprints encoding atomic environments (radius 2, length 1024)
  - **MACCS Keys**: 166-bit structural key fingerprints indicating presence/absence of specific structural features
  - **Physicochemical Descriptors**: Properties such as molecular weight, logP, polar surface area, hydrogen bond donors/acceptors [4]

- Remove zero-variance features and standardize remaining features

Step 3: Dimensionality Reduction Application

- Apply selected DR methods (PCA, t-SNE, UMAP, GTM) using optimized hyperparameters
- For PCA: No hyperparameter tuning required beyond standardization
- For t-SNE: Optimize perplexity (typically 5-50), learning rate (typically 10-1000), and number of iterations
- For UMAP: Optimize number of neighbors (typically 5-50), minimum distance (typically 0.1-0.5), and metric
- For GTM: Optimize number of latent points and regularization parameters [4]

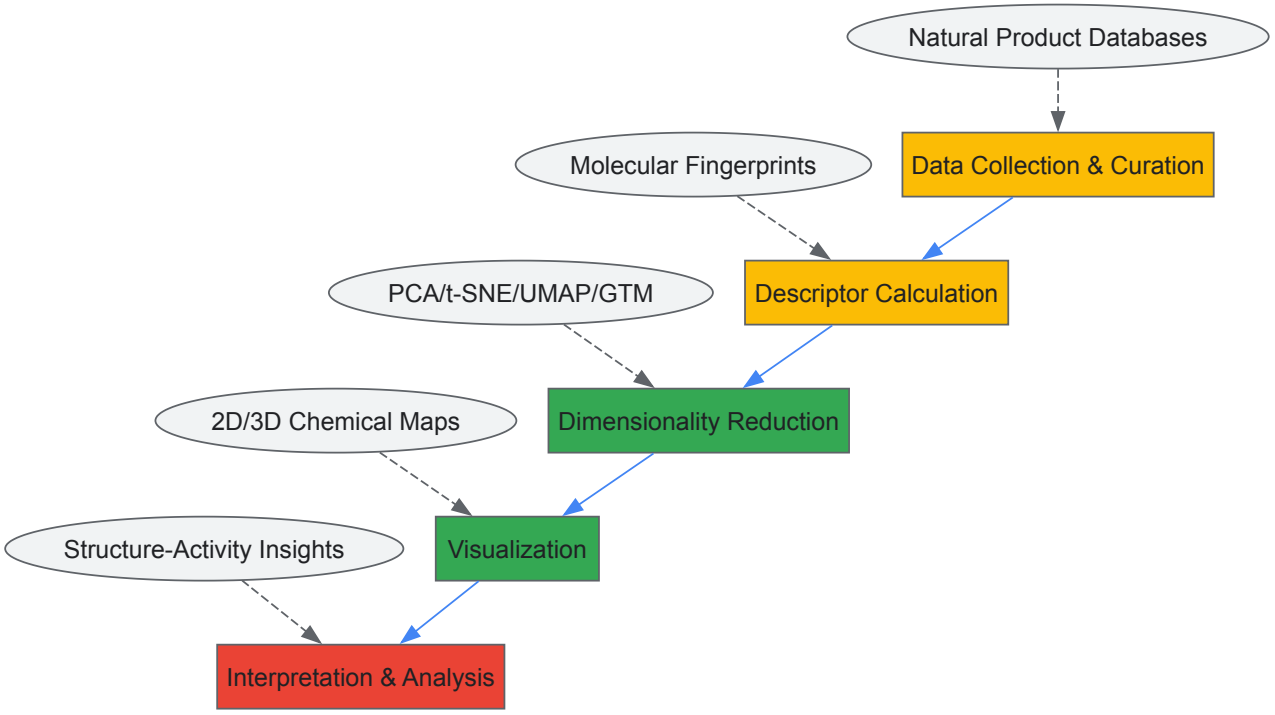
Step 4: Visualization and Interpretation

- Generate 2D or 3D scatter plots of the chemical space projections
- Color-code points based on structural properties or biological activities
- Identify clusters, outliers, and sparsely populated regions
- Corrogate chemical space positions with molecular structures to interpret patterns [4] [42]

Step 5: Validation and Analysis

- Calculate neighborhood preservation metrics to evaluate projection quality
- Perform cluster analysis to identify structurally similar compound groups
- Compare chemical space coverage across different compound collections
- Identify potential lead compounds in sparsely populated regions [4]

The following workflow diagram illustrates the key steps in chemical space analysis:



Chemical Space Analysis Workflow

Biology-Oriented Synthesis (BIOS) Protocol

Biology-Oriented Synthesis (BIOS) represents a strategic approach that uses natural product scaffolds as starting points for exploring biologically relevant chemical space [40]. The BIOS workflow involves:

Step 1: Natural Product Scaffold Selection

- Identify privileged natural product scaffolds with known biological relevance
- Use computational tools like SCONP (Structural Classification of Natural Products) for systematic scaffold simplification
- Select scaffolds that represent underrepresented regions of chemical space

Step 2: Scaffold Simplification and Analogue Design

- Apply systematic simplification rules to generate synthetically accessible core structures
- Design compound libraries around these simplified scaffolds
- Incorporate sufficient structural diversity to explore structure-activity relationships

Step 3: Library Synthesis

- Develop efficient synthetic routes enabling preparation of 30-100 derivatives
- Implement parallel synthesis approaches where possible
- Employ multistep one-pot sequences for complex scaffolds [40]

Step 4: Biological Screening and Target Identification

- Screen compound libraries in phenotypic assays or against molecular targets
- Use biotinylated analogues for target identification through pull-down assays
- Validate target engagement through competitive binding studies [40]

Successful applications of BIOS include the discovery of Wnt pathway modulators inspired by the natural product sodwanone S and hedgehog pathway inhibitors derived from sominone, demonstrating the power of this approach for identifying novel bioactive compounds [40].

Research Reagent Solutions and Essential Materials

Table 3: Essential Research Reagents and Computational Tools for Chemical Space Exploration

Tool/Resource	Type	Primary Function	Application in NP Research
ChemGPS-NP	Web-based Tool	Chemical Space Navigation	Positioning NPs in property-based chemical space [41] [38]
Scaffold Hunter	Software Framework	Visual Chemical Analytics	Scaffold-based analysis and diversity assessment [39] [42]
RDKit	Open-source Cheminformatics	Molecular Descriptor Calculation	Fingerprint generation and molecular property calculation [4]
Dictionary of Natural Products	Chemical Database	NP Structure Repository	Source of natural product structures and metadata [38]
ChEMBL Database	Bioactivity Database	Bioactive Compound Data	Source of bioactive molecules for comparative analysis [4]
αKGLib1	Enzyme Library	Biocatalytic Screening	314 α-KG-dependent enzymes for substrate compatibility testing [44]
MACCS Keys	Molecular Descriptors	Structural Fingerprinting	166-bit structural keys for molecular similarity [4]
Morgan Fingerprints	Molecular Descriptors	Circular Fingerprints	Encoding atomic environments for machine learning [4]

Case Studies and Applications in Natural Products Research

Chemical Space Analysis of Natural Products versus Synthetic Compounds

A seminal study using ChemGPS-NP to compare the chemical space coverage of natural products from the Dictionary of Natural Products (167,169 compounds) with bioactive medicinal chemistry compounds from WOMBAT (178,210 unique structures) revealed significant differences in their distribution patterns [38]. Natural products predominantly occupied regions characterized by greater structural rigidity (negative PC4 direction) and lower aromaticity (negative PC2 direction), while synthetic medicinal compounds exhibited greater flexibility and aromatic character. This analysis identified several "low-density regions" in synthetic chemical space that were populated by natural products, highlighting opportunities for novel lead discovery.

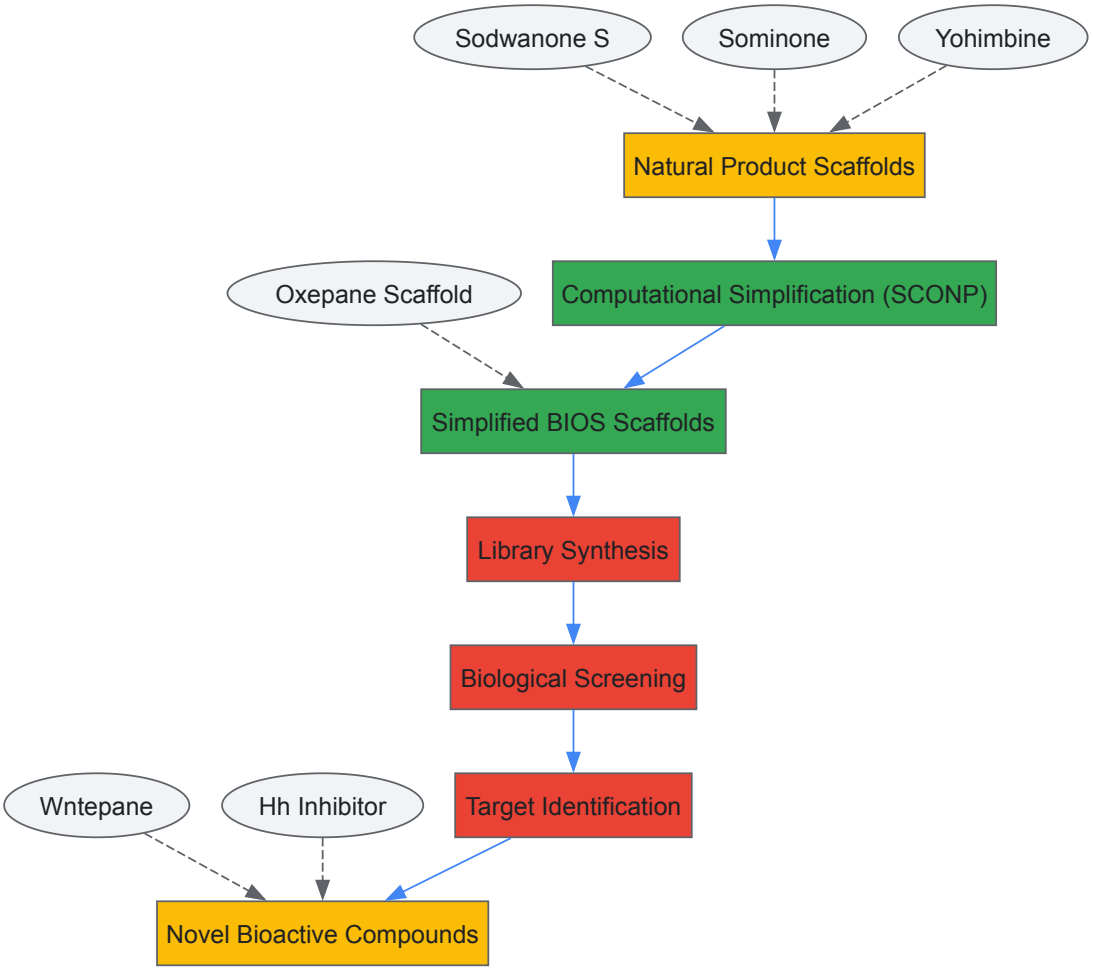
Notably, this study identified tangible lead-like natural products residing in regions of chemical space lacking representation in conventional medicinal chemistry compounds. Subsequent property-based similarity calculations identified NP neighbors of approved drugs, with several NPs confirmed to exhibit the same activity as their drug neighbors [38]. This demonstrates the utility of chemical space navigation for identifying NPs with potential as lead compounds for drug discovery.

BIOS-Directed Discovery of Wnt and Hedgehog Pathway Modulators

Biology-Oriented Synthesis (BIOS) has successfully yielded novel bioactive compounds through systematic exploration of natural product-inspired chemical space [40]. Inspired by the natural product sodwanone S, researchers designed a compound library based on a bicyclic oxepane scaffold. Using a multistep one-pot synthetic sequence, they prepared 91 derivatives, 50 of which were found to modulate the Wnt signaling pathway—a key pathway implicated in cancer and developmental disorders. Structure-activity relationship studies identified the most potent analogue, Wntepane, which was shown to activate the Wnt pathway by binding reversibly to the protein Vangl1, for which no previous small-molecule ligands had been reported [40].

In a separate application, BIOS inspired by the natural product sominone led to the preparation of 30 compounds, four of which were identified as hedgehog (Hh) pathway inhibitors. The most potent compound acted through modulation of the protein Smoothened, providing novel chemotypes for modulation of this clinically relevant pathway [40]. These successes demonstrate how natural product-informed exploration of chemical space can deliver distinctive bioactive molecules with novel mechanisms of action.

The following diagram illustrates the BIOS approach for natural product-inspired discovery:



Biology-Oriented Synthesis Workflow

Navigating Biocatalytic Reaction Space with Machine Learning

A recent innovative approach connected chemical space with protein sequence space to predict biocatalytic reactions, demonstrating the expanding applications of chemical space navigation [44]. Researchers constructed a library of 314 α-ketoglutarate (α-KG)/Fe(II)-dependent enzymes (aKGLib1) representing the sequence diversity of this protein family. By testing these enzymes against a diverse set of substrates, they generated a dataset connecting productive substrate-enzyme pairs.

This experimental data enabled the development of CATNIP, a computational tool for predicting compatible α-KG/Fe(II)-dependent enzymes for a given substrate or ranking potential substrates for a given enzyme sequence [44]. This approach demonstrates how mapping the interface between chemical space and protein sequence space can enable prediction of biocatalytic reactions, derisking the incorporation of biocatalytic steps into synthetic routes. The methodology is generalizable to other enzyme families and represents a significant advancement in predictive biocatalysis.

The visualization of chemical space using tools like ChemGPS-NP, Scaffold Hunter, and advanced dimensionality reduction techniques has transformed our ability to navigate and exploit the structural diversity of natural products. These approaches have revealed that natural products

occupy unique regions of biologically relevant chemical space that remain largely unexplored by synthetic compounds, providing exciting opportunities for the discovery of novel bioactive molecules [38].

Future developments in this field will likely include more sophisticated integration of chemical space visualization with biological activity data, enabling predictive models of compound function based on chemical space position. The incorporation of deep learning approaches for chemical space representation, such as ChemDist embeddings obtained from graph neural networks, represents another promising direction [4] [45]. Additionally, the connection of chemical space with protein sequence space, as demonstrated in biocatalytic reaction prediction, may open new avenues for understanding and exploiting the interface between small molecules and their biological targets [44].

As these tools and techniques continue to evolve, they will further enhance our ability to navigate the vastness of chemical space efficiently, focusing synthetic and discovery efforts on the most promising regions of biologically relevant chemical space. This will accelerate the identification of novel lead compounds from natural products and facilitate the development of new therapeutic agents to address unmet medical needs.

Virtual screening has emerged as a pivotal computational methodology in modern drug discovery, offering efficient identification of potential hit candidates from extensive chemical databases. When applied to natural product databases, virtual screening leverages the unique structural diversity and evolutionary-optimized bioactivity of natural compounds to address challenging biological targets. This technical guide examines current virtual screening approaches for natural product databases, focusing on target prediction and bioactivity spectra estimation within the broader context of chemical space and structural diversity research. We provide detailed experimental protocols, analyze the performance of various screening methods, and present visualization tools to aid researchers in implementing these techniques effectively.

Natural products (NPs) represent a unique sector of chemical space shaped by evolutionary selection processes rather than synthetic convenience. Through natural selection, NPs possess a vast chemical diversity optimized for interactions with biological macromolecules [46]. This evolutionary advantage makes them particularly valuable for drug discovery campaigns targeting protein-protein interactions, nucleic acid complexes, and other challenging biological systems where traditional small molecules often fail [46].

The structural complexity of natural products differs significantly from traditional combinatorial chemistry libraries. While combinatorial libraries through the 'one-synthesis/one-scaffold' approach generally show limited structural diversity, natural products occupy a broader region of chemical space with greater structural complexity and three-dimensionality [46]. This diversity makes them particularly suitable for addressing flat binding interfaces and allosteric sites that are often considered "undruggable" by conventional small molecules.

When considering the chemical space of natural products, several key characteristics emerge:

- **Structural Complexity**: NPs typically contain more stereocenters and ring systems than synthetic compounds
- **Drug-like Properties**: Evolutionary selection has optimized many NPs for bioavailability
- **Scaffold Diversity**: NPs provide a wider range of core structural frameworks than synthetic libraries
- **Target Diversity**: NPs modulate biological targets through mechanisms often distinct from synthetic compounds

Table 1: Comparison of Compound Sources for Virtual Screening

Parameter	Natural Products	Traditional Combinatorial Libraries	Synthetic Drugs
Structural Diversity	Broad chemical space coverage [46]	Limited by scaffold accessibility	Moderate to low
Structural Complexity	High (multiple stereocenters, macrocycles)	Generally low	Variable
Success in Challenging Targets	High (PPIs, nucleic acids) [46]	Limited	Low to moderate
Evolutionary Validation	Yes (optimized through natural selection) [46]	No	No
Chemical Space Coverage	Underexplored regions	Well-explored regions	Approved drug space

Virtual Screening Methodologies for Natural Products

Virtual screening methodologies can be broadly categorized into structure-based and ligand-based approaches, each with distinct advantages and limitations when applied to natural product databases.

Structure-Based Virtual Screening (SBVS)

Structure-based virtual screening, particularly molecular docking, has been increasingly applied to discover novel ligands for targets of interest in early drug discovery [47]. SBVS relies on the three-dimensional structure of the target protein to identify complementary small molecules from

virtual compound libraries.

The success of SBVS campaigns depends on several critical factors:

- **Quality of Protein Structure**: Experimental structures (X-ray, cryo-EM) generally provide more reliable results than homology models
- **Docking Software Selection**: Different programs have varying strengths for specific target classes
- **Scoring Functions**: Accurate prediction of binding affinities remains challenging
- **Chemical Library Preparation**: Proper representation of natural product stereochemistry and conformation

Recent comprehensive surveys of prospective SBVS applications indicate that molecular docking successfully identifies novel chemotypes, with approximately 25% of hits showing potency better than 1 μM [47]. The docking software landscape includes popular options such as GLIDE and the DOCK 3 series, the latter showing strong capacity for large-scale virtual screening [47].

Ligand-Based Virtual Screening (LBVS)

When structural information for the target is unavailable, ligand-based virtual screening approaches provide a valuable alternative. These methods utilize known active compounds to identify structurally or pharmacologically similar molecules from natural product databases.

Key LBVS methodologies include:

- **Pharmacophore Modeling**: Identifies partial structures responsible for biological activity
- **Similarity Searching**: Uses molecular descriptors to find compounds similar to known actives
- **Quantitative Structure-Activity Relationship (QSAR) Modeling**: Correlates structural features with biological activity

Similarity methods typically employ the Tanimoto coefficient with Morgan fingerprints to quantify molecular similarity, with a threshold below 0.4 generally indicating novel chemotypes [47]. However, these methods extrapolate poorly beyond short distances in chemical space and are often used as an initial filtering step before more advanced VS techniques [48].

AI-Enhanced Virtual Screening

Artificial intelligence has been extensively incorporated into various phases of drug discovery, including virtual screening [49]. AI approaches can effectively extract molecular structural features, perform in-depth analysis of drug-target interactions, and systematically model the relationships among drugs, targets, and diseases.

Machine learning techniques for virtual screening have increased rapidly in performance and popularity, with several paradigms proving particularly valuable:

- **Supervised Learning**: Uses labeled datasets for classification and regression tasks
- **Unsupervised Learning**: Identifies latent patterns through clustering and dimensionality reduction
- **Semi-Supervised Learning**: Leverages both labeled and unlabeled data to improve predictions
- **Reinforcement Learning**: Iteratively refines molecular design through reward-driven strategies

Table 2: Performance Analysis of Prospective SBVS Studies (419 Cases)

Performance Metric	Value/Range	Notes
Hit Potency	<1 μM (25% of studies)	Best hit from each study
	1-10 μM (30% of studies)	
	10-100 μM (35% of studies)	
Target Novelty	Least-explored targets (22%)	<10 known actives
	Less-explored targets (28%)	10-100 known actives
	Widely studied targets (50%)	>100 known actives
Structural Novelty	High (Tc <0.4)	Majority of identified hits
Software Preference	GLIDE (most popular)	
	DOCK 3 series (large-scale screening)	

Experimental Protocols for Virtual Screening

This section provides detailed methodologies for implementing virtual screening campaigns targeting natural product databases.

Structure-Based Virtual Screening Protocol

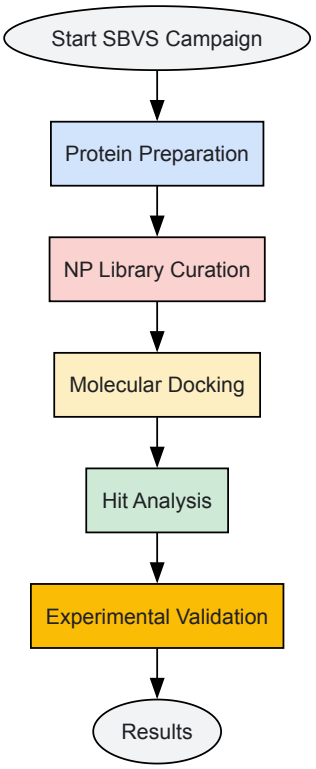
Objective: To identify potential natural product binders for a target protein using molecular docking.

Materials and Software:

- Target protein structure (PDB format)
- Natural product database (e.g., COCONUT, NPASS)
- Docking software (e.g., AutoDock Vina, GLIDE, GOLD)
- Compound preparation tools (e.g., OpenBabel, LigPrep)
- Visualization software (e.g., PyMOL, Chimera)

Procedure:

- **Protein Preparation**:
  - Obtain crystal structure from PDB or generate homology model
  - Remove water molecules and co-crystallized ligands
  - Add hydrogen atoms and optimize protonation states
  - Define binding site coordinates based on known ligands or catalytic residues
- **Natural Product Library Preparation**:
  - Download natural product structures in SDF or SMILES format
  - Generate 3D conformations using molecular mechanics methods
  - Minimize structures using appropriate force fields (MMFF94, GAFF)
  - Generate tautomers and protomers at physiological pH (7.4±0.5)
- **Molecular Docking**:
  - Convert natural products to docking-compatible format (PDBQT, MOL2)
  - Set docking grid parameters to encompass binding site
  - Execute docking simulations with appropriate sampling parameters
  - Generate multiple poses per compound (typically 10-20)
- **Post-Docking Analysis**:
  - Rank compounds by docking score or binding energy
  - Visually inspect top-ranking poses for binding mode rationality
  - Cluster results by structural similarity to identify diverse hits
  - Select compounds for experimental validation (typically 20-100 compounds)



Bioactivity Spectra Estimation Protocol

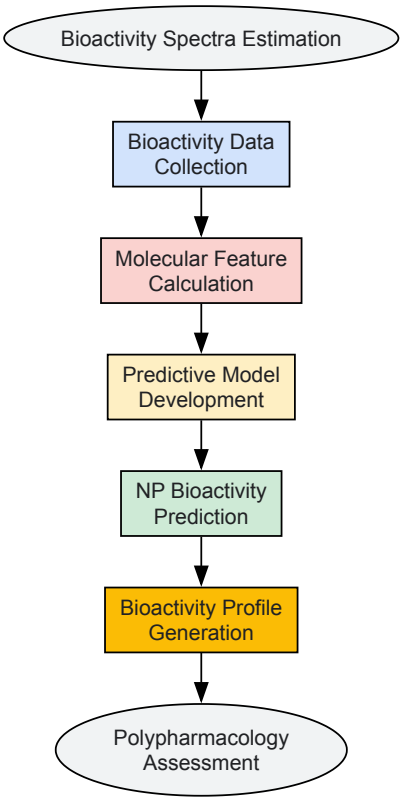
Objective: To predict the polypharmacological profiles of natural products across multiple biological targets.

Materials and Software:

- Natural product compound library
- Target annotation database (e.g., ChEMBL, GOSTAR)
- Machine learning environment (e.g., Python/R with scikit-learn, TensorFlow)
- Similarity search tools (e.g., RDKit, OpenBabel)

Procedure:

- **Data Collection and Curation**:
  - Retrieve bioactivity data from public databases (ChEMBL, GOSTAR, PubChem)
  - Filter for high-confidence measurements (IC50, Ki, Kd)
  - Standardize compound representations (canonical SMILES)
  - Resolve target nomenclature using standardized identifiers (UniProt IDs)
- **Model Training**:
  - Calculate molecular descriptors or fingerprints for known actives
  - Train multi-task classification or regression models
  - Validate models using cross-validation and external test sets
  - Optimize hyperparameters using grid search or Bayesian optimization
- **Bioactivity Prediction**:
  - Compute descriptors for natural products of interest
  - Apply trained models to predict activity across multiple targets
  - Calculate confidence estimates for predictions
  - Apply applicability domain analysis to identify reliable predictions
- **Results Interpretation**:
  - Generate bioactivity heatmaps for natural product collections
  - Identify potential off-target effects and polypharmacology
  - Prioritize compounds for specific target panels
  - Correlate structural features with bioactivity spectra



Successful implementation of virtual screening for natural products requires specialized computational tools and databases. The following table details essential resources for researchers in this field.

Table 3: Essential Research Resources for NP Virtual Screening

Resource Category	Specific Tools/Databases	Function	Key Features
Natural Product Databases	COCONUT, NPASS, UNPD	Source of natural product structures	Curated NPs with taxonomic origin
Bioactivity Databases	ChEMBL [48], GOSTAR [48], PubChem BioAssay [48]	Source of bioactivity data	Manually curated SAR data
Docking Software	AutoDock Vina, GLIDE [47], DOCK [47]	Structure-based virtual screening	Binding pose prediction and scoring
Cheminformatics Tools	RDKit [47], OpenBabel	Molecular descriptor calculation	Fingerprint generation, similarity searching
Protein Structure Resources	PDB, AlphaFold DB [47]	Source of target structures	Experimental and predicted structures
Visualization Software	PyMOL, UCSF Chimera	Analysis of docking results	3D structure visualization

Case Studies and Applications

Natural products have demonstrated significant success in virtual screening campaigns, particularly for challenging target classes. Representative examples include:

Diazonamide A and Ornithine Delta-Amino Transferase

The marine natural product diazonamide A was identified through a target identification campaign using a biotinylated derivative [46]. Initial mechanistic studies suggested tubulin binding, but direct binding measurements ruled this out. Using chemical proteomics, researchers purified ornithine delta-amino transferase (OAT) as the molecular target, revealing an unexpected role for this mitochondrial enzyme in mitotic cell division [46]. This case highlights the importance of target identification studies for natural products with unknown mechanisms.

FTY720 and S1P Receptor Modulation

FTY720, a synthetic analog of the fungal metabolite myriocin, demonstrates how natural product-inspired compounds can lead to novel mechanistic insights [46]. Originally developed as an immunosuppressive agent, FTY720 was found to produce lymphopenia by sequestering lymphocytes rather than through direct immunosuppression. Subsequent research revealed that FTY720 is phosphorylated in vivo and acts as a sphingosine 1-phosphate (S1P) receptor agonist [46]. This discovery provided novel insights into the therapeutic relevance of the S1P pathway.

AI-Driven Natural Product Discovery

Recent advances in artificial intelligence have enabled more efficient exploration of natural product chemical space. AI platforms can now decode intricate structure-activity relationships, facilitating de novo generation of bioactive compounds with optimized properties [49]. Companies including Insilico Medicine, Recursion, and Exscientia have advanced AI-discovered small molecules into clinical trials, demonstrating the potential of these approaches [49].

Table 4: Success Rates and Novelty in Prospective SBVS Studies

Target Class	Representative Targets	Average Hit Rate	Novel Chemotype Rate (Tc <0.4)
Kinases	HK-2, various tyrosine kinases	15-30%	60-75%
Proteases	Various serine proteases	10-25%	55-70%
Phosphatases	PTPσ, various dual-specificity phosphatases	5-20%	65-80%
Membrane Receptors	GPCRs, ion channels	8-22%	50-65%
Nuclear Receptors	PPARγ, RAR, RXR	12-28%	45-60%

Virtual screening of natural product databases represents a powerful strategy for exploring underexplored regions of chemical space. The unique structural features and evolutionary optimization of natural products make them particularly valuable for addressing challenging biological targets. By combining structure-based and ligand-based approaches with emerging AI technologies, researchers can effectively navigate the complex chemical space of natural products to identify novel bioactive compounds with potential therapeutic applications. The continued development of specialized databases, improved scoring functions, and better understanding of natural product biosynthesis will further enhance our ability to leverage these privileged scaffolds for drug discovery.

The concept of the **Biologically Relevant Chemical Space (BioReCS)** provides a fundamental framework for understanding the relationship between molecular structure and biological activity. BioReCS comprises the vast set of molecules—both beneficial and detrimental—that interact with biological systems, spanning therapeutic compounds, agrochemicals, natural products, and toxic substances [6]. Within this expansive universe, **natural products (NPs)** occupy a privileged position as they are inherently pre-validated by evolution to interact with biological macromolecules, binding both their biosynthetic enzymes and their target proteins [50]. This unique characteristic makes NPs exceptional starting points for exploring bioactive regions of chemical space.

The pressing challenge in modern drug discovery lies in the fact that conventional compound collections, including corporate archives and commercially available libraries, often suffer from structural redundancy and limited scaffold diversity. These collections predominantly contain structurally simple, "flat" molecules with diversity limited mainly to appendage variations around a small number of common skeletons [51]. This structural bias has contributed to the continuing decline in drug-discovery successes and has left many challenging biological targets, such as transcription factors, regulatory RNAs, and protein-protein interactions, in the "undruggable" category [51].

Diversity-Oriented Synthesis (DOS) has emerged as a powerful strategy to address these limitations by deliberately generating structural diversity in small-molecule libraries. Unlike traditional target-oriented synthesis or combinatorial chemistry focused on appendage diversity, DOS aims to efficiently create collections with significant **skeletal, stereochemical, and functional group diversity** [50] [51]. When inspired by natural product architectures, DOS enables the systematic exploration of underutilized regions of BioReCS, potentially expanding the boundaries of the ligandable proteome and uncovering novel bioactive compounds with unexpected functions [52] [53].

The Conceptual Framework: BioReCS and Natural Product Privilege

Mapping the Biologically Relevant Chemical Space

The systematic study of BioReCS requires molecular descriptors that define the dimensionality of the space, with the choice of descriptors depending on project goals and compound classes [6]. BioReCS encompasses multiple specialized subspaces (ChemSpas), including:

- **Drug-like molecules**: Extensively explored regions populated by small organic compounds in databases like ChEMBL and PubChem [6]
- **Beyond Rule of 5 (bRo5) space**: Including macrocycles, peptides, and other complex molecules [6]
- **Natural product space**: Characterized by structural complexity and pre-validated bioactivity [50]
- **Underexplored regions**: Including metal-containing compounds, proteolysis-targeting chimeras (PROTACs), and protein-protein interaction modulators [6]

Natural products reside in scientifically valuable regions of BioReCS, exhibiting structural features that distinguish them from synthetic compounds in commercial collections. NPs typically display greater **structural complexity**, characterized by higher sp3 carbon count, increased stereogenic centers, and improved shape diversity [50]. These properties enable natural products to interact with challenging biological targets that often remain intractable to conventional flat, aromatic compounds prevalent in many screening libraries [51].

The Privileged Status of Natural Products in Chemical Space

Natural products are considered "privileged" starting points for library design because they offer **validated functional diversity**—individual natural product families can selectively modulate unrelated biological targets [50]. This privileged status stems from evolutionary pressure that has optimized these molecules for specific biological interactions while maintaining structural features conducive to broader macromolecular recognition.

The table below summarizes key advantages of natural products as inspiration for DOS libraries:

Table 1: Advantages of Natural Product-Inspired Library Design

Feature	Description	Impact on BioReCS Coverage
Pre-validated bioactivity	NPs have evolved to interact with biological targets	Higher hit rates in biological screening [50]
Structural complexity	High sp3 carbon count, multiple stereocenters	Access to underexplored regions of shape space [51]
Architectural diversity	Diverse skeletal frameworks	Broader coverage of different target classes [53]
Favorable physicochemical properties	Balanced polarity, molecular weight	Enhanced compatibility with biological systems [50]
Functional group diversity	Variety of hydrogen bond donors/acceptors	Improved interactions with protein surfaces [51]

However, natural products themselves present challenges for comprehensive screening, including difficulties with purification, identification of bioactive components, and chemical modification [51]. DOS approaches inspired by—but not slavishly copied from—natural product architectures can overcome these limitations while retaining the privileged characteristics of NPs.

Strategic Implementation of NP-Inspired Diversity-Oriented Synthesis

Key Dimensions of Diversity in Library Design

Effective NP-inspired DOS libraries incorporate multiple dimensions of diversity to maximize their coverage of BioReCS. The four principal components of structural diversity include:

- **Skeletal (scaffold) diversity**: Presence of distinct molecular frameworks in the library [51]
- **Stereochemical diversity**: Variation in the three-dimensional orientation of potential macromolecule-interacting elements [51]
- **Appendage (building-block) diversity**: Variation in structural moieties around a common skeleton [51]
- **Functional group diversity**: Variation in the functional groups present that govern molecular interactions [51]

Among these, **skeletal diversity** is particularly crucial as it primarily defines the overall shape diversity of a library. Since biological macromolecules recognize their binding partners through three-dimensional complementary surfaces, scaffold diversity directly influences the range of potential biological targets a library can address [51].

Synthetic Methodologies for Skeletal Diversification

Several innovative synthetic strategies have been developed to achieve skeletal diversity in NP-inspired DOS libraries:

Late-stage diversification of natural product cores: This approach utilizes complex NP scaffolds as starting points for further structural diversification. A recent example leveraged **P450-catalyzed oxyfunctionalization** to create a skeletally diverse library of natural-product-like compounds [53]. The strategy integrated regiodivergent, site-selective P450-catalyzed C-H functionalization with divergent chemical routes for skeletal diversification and rearrangement of parent natural product scaffolds.

Build/couple/pair algorithms: This methodology involves assembling simple building blocks into linear intermediates that then undergo different cyclization pathways to generate diverse skeletons [50].

Folding pathways: Strategic placement of functional groups within linear precursors enables different cyclization modes upon activation, generating distinct ring systems from common intermediates [50].

The following workflow diagram illustrates a generalized strategy for creating NP-inspired DOS libraries:

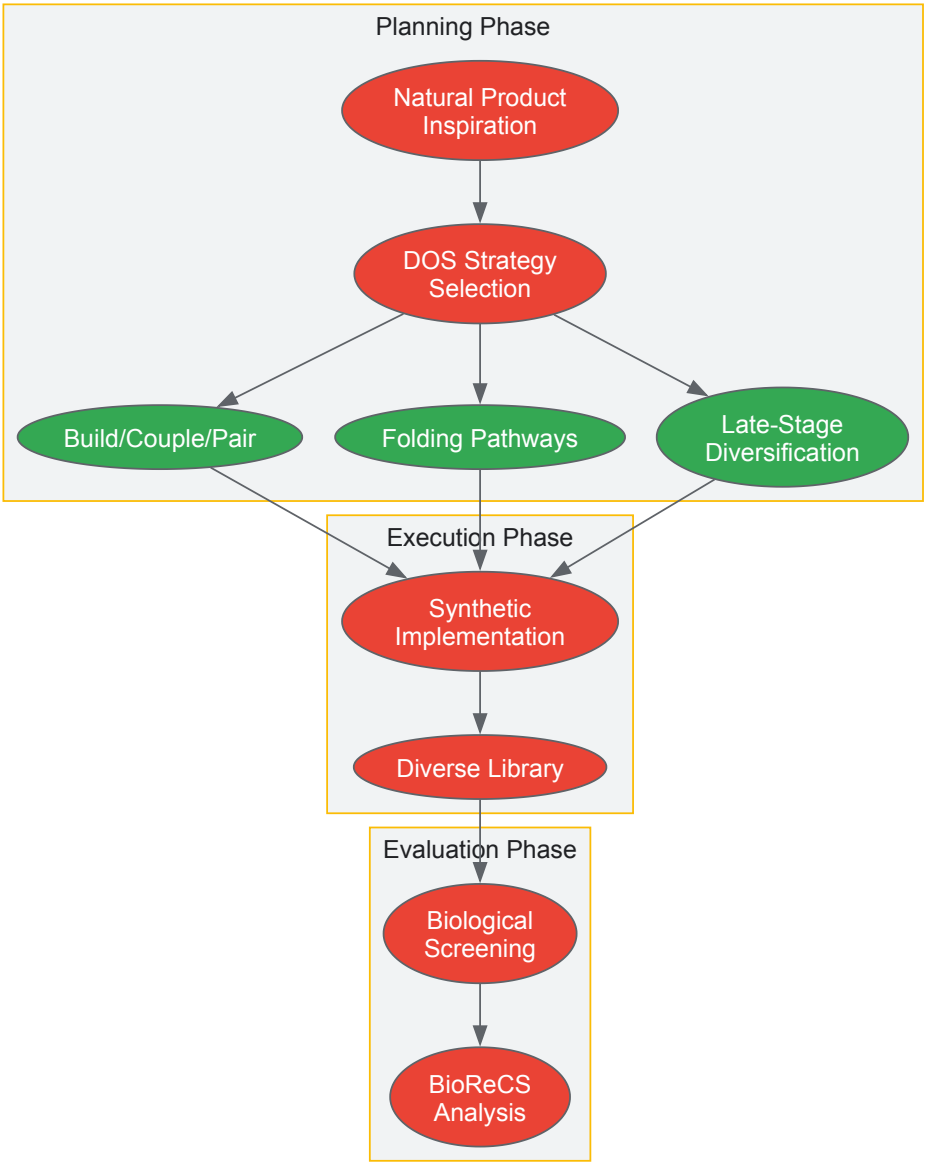


Diagram 1: NP-Inspired DOS Library Workflow

Advanced DOS Implementation: A Representative Protocol

A recent groundbreaking approach demonstrates the power of combining natural product complexity with sophisticated synthetic methodologies [53]. The protocol below outlines this strategy:

Objective: Create a skeletally diverse library of natural-product-like compounds with broad coverage of BioReCS and potential anticancer activity.

Starting Material: Parthenolide (a sesquiterpene lactone natural product) served as the core scaffold.

Key Synthetic Steps:

- **Engineering of P450 enzymes**: Developed cytochrome P450 variants with altered regio- and stereoselectivity for C-H functionalization of the parthenolide scaffold.
- **Late-stage oxyfunctionalization**: Employed regiodivergent, site-selective P450-catalyzed hydroxylation at multiple positions of the parthenolide core.
- **Skeletal diversification**: Utilized the introduced oxygen functionality as a handle for divergent synthetic transformations, including:
    - Cyclization and rearrangement reactions
    - Ring expansion/contraction
    - Introduction of additional heterocyclic systems

- **Electrophilic warhead installation**: Equipped library members with functional groups (e.g., α,β-unsaturated carbonyls) for covalent target engagement.

Library Characteristics:

- **Size**: >50 members
- **Skeletal diversity**: Multiple distinct core architectures derived from a single parent NP
- **Structural features**: High sp3 character, multiple stereocenters, diverse ring systems
- **Functional groups**: Varied hydrogen-bond donors/acceptors, electrophilic warheads for covalent targeting

This protocol successfully generated a library with broad chemical and structural diversity that included several bioactive compounds with selective cytotoxicity against cancer cells and diversified anticancer activity profiles [53].

Analytical Framework: Assessing Library Coverage of BioReCS

Quantifying Diversity and Chemical Space Coverage

As chemical libraries continue to expand rapidly, proper assessment of their diversity and BioReCS coverage becomes essential. Merely increasing the number of compounds does not necessarily translate to increased diversity [1]. Advanced cheminformatic tools are required to quantitatively evaluate the time evolution of chemical libraries in terms of chemical diversity.

The following table summarizes key metrics and tools for assessing library coverage of BioReCS:

Table 2: Analytical Methods for Assessing Library Diversity

Method	Description	Application in NP-Inspired DOS
iSIM framework	Calculates intrinsic similarity of compound sets with O(N) complexity	Global indicator of internal library diversity [1]
BitBIRCH clustering	Efficient clustering algorithm for large libraries using Tanimoto similarity	Identifies distinct molecular clusters and coverage gaps [1]
Complementary similarity analysis	Identifies medoid-like (central) and outlier (peripheral) compounds	Reveals library coverage patterns within chemical space [1]
Scaffold diversity metrics	Analysis of molecular framework distribution	Quantifies skeletal diversity relative to known NPs [51]
Shape diversity assessment	Evaluation of three-dimensional molecular geometries	Correlates with potential to modulate diverse target classes [51]

Recent analyses of evolving chemical libraries reveal that simply adding more compounds does not automatically increase diversity. The iSIM framework, which calculates the average of all distinct pairwise Tanimoto comparisons with O(N) complexity, has shown that some library expansions primarily fill existing regions of chemical space rather than exploring new territories [1]. This underscores the importance of deliberate diversity-oriented design rather than mere accumulation of compounds.

Comparative Analysis of Library Evolution

Time-evolution analysis of major chemical databases provides valuable insights for designing NP-inspired DOS libraries:

- **ChEMBL**: Analysis of releases 1 through 33 reveals how different versions contributed to diversity expansion, identifying which additions created new regions versus filling existing ones [1]
- **DrugBank**: Examination of releases from 2005 to 2022 shows the evolution of drug space and opportunities for exploring underexploited regions [1]
- **Natural product databases**: Systematic comparison of natural products versus synthetic compounds reveals distinctive structural features of NPs that can inform DOS design [1]

These analyses help guide strategic decisions about where to focus synthetic efforts to maximize coverage of underexplored BioReCS regions rather than redundant exploration of already crowded chemical territories.

Experimental Protocols and Methodologies

Chemoenzymatic DOS Protocol Based on P450 Catalysis

The following detailed protocol adapts the methodology from [53] for creating skeletally diverse NP-inspired libraries:

Materials:

- Natural product scaffold (e.g., parthenolide or other sesquiterpene lactone)
- Engineered P450 enzymes (specific variants described in [53])
- NADPH regeneration system (glucose-6-phosphate and glucose-6-phosphate dehydrogenase)
- Chemical reagents for divergent synthesis (specific to desired transformations)
- Solid-phase extraction cartridges (C18)
- Analytical and preparative HPLC systems

Procedure:

- **Initial scaffold preparation**:
  - Purify natural product starting material to >95% purity
  - Characterize fully by NMR and HRMS
  - Prepare stock solution in appropriate solvent (e.g., DMSO)
- **P450-catalyzed oxyfunctionalization**:
  - Set up reaction mixture containing:
    - Natural product substrate (0.5-1.0 mM)
    - Engineered P450 enzyme (1-5 μM)
    - NADPH regeneration system
    - Reaction buffer (optimized for specific P450 variant)
  - Incubate at optimal temperature with shaking
  - Monitor reaction progress by TLC or LC-MS
  - Terminate reaction by extraction with ethyl acetate
- **Product separation and characterization**:
  - Separate regioisomeric products by flash chromatography
  - Characterize each functionalized product by NMR and MS
  - Determine regio- and stereochemistry of hydroxylation
- **Skeletal diversification**:
  - Develop multiple divergent synthetic pathways from each hydroxylated intermediate:
    - Pathway A: Cyclization to novel ring systems
    - Pathway B: Skeletal rearrangement
    - Pathway C: Introduction of additional heterocycles
  - Optimize each pathway for yield and purity
  - Purify final compounds to >95% purity
  - Characterize all novel skeletons by comprehensive spectroscopic methods
- **Library assembly and quality control**:
  - Prepare standardized stock solutions of all library members
  - Confirm identity and purity of each compound before biological screening
  - Store library under appropriate conditions (often as DMSO stocks at -80°C)

The experimental workflow for this protocol is detailed below:

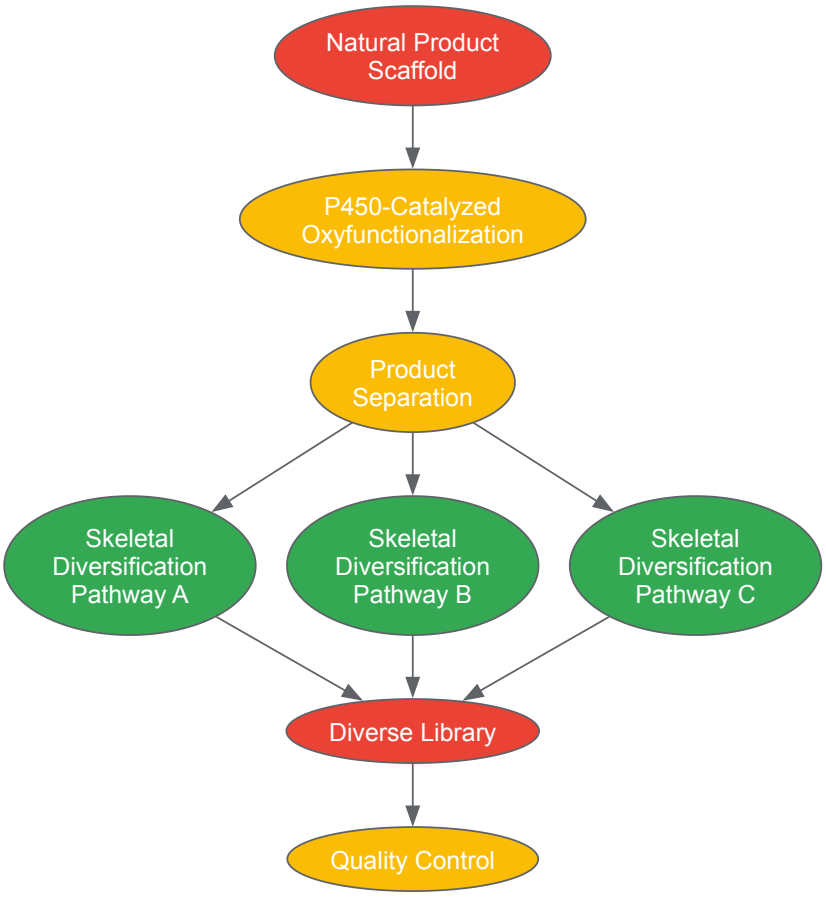


Diagram 2: Chemoenzymatic DOS Experimental Workflow

Research Reagent Solutions for NP-Inspired DOS

Table 3: Essential Research Reagents for NP-Inspired DOS

Reagent/Category	Specific Examples	Function in DOS
Natural Product Scaffolds	Parthenolide, artemisinin, rocaglamide	Complex starting materials for diversification [53]
Engineered Biocatalysts	Regiodivergent P450 variants	Site-selective C-H functionalization [53]
Diversification Reagents	Electrophiles, nucleophiles, coupling reagents	Appendage and functional group diversification [51]
Activation Reagents	Photoredox catalysts, organometallic catalysts	Enabling novel transformation pathways [50]
Purification Materials	SPE cartridges, HPLC columns, flash chromatography media	Isolation and purification of diverse library members [53]
Analytical Standards	Chiral columns, NMR reference standards	Characterization of novel skeletons and stereocenters [53]

Biological Validation: From Library Screening to Probe Discovery

Screening Strategies for Functionally Diverse Libraries

The ultimate validation of NP-inspired DOS libraries comes from their performance in biological screening. Unlike target-focused libraries, DOS libraries are particularly valuable for unbiased screening approaches where the biological target is unknown [51]. Effective screening strategies include:

- **Phenotypic screening**: Cell-based assays measuring desired phenotypic outcomes
- **Chemical proteomics**: Proteome-wide mapping of ligandable proteins directly in cells [52]
- **Protein-family focused screening**: Targeting challenging protein classes like protein-protein interactions

A key advantage of NP-inspired libraries is their suitability for **photoaffinity-based chemoproteomic methods**, which enable proteome-wide mapping of ligandable proteins directly in living cells [52]. However, due to the relatively low throughput of proteomic screening and the synthetic burden of installing photoactivatable functionality on each library member, strategic library design is paramount for success.

Case Study: NP-Inspired Probes Expand the Ligandable Proteome

A recent landmark study demonstrated the power of fully functionalized natural product probes to expand the boundaries of the chemically tractable proteome [52]. The approach involved:

- **Probe design**: Creation of a structurally unique series of diastereo/regioisomeric photoaffinity probes based on natural product scaffolds
- **Library synthesis**: Semi-synthetic modification of sp3-rich NP scaffolds to install photoactivatable groups
- **Chemoproteomic screening**: MS-based chemoproteomics to identify and quantify interactions in human cells
- **Target identification**: Discovery of topology-, regio-, and stereoselective ligands for functionally diverse proteins that previously lacked chemical probes

This approach successfully identified selective ligands for proteins that currently lack reported chemical probes, highlighting the potential of NP-inspired chemoproteomic libraries to dramatically expand the ligandable proteome [52].

Diversity-Oriented Synthesis inspired by natural products represents a powerful strategy for expanding coverage of biologically relevant chemical space. By deliberately incorporating skeletal, stereochemical, and functional group diversity into compound libraries, DOS moves beyond the limitations of traditional combinatorial approaches that primarily explore appendage diversity around common scaffolds.

The integration of innovative synthetic methodologies—including chemoenzymatic approaches using engineered P450 catalysts for late-stage diversification—with sophisticated analytical frameworks for assessing chemical space coverage enables the systematic exploration of underexplored regions of BioReCS [53]. As these strategies mature, coupled with advanced screening technologies like chemoproteomics, NP-inspired DOS promises to significantly expand the ligandable proteome and provide chemical tools for challenging biological targets [52].

Future developments in this field will likely include:

- Increased integration of biocatalysis with synthetic chemistry for unprecedented diversification
- Application of artificial intelligence for designing libraries that maximize coverage of unexplored BioReCS regions
- Enhanced chemoproteomic methods for rapid target identification of novel bioactive compounds
- Strategic focus on historically "undruggable" target classes informed by natural product biology

As the field continues to evolve, the strategic integration of natural product inspiration with diversity-oriented synthesis will play an increasingly vital role in addressing the ongoing challenges of drug discovery and chemical biology.

Natural products (NPs) are small molecules synthesized by living organisms, optimized through millions of years of evolution for optimal interactions with biological macromolecules [46]. Owing to this natural selection process, they possess a unique and vast chemical diversity, making them by far the richest source of novel compound classes for biological studies and an essential foundation for new drug discovery [46]. Between 1981 and 2002, natural products accounted for over 60% and 75% of new chemical entities for cancer and infectious diseases, respectively, highlighting their profound impact [54]. Their structural complexity and diversity allow them to modulate challenging biological targets, including protein-protein interactions and transcription factors, which are often considered "undruggable" by conventional synthetic compounds [51].

However, the intricate structures of NPs present challenges in terms of purification, identification, and synthesis, complicating their use in modern high-throughput drug discovery pipelines [51] [54]. This has created an imperative to develop computational methods that can capture the essential structural features of NPs and generate novel, synthetically accessible compounds that inhabit similar regions of chemical space. This technical guide explores how machine learning (ML) and artificial intelligence (AI) are addressing these challenges through the prediction of **natural product-likeness** and the **generation of novel scaffolds**, thereby accelerating NP-inspired drug discovery.

Quantifying Natural Product-Likeness

The NaPLeS Platform: An Open-Source Scorer

A fundamental task in virtual screening is to prioritize compounds that are structurally similar to known bioactive natural products. The **Natural Product Likeness Score (NP-Likeness Score)** is a Bayesian measure that quantifies the similarity of a molecule to the structural space covered by natural products compared to synthetic molecules [55] [56].

The open-source **NaPLeS** web application provides a portable, containerized tool for computing this score [55] [57]. Its underlying algorithm is based on the following methodology [55]:

- **Training Data Curation**: The scorer is trained on a large dataset combining molecules from multiple public databases. The training set typically consists of over 360,000 natural products and 490,000 synthetic molecules.

- **Molecular Preprocessing**: Each molecule undergoes a curation process where stereochemistry is removed to avoid fragment redundancy. Molecules smaller than 6 atoms or containing non-organic atoms are discarded. Furthermore, linear and circular sugar moieties are removed from all molecules because their repetitive and redundant nature makes them less distinctive for classification.
- **Fragment Calculation**: Atom signatures—canonical circular descriptors of an atom's environment—are calculated for each atom in the molecule. A signature height of 2 is typically used, providing a balance between structural accuracy and computational efficiency.
- **Frequency Calculation and Scoring**: For each unique fragment, its frequency in NPs versus synthetic molecules (SM) is computed. The frequency (Frag*{i}) of a fragment (i) is given by: [Frag*{i} = \log\left[\frac{NP*{i}}{SM*{i}} \times \frac{SM*{t}}{NP*{t} \right]] where (NP*{i}) and (SM*{i}) are the number of occurrences of the fragment (i) in natural products and synthetic molecules, respectively, and (NP*{t}) and (SM*{t}) are the total number of these molecules.
- **Molecule-Level Score**: The final NP-likeness score for a molecule is the sum of the frequencies of all its fragments, normalized by the number of fragments (N): [NPls = \frac{\sum*{i=o}^{N} Frag*{i}}{N}]

This score efficiently separates NPs from synthetic molecules in validation experiments and can be used to prioritize compound libraries, guide the design of NP-like combinatorial libraries, and perform virtual screening [55] [56].

Experimental Protocol: Calculating NP-Likeness for a Compound Set

Table 1: Experimental workflow for using the NaPLeS scorer.

Step	Action	Details and Considerations
1. Input Preparation	Prepare molecular structures.	Ensure structures are in an accepted format (SMILES, SDF, MOL). The web application accepts a maximum of 1000 molecules per file [57].
2. Data Submission	Submit structures to NaPLeS.	Use one of the four available methods: file upload, SMILES string paste, molecular drawing, or database exploration [57].
3. Score Computation	The application computes scores.	The backend processes molecules by removing sugars, calculating atom signatures, and summing fragment frequencies [55].
4. Result Analysis	Analyze and interpret scores.	A higher positive score indicates a higher similarity to natural products. Scores can be used to rank and filter virtual libraries [55] [56].

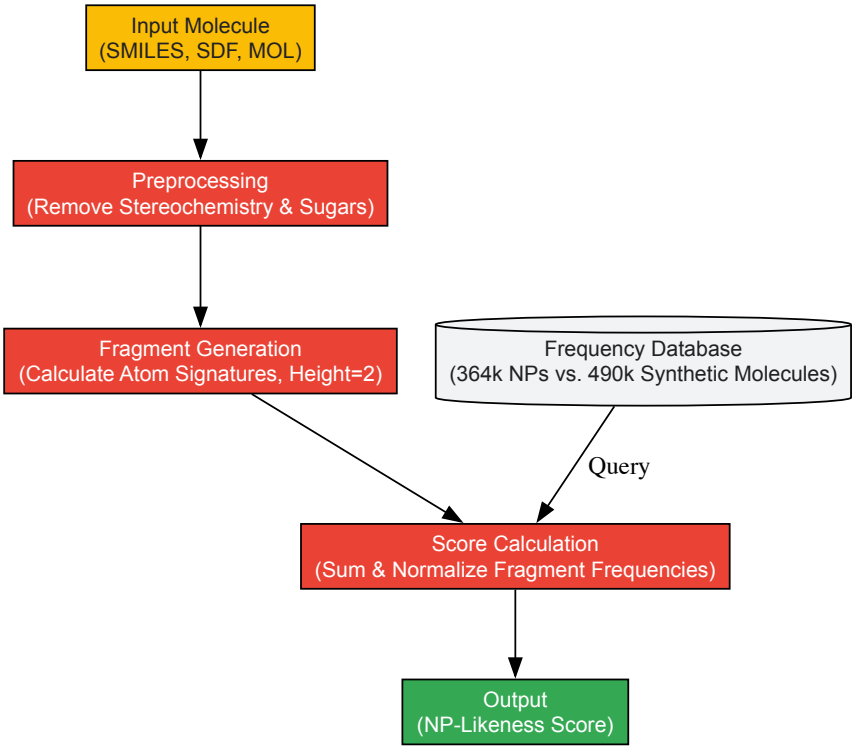


Figure 1: NP-likeness score computation workflow in NaPLeS, from molecular input to final score output.

Generating Novel Natural Product-Like Scaffolds

While predicting NP-likeness helps in filtering existing libraries, generating novel compounds that inhabit the underrepresented regions of NP chemical space is a more ambitious goal. AI-driven generative models have emerged as powerful tools for this task, with two primary architectures

leading the way: **Variational Autoencoders (VAEs)** and **GPT-based Chemical Language Models**.

ScaffoldGVAE: A Multi-View Graph Neural Network for Scaffold Hopping

Scaffold hopping is a key strategy in drug design to discover compounds with distinct core structures but similar biological activities, potentially leading to improved properties and intellectual freedom-to-operate [58]. **ScaffoldGVAE** is a sophisticated VAE designed explicitly for scaffold generation and hopping [58].

Table 2: Key components and experimental protocol for ScaffoldGVAE.

Component/Step	Description	Function/Rationale
Architecture	Variational Autoencoder (VAE) based on Multi-View Graph Neural Networks.	Separately encodes node (atom) and edge (bond) information for a richer molecular representation [58].
Data Preparation	Pre-training on ~800,000 molecule-scaffold pairs from ChEMBL. Scaffolds are extracted using ScaffoldGraph, filtering for 1+ ring (excl. benzene), ≤20 heavy atoms [58].	Ensures model learns diverse, drug-like scaffold foundations. Filtering removes overly common and complex structures.
Encoder	Maps input molecule to a latent representation. Separates embedding into side-chain and scaffold components. Projects scaffold embedding onto a Gaussian Mixture Model (GMM).	GMM prior encourages a structured, continuous latent space, facilitating smooth scaffold interpolation and generation [58].
Decoder	Uses a Recurrent Neural Network (RNN). Concatenates the sampled scaffold vector with the original side-chain vector to reconstruct the scaffold SMILES.	Ensures generated scaffolds are compatible with the original molecule's side chains, preserving key functional elements [58].
Fine-Tuning	Model can be fine-tuned on small, target-specific datasets (e.g., active kinase inhibitors).	Tailors the generative process to produce scaffolds with a higher probability of activity against a specific target [58].
Evaluation	Uses metrics like validity, uniqueness, novelty, and FCD. Docking (LeDock) and binding free energy calculations (MM/GBSA) validate bioactivity.	Assesses both the chemical quality of generated molecules and their potential biological activity [58].

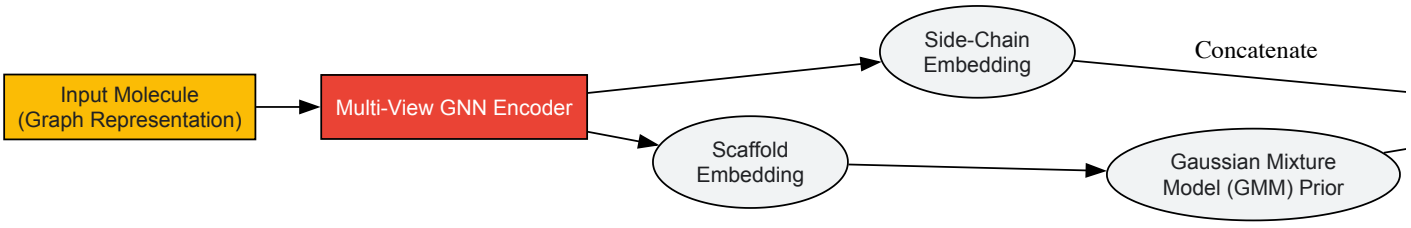


Figure 2: ScaffoldGVAE architecture illustrating the separation of side-chain and scaffold information for targeted scaffold generation [58].

NPGPT: GPT-Based Chemical Language Models for de novo Generation

Another powerful approach treats molecular generation as a language modeling task. **NPGPT** involves fine-tuning Generative Pre-trained Transformers (GPT) on large datasets of natural products to generate novel, NP-like compounds [54].

Experimental Protocol for NPGPT:

- **Dataset Curation**: A dataset of approximately 400,000 natural products is sourced from the COCONUT database. SMILES strings are standardized, and very large compounds are filtered out. Data augmentation is performed by randomizing SMILES traversal order, expanding the dataset to about 3.6 million entries [54].
- **Model Selection and Fine-Tuning**: Pretrained chemical language models (e.g., **smiles-gpt** using SMILES and **ChemGPT** using SELFIES) are selected. These decoder-only Transformer models, initially trained on large general molecular databases like PubChem, are subsequently fine-tuned on the curated NP dataset. This transfers their generative capabilities to the chemical space of natural products [54].
- **Generation and Evaluation**: The fine-tuned model generates new molecular strings. The generated compounds are evaluated using standard metrics:
  - **Validity**: The ratio of grammatically correct and chemically feasible structures.
  - **Uniqueness**: The ratio of non-duplicate molecules in the generated set.
  - **Novelty**: The ratio of molecules not found in the training database.

- **Fréchet ChemNet Distance (FCD)**: Measures the similarity between the distributions of generated molecules and the training NP dataset. A lower FCD indicates better capture of the NP chemical space [54].

Table 3: Performance comparison of AI models for NP-like compound generation.

Model	Architecture	Validity (%)	Uniqueness (%)	Novelty (%)	FCD (vs. COCONUT)	Key Findings
Tay et al. (RNN) [54]	RNN with LSTM	93.70	98.90	99.99	12.19	Established baseline for NP generation.
NPGPT (smiles-gpt) [54]	GPT (SMILES)	95.94	99.97	99.99	6.75	**Generated distribution is closer to NPs** than the RNN baseline.
NPGPT (ChemGPT) [54]	GPT (SELFIES)	**99.90**	**100.0**	**100.0**	223.16	High validity but failed to capture NP distribution effectively.
ScaffoldGVAE [58]	Graph-based VAE	Reported high scores on specialized scaffold-hopping metrics.	N/A	N/A	N/A	Generated molecules validated by molecular docking (LeDock), demonstrating practical utility for drug design.

Table 4: Key resources for AI-driven natural product research.

Resource Name	Type	Function in Research
NaPLeS [55] [57]	Web Application / Database	Computes the Natural Product-likeness score for input molecules to prioritize compounds for screening.
COCONUT Database [54]	Chemical Database	A comprehensive collection of natural product structures used for training generative AI models.
ChEMBL Database [58]	Chemical Database	A curated database of bioactive molecules with drug-like properties, used for pre-training generative models.
ScaffoldGraph [58]	Software Library	A tool for scaffold network analysis and extracting molecular scaffolds from compound sets for model training.
RDKit [54]	Cheminformatics Toolkit	An open-source toolkit for cheminformatics, used for molecule manipulation, descriptor calculation, and fingerprint generation.
Chemistry Development Kit (CDK) [55]	Cheminformatics Toolkit	Provides open-source algorithms for structural chemo-informatics, used in NaPLeS for molecular curation and fragment calculation.

Machine learning and artificial intelligence are fundamentally reshaping the exploration of natural product chemical space. By providing quantitative metrics like the NP-likeness score, tools such as NaPLeS enable researchers to efficiently filter vast virtual libraries for compounds with NP-like characteristics [55] [57]. Furthermore, advanced generative models like ScaffoldGVAE and NPGPT move beyond filtering to active design, creating novel, structurally diverse, and complex scaffolds that are either inspired by natural products or are predicted to reside in the under-explored regions of NP chemical space [58] [54]. These AI-driven strategies effectively bridge the gap between the rich bioactivity of natural products and the practical demands of modern drug discovery, offering a powerful, in-silico toolkit for researchers to accelerate the discovery of next-generation therapeutics.

Overcoming Challenges in Natural Product Research and Development

Within the framework of chemical space and structural diversity, natural products (NPs) represent an unsurpassed source of leading structures in drug discovery, possessing unique chemical diversity that complements synthetic collections [46] [3]. Through evolutionary selection, these molecules have been optimized for interactions with biological macromolecules, making their precise structural characterization essential for understanding function. Stereochemistry—the three-dimensional arrangement of atoms in molecules—forms a fundamental aspect of this structural identity, profoundly influencing biological activity, target affinity, and specificity [46]. Despite advances in analytical technologies, stereochemical inaccuracies and annotation gaps persist in chemical databases, creating significant downstream challenges for research reproducibility, drug discovery, and chemical biology.

The biosynthetic machinery of producing organisms generates natural products with specific stereochemical configurations that are often essential for their bioactivity. **Nature has settled for L-chirality for proteinogenic amino acids and D-chirality for the carbohydrate backbone of nucleotides**, suggesting that similar stereochemical patterns might exist among broader classes of natural products [59]. This precise

stereochemistry enables natural products to effectively modulate challenging biological targets, including protein-protein interactions, nucleic acid complexes, and antibacterial targets [46]. However, when stereochemical data is incompletely recorded or inaccurately annotated in databases, this valuable structural information is compromised, potentially leading to misinterpretation of structure-activity relationships and failed research replication.

The Scope of the Problem: Quantitative Assessment of Data Quality Issues

Stereochemical and Structural Error Rates

Chemical data quality issues extend beyond stereochemistry to encompass broader structural inaccuracies that significantly impact research outcomes. Systematic analyses of public databases and published literature reveal alarming error rates that undermine research reproducibility and model development.

Table 1: Chemical Data Error Rates in Public Databases and Literature

Data Source	Error Type	Error Rate	Impact
Medicinal Chemistry Publications [60]	Erroneous chemical structures	8% of compounds in WOMBAT database	Incorrect structure-activity relationships
Public & Commercial Databases [60]	Structural errors	0.1-3.4% (varies by database)	Compromised database reliability
Bioactivity Data [60]	Experimental variability	Mean error: 0.44 pKi units	Reduced model accuracy
Natural Products [59]	Undefined stereochemistry	~20% of stereocenters incorrectly assigned	Loss of bioactive configurations

The annotation gaps extend beyond structural errors to encompass significant limitations in bioactivity data quality. An assessment of >1.8 million compounds in public databases revealed that only 10.5% have acceptable biochemical activity (<10 μM) against human proteins, and this percentage drops dramatically when considering stricter quality filters [61]. When minimal criteria for chemical probes (potency ≤100 nM, selectivity ≥10-fold, and cellular activity ≤10 μM) are applied, only 0.7% of human-active compounds qualify as high-quality research tools [61]. This deficit in well-characterized chemical probes significantly impedes target validation and mechanistic studies, particularly for understudied proteins.

Impact on Research Reproducibility and Predictive Modeling

The consequences of poor data quality extend throughout the research lifecycle, affecting everything from initial discovery to development of predictive models. Studies have demonstrated that the prediction performance of Quantitative Structure-Activity Relationship (QSAR) models can be significantly affected by inaccurate and inconsistent representations of chemical structures [60]. The presence of structural duplicates with conflicting bioactivity data further complicates model development, artificially skewing predictivity and leading to either over-optimistic or inaccurately low performance estimates [60].

The reproducibility crisis in chemical genomics is underscored by analyses showing that only 20-25% of published assertions concerning purported biological functions for novel deorphanized proteins were consistent with pharmaceutical companies' in-house findings, with one analysis at Amgen yielding an even lower reproducibility rate of 11% [60]. These findings highlight the urgent need for improved data curation practices, particularly regarding stereochemical assignment and annotation.

Curation Workflows: Integrated Approaches for Chemical Data Quality

Addressing stereochemical inaccuracies requires systematic, integrated curation workflows that encompass both chemical structures and associated bioactivity data. These workflows should be implemented prior to or in conjunction with data deposition in public repositories to prevent the proliferation of irreproducible data.

Chemical Structure Curation Protocol

A comprehensive chemical curation workflow involves multiple steps to identify and correct structural errors [60]:

- **Compound Filtering**: Removal of incomplete or confusing records, including inorganics, organometallics, counterions, biologics, and mixtures that most molecular descriptor programs cannot handle effectively.
- **Structural Cleaning**: Detection and correction of valence violations, extreme bond lengths and angles, followed by ring aromatization and normalization of specific chemotypes.

- **Tautomer Standardization**: Application of empirical rules to consistently treat and represent tautomers, accounting for the most populated forms of a given chemical.
- **Stereochemical Verification**: Specialized checking of asymmetric carbons, with comparison to similar compounds in online databases to detect incorrect assignments.
- **Manual Inspection**: Despite automated tools, manual curation remains critical as some errors obvious to chemists are not detected by computers. Checking complex structures or compounds with numerous atoms is recommended.

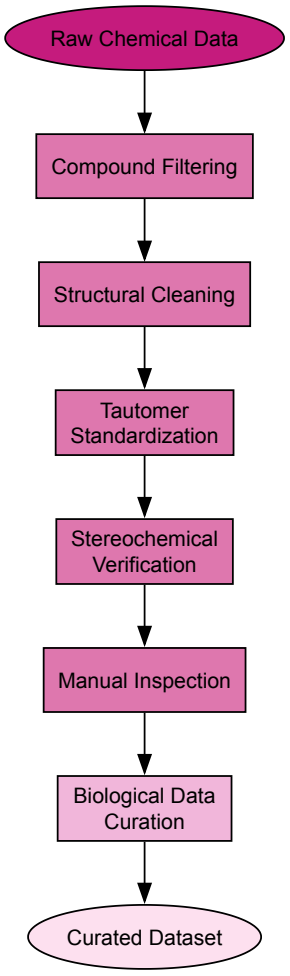
Table 2: Software Tools for Chemical Data Curation

Tool	Functionality	Access
Chemaxon JChem [60]	Molecular checker/standardizer	Free for academic organizations
RDKit [60]	Programmatic cheminformatics tools	Open-source
Schrodinger LigPrep [60]	Structure preparation and optimization	Commercial license
KNIME [60]	Workflow integration and automation	Open-source and commercial
NPstereo [59]	Stereochemical assignment for natural products	Open access

Biological Data Curation Protocol

Biological data curation presents unique challenges, as unlike chemical structures, there are no definitive rules defining the "true" value of a biological measurement [60]. However, suspicious entries in large chemogenomics datasets can be flagged through several approaches:

- **Duplicate Compound Processing**: Detection of structurally identical compounds tested in the same assay, often resulting in different internal substance IDs and experimental responses.
- **Activity Consistency Checking**: Identification of conflicting bioactivity measurements for the same compound-target pairs across different experimental setups.
- **Experimental Context Annotation**: Capturing critical methodological details that influence results, such as biological screening technologies (e.g., tip-based versus acoustic dispensing), which have been shown to significantly influence experimental responses [60].
- **Selectivity Assessment**: Evaluation of compound activity across multiple targets to identify potential off-target effects and assess selectivity profiles.



Integrated Chemical and Biological Data Curation Workflow

Advanced Methodologies for Stereochemical Assignment

Machine Learning Approaches

Recent advances in machine learning offer promising approaches for addressing stereochemical assignment challenges, particularly for natural products. The NPstereo language model demonstrates that stereochemical patterns exist among natural products that can be leveraged for automatic assignment [59]. This model translates natural product structures written as absolute SMILES (simplified molecular-input line-entry system) into corresponding isomeric SMILES notation containing stereochemical information, achieving **80.2% per-stereocenter accuracy for full assignments and 85.9% per-stereocenter accuracy for partial assignments** across various natural product classes [59]. These classes include secondary metabolites such as alkaloids, polyketides, lipids, and terpenes, highlighting the broad applicability of this approach.

The implementation of machine learning models like NPstereo can potentially correct or assign stereochemistry for newly discovered natural products, addressing a significant gap in current annotation practices. By training on the open-access COCONUT natural products database, this approach leverages existing curated knowledge to predict stereochemical configurations for incompletely characterized compounds.

Analytical and Chemical Solutions

For experimental verification of stereochemistry, several advanced methodologies provide robust solutions:

Chiral LC-MS Strategy: A recently developed approach combines chemical degradation, synthesis, and chiral liquid chromatography-mass spectrometry to determine absolute configuration with high accuracy and sensitivity [62]. This method addresses the challenging 3-methylpent-4-en-2-ol (MPO) moiety common in various polyketide natural products. The protocol involves:

- **Chemical Derivatization**: p-Nitrobenzoylation of hydroxy groups to enable sensitive detection
- **Oxidative Cleavage**: RuO_4 oxidation to cleave specific olefins and generate characteristic fragments
- **Stereoselective Synthesis**: Preparation of all possible stereoisomers for comparison
- **Chromatographic Separation**: Using a CHIRALPAK ID-3 column with aqueous MeOH gradient elution
- **Mass Spectrometric Detection**: Identification based on retention time matching with synthesized standards

This approach successfully assigned the absolute configuration of capsulactone, demonstrating its utility for microgram-scale natural products where traditional methods like X-ray crystallography or computational analysis of NMR data are challenging due to conformational flexibility or limited sample availability [62].

Chemical Engineering of Extracts: For enhancing structural diversity and verifying structure-activity relationships, chemical engineering of natural extracts provides a valuable strategy [3]. This approach applies chemical reactions that remodel molecular scaffolds directly on extracts of natural resources, generating analogs with new molecular skeletons. When applied to sesquiterpene lactones from Ambrosia tenuifolia using acid media conditions (p-toluenesulfonic acid), this method yielded derivatives containing unprecedented skeletons with potential anti-glioblastoma activity [3].

Research Reagent Solutions for Stereochemical Analysis

Table 3: Essential Reagents and Tools for Stereochemical Assignment

Reagent/Tool	Function	Application Example
NPstereo [59]	Machine learning-based stereochemical assignment	Predicting natural product stereochemistry from structural data
Chiral HPLC Columns (e.g., CHIRALPAK ID-3) [62]	Separation of stereoisomers	Resolving MPO moiety derivatives for absolute configuration assignment
p-Nitrobenzoyl Chloride [62]	Derivatizing agent for sensitive detection	Enhancing LC-MS detection of stereochemical fragments
RuO_4 Oxidation System [62]	Selective oxidative cleavage	Generating characteristic fragments from natural product scaffolds
p-Toluenesulfonic Acid [3]	Acid-catalyzed scaffold remodeling	Generating structural diversity in natural product extracts
COCONUT Database [59]	Natural product resource	Training data for machine learning approaches

Future Directions and Implementation Guidelines

Addressing stereochemical inaccuracies and annotation gaps requires coordinated efforts across multiple stakeholders in the research community. Based on current challenges and emerging solutions, several key directions emerge:

Enhanced Database Curation Practices: Implementation of standardized curation workflows should become a prerequisite for data deposition in public repositories. The structural standardization workflow implemented by PubChem provides a model for ensuring consistent representation of chemical structures across databases [60]. Community-engaged curation efforts, exemplified by ChemSpider's crowd-curated approach, can leverage collective expertise to improve data quality [60].

Integrative Approaches for Stereochemical Assignment: Combining computational predictions with experimental verification provides a powerful strategy for addressing stereochemical gaps. Machine learning models like NPstereo can provide initial assignments, which can then be verified through targeted analytical approaches such as chiral LC-MS for high-priority compounds [59] [62].

Metadata Capture and Contextual Annotation: Critical to addressing annotation gaps is the comprehensive capture of experimental context and metadata. As emphasized in data curation guidelines, **"the single most important function of metadata is to capture context"** [63]. Problems arising from incomplete contextual information can be minimized by capturing metadata at the time of data generation, rather than attempting to reconstruct it later.

Development of Minimum Information Standards: Establishment of community-accepted minimum information standards for natural product characterization would significantly improve data quality and reproducibility. These standards should explicitly include stereochemical metadata, experimental conditions for bioactivity testing, and methodological details critical for interpretation.

The movement toward improved data quality and curation in the chemical sciences represents both a challenge and opportunity for enhancing research reproducibility and accelerating discovery. As the field becomes increasingly data-driven, adopting robust practices for addressing stereochemical inaccuracies and annotation gaps will be essential for fully leveraging the structural diversity of natural products and their potential as probes of biological function and sources of therapeutic agents.

The exploration of the **biologically relevant chemical space (BioReCS)**, particularly the region inhabited by **Natural Products (NPs)**, is a cornerstone of drug discovery and biomedical research [6]. These compounds, evolved over millennia to interact with biological systems, represent an invaluable source of structural diversity and novel bioactivity [1]. However, the full potential of this chemical universe remains largely untapped due to profound **supply chain vulnerabilities** that create critical bottlenecks in natural product research [64] [65]. This whitepaper examines the

complex supply challenges facing researchers and presents a strategic framework for securing access to rare and renewable natural product sources, ensuring the continued expansion of explored chemical space for therapeutic development.

The chemical space of natural products constitutes a distinctive region within the broader bioactive chemical universe, characterized by unique structural complexity and three-dimensionality often lacking in purely synthetic libraries [6] [1]. This structural diversity translates to privileged binding characteristics with biological targets, making NPs particularly valuable for targeting challenging protein-protein interactions and other complex therapeutic targets. However, this same complexity often renders them difficult to synthesize economically, creating a fundamental dependency on biological sourcing with all its inherent supply chain uncertainties [6].

The Supply Chain Crisis: Systemic Vulnerabilities in Natural Product Sourcing

Geographic Concentration and Geopolitical Risks

The natural product supply chain exhibits **extreme geographical concentration** that mirrors the vulnerabilities seen in the rare earth elements market, where China controls 60-70% of mining and 85-90% of processing capacity [64]. This concentration creates unprecedented systemic risks for research institutions and pharmaceutical companies dependent on consistent access to rare natural materials. Similar to the **95-97% Chinese dominance in rare earth elements** critical for renewable energy technologies [65], the sourcing of many plant-derived natural products remains concentrated in politically and climatically vulnerable regions, creating single points of failure in the research pipeline [66].

Material Shortages and Infrastructure Limitations

Research institutions face **project-crippling shortages** of key natural materials due to intersecting challenges across the supply landscape:

- **Climate change impacts** on natural habitats where source organisms grow, with billion-dollar natural disasters increasing from 16 to 23 annually on average [66]
- **Processing and refinement bottlenecks** similar to those affecting rare earth elements, where limited specialized facilities create dependencies on single processing centers [64]
- **Transportation vulnerabilities** exemplified by the 40% drop in Suez Canal shipping during regional conflicts, critically delaying sample deliveries [65]

Regulatory and Compliance Complexities

The **Nagoya Protocol** and related access and benefit-sharing frameworks, while ethically crucial, have introduced significant administrative hurdles for researchers seeking to legally access genetic resources and traditional knowledge. Compliance requires meticulous documentation and benefit-sharing agreements that can take years to negotiate, dramatically slowing research initiation [65]. Additionally, **increasing trade restrictions** on biological materials, which have multiplied five-fold since 2009, further complicate international collaboration and material transfer [65].

Table 1: Quantitative Analysis of Key Supply Chain Vulnerabilities Affecting Natural Product Research

Vulnerability Factor	Current Impact Level	Projected 2030 Outlook	Primary Research Implications
Geographic Concentration	China controls 60-70% of rare earth mining [64]	China's market share reduced to 60-70% (from 85-90%) [64]	Limited sourcing options for mineral-rich NP sources
Material Shortages	50-60% shortage of rare earth metals predicted by 2030 [65]	6 key materials face significant supply gaps [65]	Critical research material unavailability
Transportation Disruptions	40% drop in Suez Canal shipping capacity [65]	Increasing climate and conflict-related disruptions [66]	Project delays and sample degradation
Regulatory Restrictions	Export restrictions affect 10% of critical material trade [65]	Increasing resource nationalism measures [65]	Legal barriers to international collaboration

Strategic Framework for Supply Resilience in Natural Product Research

Supply Chain Diversification Strategies

Building resilient natural product supply chains requires **systematic diversification** across multiple dimensions:

- **Geographic Diversification**: Developing sourcing partnerships across distinct biogeographic regions to minimize region-specific disruption impacts, similar to Japan's success in reducing Chinese rare earth imports from 90% to 60% through strategic alliances [65]
- **Taxonomic Diversification**: Expanding research beyond traditionally studied organisms to include extremophiles, marine invertebrates, and microbial symbionts with similar bioactivity potential but reduced sourcing challenges

- **Supplier Tiering**: Implementing multi-tier supplier relationships that provide redundancy for critical natural materials, creating alternative sourcing options during supply disruptions [66]

Technological Substitution and Innovation

Advanced technologies offer pathways to reduce dependency on wild-harvested natural materials:

- **Cultivation Advances**: Development of **controlled environment agriculture** specifically for medicinal plants, ensuring consistent quality and yield independent of climate variability
- **Synthetic Biology**: Engineering microbial hosts to produce complex natural products through heterologous expression of biosynthetic gene clusters, eliminating wild harvesting needs
- **Bioinspired Synthesis**: Developing efficient synthetic routes inspired by natural architectures but using commercially available starting materials [6]

Circular Economy Applications in Natural Product Research

Implementing **circular economy principles** within natural product research can dramatically reduce primary material requirements:

- **Sample Banking and Redistribution**: Establishing centralized repositories for research quantities of rare natural products, enabling redistribution to multiple research groups
- **Extraction Process Optimization**: Implementing sequential extraction protocols that maximize information and material yield from single samples
- **Waste Stream Valorization**: Identifying bioactive compounds in agricultural and processing byproducts traditionally considered waste [65]

Table 2: Strategic Solutions for Natural Product Supply Chain Resilience

Solution Category	Implementation Timeline	Key Advantage	Technical Requirements
Dual Sourcing	Immediate (0-6 months)	Reduces single-point failure risk	Supplier qualification protocols
Synthetic Biology Production	Long-term (3-5 years)	Eliminates wild harvesting	Metabolic engineering expertise
Circulating Sample Libraries	Short-term (6-18 months)	Maximizes utilization of rare materials	Standardized curation protocols
Agricultural Cultivation	Medium-term (2-4 years)	Quality and supply consistency	Horticultural optimization

Experimental Framework: Methodologies for Sustainable Natural Product Sourcing

Microscale Extraction and Analysis Protocol

This optimized workflow maximizes information yield from minimal natural product starting material, critical when working with rare or endangered species.

Materials and Reagents:

- **Enhanced Solvent System**: Dichloromethane-Methanol-Water (CH_2Cl_2-MeOH-H_2O, 2:1:0.5 v/v/v) for comprehensive metabolite extraction
- **Solid Phase Extraction (SPE) Cartridges**: C18-functionalized silica (500 mg/6 mL capacity) for fractionation
- **Analytical UHPLC-ESI-QTOF System**: Equipped with C18 column (100 × 2.1 mm, 1.7 µm) for high-resolution separation
- **Multiplicity-Edited HSQC NMR**: 600 MHz spectrometer with cryoprobe for structure elucidation at nanomole levels

Procedure:

- **Minimal Biomass Processing**: Homogenize 10-50 mg of dried plant/ microbial material with 1 mL extraction solvent using tissue lyser (2 × 30 s bursts)
- **Sequential Fractionation**: Apply total extract to SPE cartridge, elute with step gradient (5%, 25%, 50%, 75%, 100% MeOH in H_2O)
- **High-Throughput Screening**: Subject fractions to biological assays (minimum 5 µg per well for 384-well format)
- **Microscale Structure Elucidation**: For active fractions, concentrate to 50 µL for NMR analysis using 1.7 mm microcapillary tubes
- **Digital Archiving**: Complete LC-MS and NMR data uploaded to institutional repository with unique compound identifiers

Microbial Cultivation and Natural Product Production Protocol

This methodology enables sustainable production of microbial natural products through optimized fermentation.

Materials and Reagents:

- **Modified ISP-2 Medium**: Malt extract (10 g/L), yeast extract (4 g/L), glucose (4 g/L), pH 7.2 before sterilization
- **Adsorbent Resin**: XAD-16N (20 g/L) for in-situ product capture
- **Cryopreservation Solution**: Glycerol (20% v/v) in deionized water for strain preservation
- **Analytical Standards**: Commercially available natural products for HPLC calibration

Procedure:

- **Inoculum Preparation**: Transfer cryopreserved production strain to 50 mL seed medium in 250 mL baffled flask, incubate 48 h (28°C, 200 rpm)
- **Production Fermentation**: Transfer 10% (v/v) inoculum to production medium, incubate 5-7 days with XAD resin addition at 24 h
- **Product Recovery**: Separate resin by filtration, elute with acetone (2 × 100 mL), concentrate under reduced pressure
- **Chemical Profiling**: Analyze crude extract by UHPLC-PDA-ELSD-MS against standard compound library
- **Scale-Up Potential Assessment**: Correlate titer with cultivation parameters for process optimization

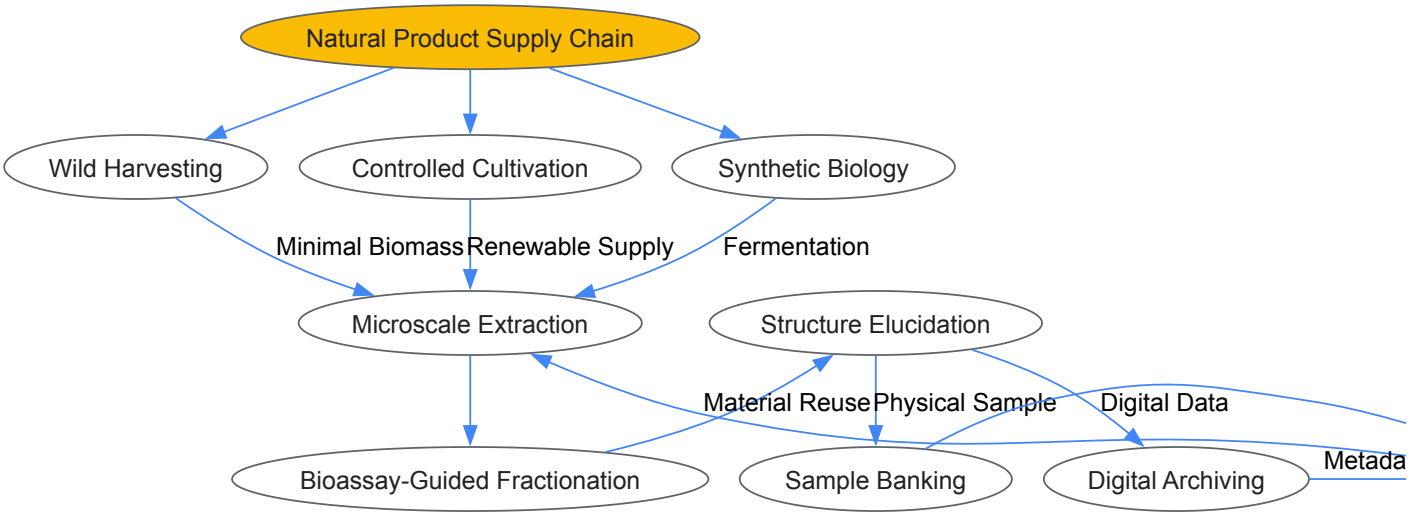


Diagram 1: Sustainable Natural Product Sourcing Workflow. This framework integrates traditional sourcing with modern sustainability practices, creating closed-loop material utilization.

The Researcher's Toolkit: Essential Solutions for Natural Product Supply Challenges

Table 3: Research Reagent Solutions for Natural Product Supply Chain Challenges

Reagent/Resource	Primary Function	Strategic Application	Sustainability Advantage
Diversity-Oriented Synthetic Libraries	Provides synthetic analogs of complex NPs	Enables SAR studies without resourcing	Eliminates harvesting of rare species [6]
Cryopreserved Culture Collections	Long-term storage of producing strains	Ensures genetic stability and access	Prevents genetic drift and loss [1]
Immobilized Enzyme Systems	Biocatalytic production of NP scaffolds	Semisynthesis from common precursors	Reduces multi-step synthesis waste [6]
Molecularly Imprinted Polymers	Selective capture of target chemotypes	Enriches minor constituents from complex mixtures	Maximizes information from limited biomass [1]
Annoticated Fraction Libraries	Pre-fractionated natural extracts	High-throughput screening resource	Enables multiple assays from single extraction [1]

The **structural diversity inherent in natural products** represents an irreplaceable region of chemical space that must be preserved and expanded despite significant supply challenges [6] [1]. By implementing the strategic framework outlined in this whitepaper—combining **supply chain diversification**, **technological innovation**, and **circular economy principles**—the research community can overcome current bottlenecks while building a more resilient and sustainable foundation for future discovery. The integration of traditional knowledge with cutting-edge sourcing and analytical methodologies will enable continued exploration of nature's chemical bounty while ensuring its preservation for generations of researchers to come.

The future of natural product research depends on transforming sourcing strategies from **linear, extractive models** to **circular, sustainable ecosystems** that maximize information yield while minimizing environmental impact. Through collaborative efforts across academic, industrial,

and governmental sectors, the research community can secure access to the rare and renewable natural resources that will fuel the next generation of therapeutic innovations.

The exploration of chemical space, particularly the region inhabited by natural products (NPs), is fundamental to modern drug discovery. NPs are the result of millions of years of evolutionary refinement, exhibiting unparalleled structural diversity, complex pharmacophores, and rich stereochemistry [67]. This inherent **structural complexity** is a double-edged sword: it underpins their high biological activity and ability to engage multiple targets—a valuable feature for overcoming drug resistance—but also presents significant **synthetic accessibility** and **optimization hurdles** for chemists [67]. The concept of the "Biologically Relevant Chemical Space" (BioReCS) encompasses all molecules with biological activity, and within this vast space, NPs represent a critically important, yet underexplored, region [6]. The structural complexity of NPs often surpasses the limits of contemporary synthetic chemistry, creating a formidable barrier between their discovery and practical application in therapeutics [68]. This whitepaper provides an in-depth examination of these challenges, framed within the context of chemical space and NP research, and details the computational and experimental strategies being developed to navigate them.

Quantifying Synthetic Accessibility: The SAscore Framework

For drug discovery professionals, the early and rapid assessment of a compound's synthetic tractability is crucial for prioritizing leads. The **Synthetic Accessibility Score (SAscore)** is a computational tool designed to meet this need, providing a quantitative estimate of ease of synthesis on a scale from 1 (easy) to 10 (very difficult) [69].

Calculation Methodology of the SAscore

The SAscore is a hybrid metric that combines two components:

- **Fragment Score (`fragmentScore`):** This component captures historical synthetic knowledge by statistically analyzing common structural features in a vast number of already synthesized molecules. The method involves:
  - **Training Set:** A representative set of nearly one million molecules from the PubChem database was used as a training set, as it encapsulates the collective output of synthetic efforts [69].
  - **Fragmentation:** The molecules are fragmented using the Extended Connectivity Fingerprints (ECFC_4#) method in Pipeline Pilot. This generates fragments that include a central atom and several levels of its neighbors [69].
  - **Contribution Calculation:** Each fragment's contribution to the overall score is derived from its frequency and context within the training set. The final `fragmentScore` is the sum of contributions from all fragments in the molecule, normalized by the total number of fragments [69].
- **Complexity Penalty (`complexityPenalty`):** This component directly penalizes structurally complex features that are synthetically challenging, including:
  - Presence of large rings and non-standard ring fusions.
  - High stereochemical complexity (e.g., numerous stereocenters).
  - Overall molecular size and the presence of non-standard structural features [69].

The final SAscore is a weighted combination of the `fragmentScore` and the `complexityPenalty`. This method has been validated against the estimations of experienced medicinal chemists, showing a strong correlation ($r^2 = 0.89$) for a set of 40 test molecules [69].

SAscore in Practice: A Quantitative Guide

The following table summarizes the SAscore scale and its interpretation, which is vital for decision-making in lead optimization.

Table 1: Interpretation of the Synthetic Accessibility Score (SAscore)

SAscore Range	Interpretation	Implications for Synthesis
1 - 3	Easy	Readily accessible using common reagents and reactions. Ideal for high-throughput synthesis.
4 - 6	Moderate	Requires specialized synthetic routes or reagents. Feasible with dedicated effort.
7 - 8	Difficult	Involves complex, multi-step syntheses with challenging purifications. Significant resource investment needed.
9 - 10	Very Difficult	Approaches the limits of synthetic chemistry. May require novel methodologies or be considered currently inaccessible [69].

The Structural Complexity of Natural Products and Associated Hurdles

Natural products occupy a region of chemical space characterized by high structural complexity. This complexity, while the source of their bioactivity, creates specific optimization hurdles.

Defining Features of Natural Product Complexity

- **Diverse and Novel Skeletons:** NPs often contain unique, polycyclic carbon skeletons that are not represented in common synthetic libraries. For example, genome mining of bacterial diterpene synthases has revealed previously unreported skeletons like tetraisoquinene (a 5/5/5/5-fused tetracyclic system) and salbirenol (a 7/5/6-tricyclic system) [70].
- **Rich Stereochemistry:** NPs frequently possess multiple stereocenters with specific absolute configurations that are essential for their biological activity. This stereochemical complexity is a major synthetic challenge [68].
- **Macrocycles and Non-standard Ring Systems:** Many NPs, such as macrocycles and compounds beyond the Rule of 5 (bRo5), are considered "underexplored chemical subspaces" due to the difficulty in their synthesis and modeling [6].

Optimization Hurdles in NP-Derived Drug Discovery

The following table outlines common hurdles encountered when optimizing natural product lead compounds.

Table 2: Common Optimization Hurdles for Natural Product Lead Compounds

Hurdle	Description	Impact on Development	Example from Literature
Narrow Antimicrobial Spectrum	The lead compound is only effective against a limited range of pathogens.	Limits therapeutic applicability.	A common limitation of many natural product leads, requiring structural modification to broaden activity [67].
Poor Solubility & Bioavailability	Unfavorable physicochemical properties hinder absorption and distribution.	Reduces in vivo efficacy.	Addressed through semi-synthetic modification, as seen in the development of more soluble pleuromutilin derivatives like Lefamulin [67].
Chemical Instability	The compound degrades under physiological or storage conditions.	Shortens shelf-life and in vivo half-life.	A key driver for the structural modification of many NPs to improve stability [67].
Complexity-Driven Low Yields	The synthetic route is long and produces a low overall yield.	Makes production economically unviable and limits material supply for testing.	The total synthesis of complex NPs like Taxol is a feat of organic chemistry but is not a practical source of the drug [70].

Experimental Protocols for Navigating Complexity

To overcome these challenges, researchers employ a suite of advanced experimental protocols that blend traditional chemistry with modern biotechnology and data science.

Genome Mining for Novel Natural Product Discovery

Genome mining allows for the discovery of novel NPs without prior cultivation of the source organism, directly tapping into the untapped regions of BioReCS.

Detailed Protocol:

- **Genome Sequencing and Analysis:** Sequence the genome of the target organism (e.g., bacteria from underexplored phyla like Myxococcota or Cyanobacteriota) [70].
- **Identification of Biosynthetic Gene Clusters (BGCs):** Use computational tools (e.g., antiSMASH) to identify BGCs encoding for terpene synthases, polyketide synthases, or non-ribosomal peptide synthetases.
- **Phylogenetic and Sequence Similarity Network (SSN) Analysis:** Construct phylogenetic trees and SSNs to prioritize BGCs that are evolutionarily distant from known clusters, indicating potential novelty [70].
- **Heterologous Expression:** Clone the codon-optimized gene of the target enzyme (e.g., a putative diterpene synthase) into an engineered heterologous host like *E. coli* that is equipped with the necessary biosynthetic precursors (e.g., geranylgeranyl diphosphate (GGPP) overproduction) [70].
- **Product Detection and Isolation:**
  - **Initial Screening:** Use thin-layer chromatography (TLC) or high-performance liquid chromatography (HPLC) of organic extracts to detect new products.
  - **Large-Scale Fermentation:** Culture the expressing strain in large volumes to produce sufficient material.

    - **Extraction and Purification:** Extract the culture with organic solvents (e.g., ethyl acetate) and purify the compound using techniques like flash chromatography and HPLC.
- **Structural Elucidation:** Determine the structure of the isolated compound using a combination of:
    - **Nuclear Magnetic Resonance (NMR):** 1D and 2D NMR experiments to establish the planar structure and relative configuration.
    - **Mass Spectrometry (MS):** High-resolution MS to confirm the molecular formula.
    - **Vibrational Circular Dichroism (VCD):** A powerful technique for determining the absolute configuration of complex molecules like diterpenes in solution without derivatization [70].

Structural Modification of Natural Product Leads

This protocol aims to improve the drug-like properties of a bioactive NP.

Detailed Protocol:

- **Structure-Activity Relationship (SAR) Analysis:** Synthesize or isolate a small library of analogues of the lead compound (e.g., through minor structural changes or sourcing from related species). Test these analogues for bioactivity to identify key pharmacophores.
- **Semi-Synthesis:** Use the naturally sourced lead compound as a starting material for chemical modification. Common reactions include:
    - **Acylation/Esterification:** To modify hydroxyl groups and improve lipophilicity or stability.
    - **Alkylation:** To alter steric and electronic properties.
    - **Introduction of Heteroatoms:** To modulate electronic character and binding interactions.
- **Biosynthetic Engineering:** Genetically engineer the native or heterologous host's BGC to produce novel "unnatural" natural products by altering the substrate specificity of key enzymes [68].
- **In vitro Evaluation of Modified Compounds:**
    - **Potency Assay:** Determine the IC~50~ or MIC against the target pathogen or cell line.
    - **Solubility and Metabolic Stability Measurement:** Assess improvements in key physicochemical and ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) properties.
    - **Cytotoxicity Assay:** Ensure selectivity against human cell lines.

Visualizing Workflows and Pathways

The following diagrams, generated using DOT language and adhering to the specified color and contrast guidelines, illustrate key experimental and conceptual workflows.

Diagram 1: SAscore Calculation Workflow

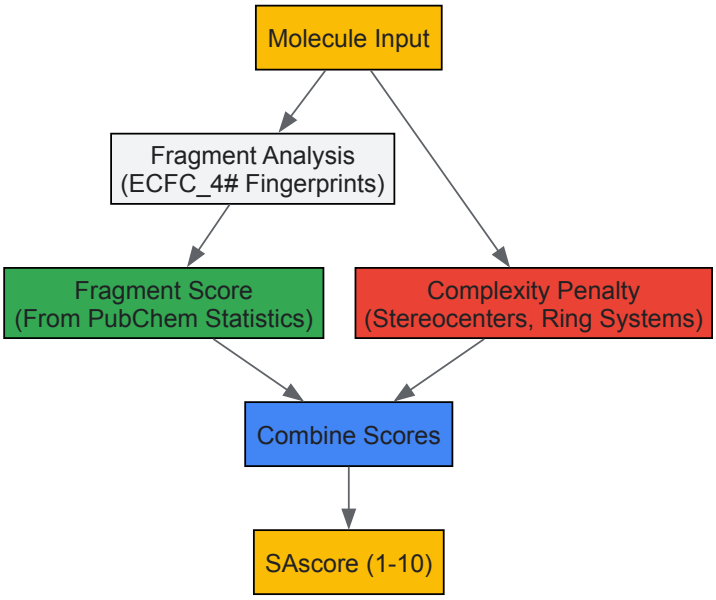
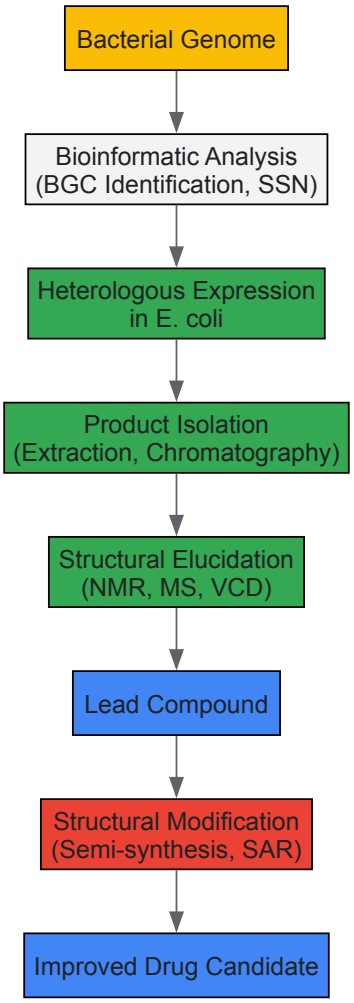


Diagram 2: Genome Mining & Modification Pipeline



The Scientist's Toolkit: Key Research Reagent Solutions

Table 3: Essential Reagents and Materials for Natural Product Research and Synthesis

Item	Function/Brief Explanation	Application Example
Cod-Optimized Synthetic Genes	Genes synthesized with codon usage optimized for the heterologous host to maximize protein expression.	Heterologous expression of bacterial terpene synthases in *E. coli* [70].
GGPP-producing E. coli Strain	An engineered *E. coli* strain that overproduces geranylgeranyl diphosphate (GGPP), the universal substrate for diterpene synthases.	Functional screening of putative diterpene synthase (di-TS) libraries without substrate synthesis [70].
Stereodivergent Enzymes	Engineered or discovered enzymes that catalyze the same reaction but produce products with different stereochemistry.	Accessing diverse stereoisomers of a complex natural product scaffold from a single precursor [68].
Vibrational Circular Dichroism (VCD)	An analytical technique for determining absolute configuration of chiral molecules in solution without crystal growth or chemical derivation.	Establishing the absolute stereochemistry of novel diterpenes like tetraisoquinene [70].
Deep Learning-Based Encoders (e.g., Chemformer)	Neural network models that convert molecular structures (SMILES) into numerical vectors (embeddings) that capture chemical meaning.	Mapping the chemical space of NPs and exploring evolutionary relationships [71].

The structural complexity of natural products presents a persistent challenge in drug discovery, directly impacting synthetic accessibility and optimization. However, the integration of computational tools like the SAscore with advanced experimental paradigms—such as genome mining, heterologous expression, and structure-based engineering—is providing a robust framework to navigate these hurdles. The ongoing development of "universal descriptors" and deep learning models for chemical space analysis [6] [71], coupled with the discovery of novel, promiscuous, or engineered enzymes [68] [70], is steadily expanding the accessible regions of BioReCS. By embracing these interdisciplinary strategies, researchers

can systematically deconstruct the complexity of NPs, overcome optimization hurdles, and unlock their full potential as next-generation therapeutics against pressing global health challenges.

The exploration of the **biologically relevant chemical space (BioReCS)** is a fundamental objective in modern drug discovery, particularly within natural products research [6]. Natural products (NPs) are invaluable resources, characterized by intricate scaffolds and diverse bioactivities, yet their clinical translation is often hindered by inadequate pharmacokinetic profiles and unforeseen toxicities [18] [72]. **In silico ADME/Tox (Absorption, Distribution, Metabolism, Excretion, and Toxicity) profiling** has emerged as a transformative discipline, enabling researchers to predict the behavior and safety of molecules within a biological system prior to synthesis and costly wet-lab experimentation [73]. This guide provides an in-depth technical overview of the computational strategies used to profile the pharmacokinetic and safety properties of chemical compounds, with a specific emphasis on navigating the unique and complex region of BioReCS occupied by natural products.

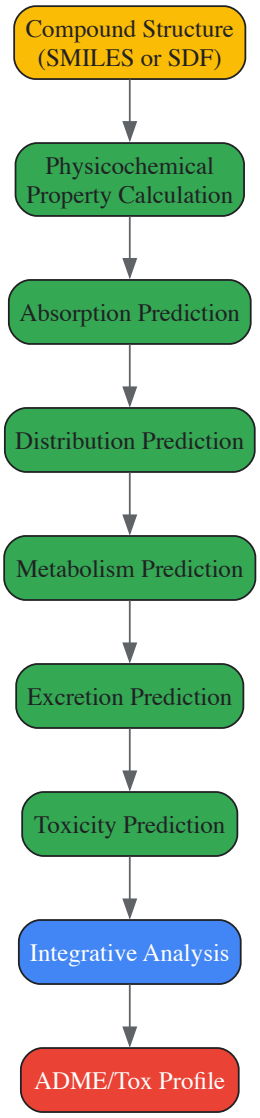
The Chemical Space of Natural Products and ADME/Tox Challenges

Natural products inhabit a distinct and privileged region of the chemical universe. Comparative chemoinformatic analyses reveal that NPs generally possess greater **molecular size, complexity**, and **structural diversity** compared to synthetic compounds (SCs) [13]. They tend to have more rings and non-aromatic rings, while SCs are characterized by a greater prevalence of aromatic rings [13].

However, this structural sophistication presents specific challenges for ADME/Tox profiling. Key issues include:

- **Instability and Low Solubility**: Many NPs, such as curcumin, suffer from instability and low water solubility, leading to inadequate oral bioavailability [72].
- **Complex Metabolism**: The intricate structures of NPs can lead to complex metabolic pathways that are difficult to predict without specialized tools [74].
- **Underexplored Territories**: Certain classes of NPs, such as macrocycles and peptides beyond the Rule of 5 (bRo5), represent underexplored regions of BioReCS where ADME/Tox prediction models are less established [6].

The following workflow outlines the core process of integrative in silico ADME/Tox profiling for a novel compound:



Key In Silico Methodologies and Properties

Fundamental Physicochemical Properties

The foundation of all in silico ADME/Tox predictions lies in the calculation of fundamental physicochemical properties. These descriptors define a compound's position in chemical space and its potential interactions with biological systems [6] [75].

Table 1: Key Physicochemical Properties for ADME/Tox Profiling

Property	Description	ADME/Tox Relevance	Ideal Range (Drug-like)
LogP	Partition coefficient (octanol/water)	Measures lipophilicity; affects membrane permeability and solubility [75]	Typically 0-5
LogS	Aqueous solubility	Critical for oral bioavailability and absorption [74]	> -4 logS
Molecular Weight (MW)	Mass of molecule	Impacts permeability; part of Rule of 5 [75]	< 500 Da
Polar Surface Area (PSA)	Surface area of polar atoms	Predicts passive molecular transport through membranes [75]	< 140 $Å^2$
H-bond Donors/Acceptors	Count of H-bond donors/acceptors	Affects solubility and permeability; part of Rule of 5 [75]	Donors: <5, Acceptors: <10

Core ADME/Tox Parameters and Prediction Models

In silico tools predict a wide array of biological and pharmacokinetic endpoints by leveraging machine learning and QSAR models trained on large experimental datasets [73].

Table 2: Core ADME/Tox Parameters and Predictive Assays

Parameter Category	Specific Endpoints	Common In Silico Models / Assays
Absorption	Gastrointestinal absorption, Caco-2 permeability, P-glycoprotein substrate/inhibition [74]	PAMPA, Caco-2 cell models [76]
Distribution	Plasma Protein Binding (PPB), Volume of Distribution (Vd), Blood-Brain Barrier (BBB) penetration [74]	PPB assays, BBB prediction models [76]
Metabolism	Cytochrome P450 (CYP) enzyme inhibition/induction/substrate specificity, metabolic stability, metabolite identification [74]	CYP inhibition assays, liver microsome stability assays [76]
Excretion	Total clearance, half-life, renal transporter interactions (e.g., OCT2) [74]	Renal and biliary excretion studies [76]
Toxicity	hERG inhibition (cardiotoxicity), hepatotoxicity, genotoxicity (Ames test), acute toxicity [74] [72]	hERG models, Ames test, in vivo toxicity studies [76]

Experimental Protocols for In Silico Profiling

This section details a standard operational protocol for conducting a comprehensive in silico ADME/Tox analysis, as exemplified by recent studies on novel compounds [74] [72].

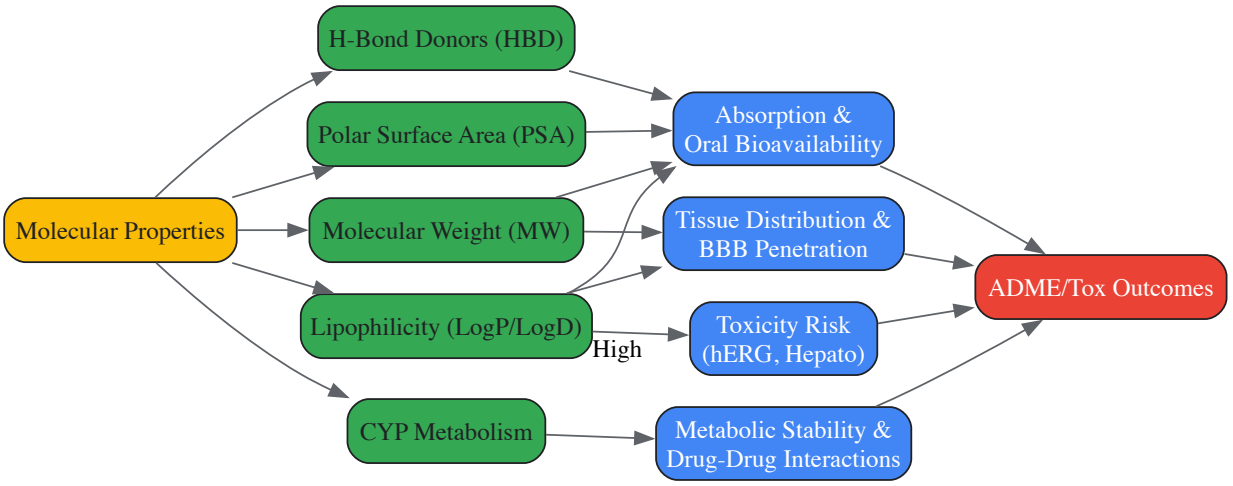
Protocol: Multifaceted ADME Profiling of a Novel Compound

Objective: To generate a comprehensive ADME profile for a novel chemical entity using an integrative computational approach.

Methodology:

- **Compound Structure Preparation**:
  - Obtain the canonical SMILES string or SDF file of the compound.
  - For ionizable compounds, generate the major microspecies at physiological pH (7.4), as ionization state profoundly impacts properties like solubility and permeability [6].
- **Software Suite Selection**:
  - Employ a diverse set of computational tools to ensure robustness and cross-validation. A recommended toolkit includes:
    - **ADMETlab 3.0**: Open-access platform with >100 predicted endpoints [74] [72].
    - **SwissADME**: Web tool for fast physicochemical and pharmacokinetic profiling.
    - **ACD/Percepta** or **ADMET Predictor**: Commercial software for advanced metabolic and toxicological predictions [74].
    - **XenoSite**: For detailed metabolite identification [74].
- **Data Generation and Extraction**:
  - Input the prepared structure into each software.
  - Execute predictions for all core ADME/Tox parameters listed in Table 2.
  - Systematically record quantitative predictions (e.g., numerical values for LogP, LogS, % absorption) and qualitative classifications (e.g., "Yes/No" for BBB penetration, "High/Low" solubility).
- **Integrative Analysis and Data Reconciliation**:
  - Compare results across different software tools. Discrepancies are common and should be analyzed based on the underlying algorithms and training data of each tool.
  - Generate a consensus profile, noting any parameters with high prediction uncertainty.
  - For metabolism, compile a list of predicted major metabolites and the CYP enzymes responsible for their formation [74].

The relationships between the key molecular properties predicted in this protocol and the resulting ADME/Tox outcomes can be visualized as follows:



The Scientist's Toolkit: Essential Research Reagents and Computational Solutions

Table 3: Key Research Reagent Solutions for ADME/Tox Profiling

Tool / Resource	Type	Function in ADME/Tox Profiling
ADMETlab 3.0	Open-access computational platform	Systematically evaluates >100 ADMET endpoints based on chemical structure, providing a comprehensive profile for drug design [74] [72].
SwissADME	Free web tool	Provides fast predictions of key physicochemical properties, pharmacokinetics, and drug-likeness for early-stage compound prioritization [74].
ACD/Percepta	Commercial software suite	Predicts physicochemical properties, pKa, logP, and spectroscopic data, offering high accuracy with reliability indicators for results [74].
ADMET Predictor	Commercial machine learning software	Generates robust QSAR models for over 175 ADMET endpoints, including identification of metabolic hotspots and metabolites [74].
XenoSite	Web server	Predicts the sites and pathways of small molecule metabolism, particularly by Cytochrome P450 enzymes, aiding in metabolite identification [74].
Caco-2 Cell Line	*In vitro* assay system	Models human intestinal absorption; used to generate experimental data for validating in silico permeability predictions [76].
Human Liver Microsomes	*In vitro* assay system	Used to study Phase I metabolic stability and identify major metabolic pathways; data validates computational metabolism models [76].

In silico ADME/Tox profiling has matured into an indispensable component of the drug discovery workflow, effectively bridging the gap between the vast potential of chemical space, particularly that of natural products, and the practical requirements for developing safe and effective therapeutics [6] [73]. The integrative application of diverse computational tools allows for the rapid and resource-efficient triaging of candidate molecules, guiding the optimization of natural product-inspired analogs towards favorable pharmacokinetic and safety profiles [18] [72].

The future of this field is being shaped by several key trends. The rise of **generative AI and deep learning models** enables the simultaneous optimization of bioactivity and multiple ADME/Tox properties during molecular design [73] [18]. Furthermore, there is a growing emphasis on making toxicological models **FAIR (Findable, Accessible, Interoperable, and Reusable)** to foster regulatory acceptance [73]. As the chemical space continues to expand with novel modalities like PROTACs and macrocycles, in silico models must also evolve. The development of **universal molecular descriptors** capable of handling the immense structural diversity of natural products and other complex molecules will be crucial for comprehensively mapping and exploiting the biologically relevant chemical space for therapeutic innovation [6] [73].

The exploration of natural products represents a cornerstone of modern therapeutic discovery, offering a vast reservoir of chemical diversity honed by millions of years of evolution. The fundamental challenge in leveraging this resource lies in efficiently navigating its immense structural complexity to identify genuine bioactive leads. Research on fungal isolates, such as those from the *Alternaria* genus, reveals that a surprisingly modest number of isolates (195) can capture nearly 99% of observable chemical features, yet a significant proportion (17.9%) of features appear in single isolates, underscoring the extensive, untapped diversity available [21]. This reality necessitates robust strategies that can bridge the gap between in silico prediction and experimental confirmation. The integration of computational and experimental methodologies creates a powerful synergy, accelerating the transition from hypothesis to validated candidate and maximizing the return on research investment. This guide details the protocols and platforms that enable this integration, with a specific focus on applications within natural product-based drug discovery.

Computational Prediction Frameworks

Machine Learning and Cheminformatic Models

Computational prediction has moved beyond simple filtering to sophisticated, learning-based models that can prioritize compounds with a high probability of success.

- **Bayesian Classification for Medicinal Chemistry Due Diligence**: Bayesian models can be trained to predict the subjective quality assessments of experienced medicinal chemists. These models analyze molecular properties and structural features to flag compounds with potential chemical reactivity, promiscuity, or other undesirable characteristics. Analysis has shown that such models achieve accuracy comparable to other established measures of drug-likeness [77]. Key molecular properties considered often include pKa, molecular weight, heavy atom count, and rotatable bond number.
- **Network-Based Lead Identification**: This approach leverages the intrinsic similarity between molecules within large chemical spaces. One framework involves constructing an ensemble of chemical similarity networks based on multiple fingerprint types. Given a target protein, a deep learning-based drug-target interaction model first narrows the candidate pool. Network propagation then prioritizes candidates highly correlated with experimental drug activity scores (e.g., IC50). This method has demonstrated success in identifying active compounds for targets like CLK1, with experimental validation rates of two out of five synthesizable candidates in a case study [78].
- **Ultra-Large Virtual Screening**: The advent of gigascale chemical libraries, containing billions of readily accessible virtual compounds, has transformed structure-based screening. Docking these vast libraries is now computationally feasible, allowing for the discovery of entirely new chemotypes. This is further accelerated by iterative screening approaches and active learning, which focus computational resources on the most promising regions of chemical space [79].

Quantitative Diversity Assessment for Natural Product Libraries

For natural products, a key computational step is assessing and guiding the construction of the library itself. A bifunctional approach combining genetic barcoding and metabolomics provides actionable metrics.

- **Methodology**: Internal transcribed spacer (ITS) sequencing is used to establish gene-based clades of source organisms, such as fungi. Concurrently, liquid chromatography-mass spectrometry (LC-MS) profiling generates data on the chemical features present in each isolate. Principal-coordinate analysis (PCoA) of this metabolomic data can reveal distinct chemical clusters [21].
- **Analysis**: Feature accumulation curves, which plot the number of unique chemical features detected against the number of isolates sampled, are generated. This quantitative measurement allows researchers to determine when a library is approaching adequate coverage for a given clade and to identify under- or oversampled pools of secondary-metabolite scaffolds, thus enabling real-time adjustment of collection strategies [21].

Table 1: Key Computational Platforms and Their Applications in Natural Product Research

Platform/Method	Primary Function	Key Advantage	Reference
Bayesian Classifiers	Predicts medicinal chemistry quality	Learns from expert evaluation to filter undesirable compounds	[77]
Network Propagation	Identifies lead compounds in large databases	Uses ensemble similarity networks for high correlation with activity	[78]
Ultra-Large Docking	Structure-based virtual screening of gigascale libraries	Discovers novel chemotypes from billions of compounds	[79]
LC-MS Metabolomics + ITS Barcoding	Measures chemical diversity of natural product libraries	Guides rational library design to maximize metabolite coverage	[21]

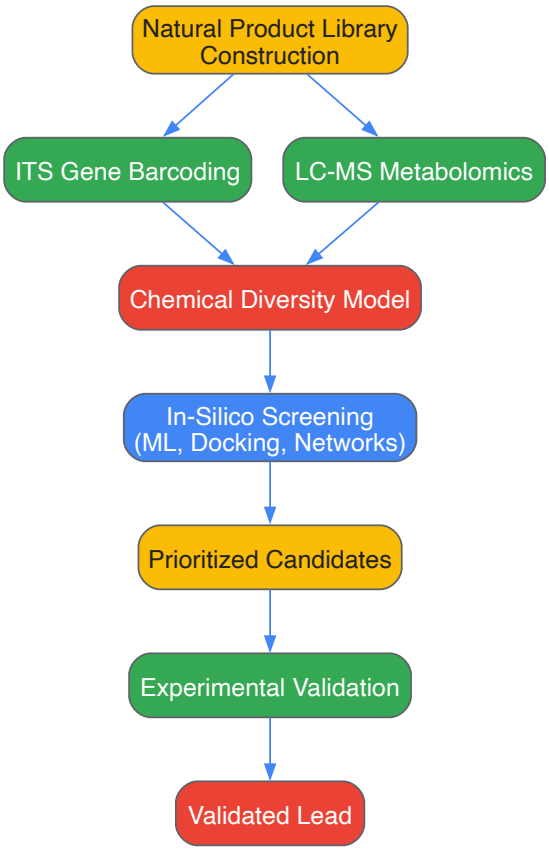


Diagram 1: Integrated discovery workflow from library building to validated lead.

Experimental Validation Protocols

Computational predictions are only as valuable as their experimental confirmation. The following protocols are essential for validating putative hits.

Compound Sourcing and Preparation

- **Sourcing from On-Demand Libraries**: For synthetic compounds identified from virtual libraries like ZINC20, many are available for purchase from commercial vendors as powder or DMSO stock solutions [79].
- **Natural Product Isolation**: For novel natural products, follow bioactivity-guided fractionation. The crude extract is subjected to chromatographic separation (e.g., HPLC, flash chromatography). Individual fractions are tested for activity, and active fractions are further purified until a pure, active compound is isolated. Structure elucidation is performed via NMR, HR-MS, and other spectroscopic techniques [21] [80].

In Vitro Binding and Potency Assays

Objective: To confirm direct binding to the target and quantify biological activity. **Protocol for a Kinase Target (e.g., CLK1)**:

- **Expression and Purification**: Express the recombinant protein target in a suitable system (e.g., *E. coli*, insect cells) and purify it using affinity chromatography.
- **Binding Assay**: Use a technique like Surface Plasmon Resonance (SPR) or a thermal shift assay. For SPR, immobilize the target on a sensor chip. Inject compound candidates at a range of concentrations and measure the binding response in Resonance Units (RU). A dose-dependent binding signal confirms direct interaction. Calculate the equilibrium dissociation constant (K_D) from the sensorgrams [78].
- **Activity Assay**: Perform a biochemical assay to determine IC50 values. For a kinase, this typically involves incubating the kinase with its substrate, ATP (including radiolabeled or ATP-conjugated), and varying concentrations of the test compound. Quantify the phosphorylated product. Fit the dose-response data to a sigmoidal curve to calculate the IC50, the concentration that inhibits 50% of the kinase activity [78].

Selectivity and Specificity Profiling

Objective: To ensure the compound is not a promiscuous binder or a pan-assay interference compound (PAINS). **Protocol**:

- **Counter-Screening**: Test the compound against a panel of related and unrelated targets (e.g., other kinases, GPCRs, ion channels) at a single concentration (e.g., 10 μM).

- **Specificity Assessment**: A compound with good specificity will show significant activity only against the intended target. Tools like the BadApple (Bioactivity Data Associative Promiscuity pattern Learning Engine) can be used in silico to flag scaffolds known for promiscuous behavior [77]. Furthermore, compounds should be evaluated against a set of PAINS filters to rule out common assay interference motifs [77].

Initial ADMET and Physicochemical Property Assessment

Objective: To evaluate early developability of the validated hit. **Protocol**:

- **Solubility**: Measure kinetic solubility in PBS (pH 7.4) using nephelometry or UV/vis spectrometry.
- **Metabolic Stability**: Incubate the compound with liver microsomes (human or other species) and measure the half-life of parent compound depletion over time using LC-MS/MS.
- **Cytotoxicity**: Perform a cell viability assay (e.g., MTT, CellTiter-Glo) on a standard cell line (e.g., HEK293, HepG2) to determine a preliminary cytotoxicity profile (CC50).
- **Property Calculation**: Calculate key physicochemical properties such as LogP, molecular weight, hydrogen bond donors/acceptors, and topological polar surface area (TPSA) to assess compliance with rules of drug-likeness [77].

Table 2: Essential Research Reagent Solutions for Validation

Reagent / Material	Function in Validation	Application Example
Recombinant Target Protein	The macromolecular target for binding and activity assays.	Purified CLK1 kinase for SPR and enzymatic assays [78].
LC-MS/MS Systems	Separation, quantification, and structural characterization of compounds and metabolites.	Purity assessment, metabolic stability testing, and metabolomics [21].
Surface Plasmon Resonance (SPR) Chip	Immobilizes the target protein to measure real-time binding kinetics of candidate compounds.	Determining association/dissociation rate constants (*kon*, *koff*) and K_D [78].
Liver Microsomes	A subcellular fraction containing drug-metabolizing enzymes (CYPs, UGTs).	In vitro assessment of metabolic stability (T_{1/2}) and metabolite identification [77].
Cell-Based Assay Kits (e.g., MTT, CellTiter-Glo)	Measure cell viability and proliferation through colorimetric or luminescent signals.	Determining cytotoxic concentration (CC50) in preliminary safety profiling.

Integrated Workflows: From Prediction to Proof

The true power of integration is realized when computational and experimental phases are interlinked in an iterative cycle. The case study on CLK1 inhibitors provides a concrete example [78]. A network-based lead identification framework was applied to a large compound database, prioritizing 24 candidates. Five of these were selected for synthesis, and two were experimentally validated in binding assays, demonstrating a high success rate. This validates the computational model and provides a shortlist of high-quality starting points for further medicinal chemistry optimization.

This integrated approach is also revolutionizing natural products research. The metabolomics-driven strategy for building fungal natural product libraries ensures that screening collections are optimally diverse, increasing the likelihood of discovering novel bioactive compounds [21]. Once a library is built, ultra-large virtual screening can be applied to natural product-inspired or dereplicated chemical spaces, with active learning models using initial assay results to refine subsequent selection of natural product fractions for testing [79].

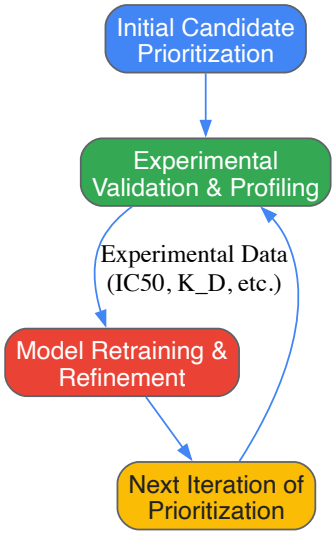


Diagram 2: Iterative lead optimization feedback cycle.

The integration of computational predictions with experimental validation is no longer a luxury but a necessity for efficient drug discovery, particularly within the rich and complex landscape of natural products. Frameworks that combine quantitative diversity assessment, machine learning-based prioritization, and robust experimental protocols create a powerful, iterative engine for lead identification. As computational power grows and algorithms become more sophisticated, the ability to navigate chemical space predictively will only improve. The future lies in tighter, more automated feedback loops where experimental data continuously refines computational models, enabling the systematic and accelerated transformation of nature's chemical diversity into the next generation of therapeutic agents.

Validating Success: Natural Products in Clinical Development and Beyond

Natural products (NPs) and their structural analogues have historically been, and continue to be, a major source of new therapeutic agents, especially in the critical areas of cancer and infectious diseases [14]. These compounds, which are naturally synthesized by organisms such as plants, bacteria, animals, and fungi, possess a unique and vast chemical diversity that has evolved through natural selection for optimal interactions with biological macromolecules [46]. The structural complexity and bio-relevance of NPs allow them to occupy a broad and privileged region of the **biologically relevant chemical space (BioReCS)**, enabling them to modulate challenging targets such as protein-protein interactions and nucleic acid complexes [46] [6]. Within the context of chemical space, NPs exhibit greater steric complexity and a wider variety of ring systems compared to synthetic and combinatorial chemistry compounds, filling regions that are often underexplored by synthetic approaches [3]. This review quantifies the success rates of natural products in the drug development pipeline from 1981 to the present, examining the structural and evolutionary drivers behind their continued prominence and their critical role in populating the bioactive regions of chemical space with successful drugs.

Quantitative Analysis of Natural Product Contributions

Success Rates Across Clinical Development Phases

Recent analysis of clinical trial progression reveals a telling trend: natural products and their derivatives demonstrate increasing success rates as they advance through clinical phases, in contrast to the declining success rates of purely synthetic compounds. This trend provides a key explanation for the over-representation of NP-derived structures among approved drugs despite their under-representation in early-stage discovery [81].

Table 1: Clinical Trial Success Rates by Compound Origin

Clinical Trial Phase	Natural Products	Natural Product-Derived (Hybrid)	Synthetic Compounds
Phase I	~20% (940/4749)	~15% (724/4749)	~65% (3085/4749)
Phase III	~26% (860/3356)	~19% (632/3356)	~55.5% (1863/3356)
FDA Approved Drugs	~25% (1149/4749)	~20% (895/4749)	~55% (Approx.)

The data demonstrates a steady increase in the proportion of NP and NP-derived compounds from clinical trial phases I to III (from approximately 35% combined in phase I to 45% in phase III), with a corresponding inverse trend observed for synthetics [81]. This increasing success rate culminates in NPs and NP-derivatives constituting approximately 45% of approved small molecule drugs, aligning with the proportions observed in late-stage clinical trials [81].

Historical Contributions from 1981 to Present

Analysis of approved drugs over a 30-year period from 1981 to 2010 provides robust quantitative evidence of the sustained importance of natural products in drug discovery. During this period, nearly half of all approved drugs traced their origins back to natural products, either as unaltered NPs, derivatives of NPs, or synthetic compounds with pharmacophores resembling NPs [81] [82].

Table 2: Natural Product Contributions to Approved Drugs (1981-2010)

Drug Category	Percentage of Approved Drugs	Key Therapeutic Areas
Unaltered Natural Products, Derivatives, or NP-like Pharmacophores	~50%	Anti-infectives, Anticancer
Purely Synthetic Small Molecules	~25%	Various
Biologics, Vaccines, and Others	~25%	Various

The influence is particularly pronounced in specific therapeutic areas. In the domain of **cancer therapeutics**, the contribution of natural products is even more substantial. From the 1940s to the present, of the 175 small molecule anticancer drugs approved, 131 (74.8%) are other than purely synthetic, with 85 (48.6%) being either natural products or directly derived therefrom [82]. This data underscores the critical role of natural products as sources of novel structures, if not always the final drug entity.

The Chemical Space and Structural Diversity of Natural Products

Exploring the Biologically Relevant Chemical Space (BioReCS)

The concept of **chemical space (CS)** represents a multidimensional descriptor where molecular properties define coordinates and relationships between compounds [6]. Within this framework, the **biologically relevant chemical space (BioReCS)** comprises molecules with biological activity, both beneficial and detrimental. Natural products occupy a strategic region within BioReCS, characterized by structural features optimized through evolutionary pressure for biological interactions [46] [6].

Analysis of microbial natural products alone reveals extensive chemical diversity, with the Natural Products Atlas database containing 36,454 compounds (version v2024_09) forming 4,148 structural clusters based on molecular fingerprint similarity [8]. This clustering demonstrates that NP scaffolds often form distinct "islands of chemical diversity" with high interconnectivity within clusters but significant structural distinction between different scaffold classes [8]. This distribution suggests evolutionary drivers have selected for specific structural frameworks that effectively interact with biological targets.

Structural Drivers of Success

Several structural and physicochemical properties contribute to the higher success rates of natural products in drug development:

- **Enhanced Drug-like Properties**: NPs frequently exhibit desirable Absorption, Distribution, Metabolism, and Excretion (ADME) properties, despite often falling outside the traditional "Rule of 5" guidelines for drug-likeness [81] [14].
- **Structural Complexity**: NPs possess greater steric complexity and a wider variety of ring systems compared to synthetic compounds, with less than one-fifth of the ring systems found in nature represented in current commercial drugs [3].
- **Target Selectivity**: Through evolutionary pressure, NPs are intrinsically endowed with ligand-protein binding motifs that serve a broad range of biological functions, resulting in molecular scaffolds and pharmacophore patterns that are especially selective to cellular targets [81].
- **Reduced Toxicity Profile**: In vitro and in silico toxicity studies indicate that NPs and their derivatives generally present less toxic alternatives to their synthetic counterparts, contributing to their higher success rates in late-stage clinical development where toxicity is a major cause of attrition [81].

Research Methodologies and Experimental Protocols

Quantifying Clinical Success Rates

Objective: To determine the success rates of natural products, natural product-derived (hybrid), and synthetic compounds through stages of clinical development and their representation in approved drugs.

Methodology:

- **Data Collection**: Compile comprehensive datasets of compounds entering clinical trial phases I, II, and III from publicly available sources and proprietary databases [81].
- **Compound Classification**: Categorize compounds into:
  - **Natural Products (NPs)**: Unmodified natural compounds.
  - **Hybrid Compounds**: Semisynthetic derivatives of natural products.
  - **Synthetic Compounds**: Purely synthetic compounds without natural product inspiration.
- **NP-likeness Assessment**: Implement chemical similarity metrics based on structural fingerprints to classify compounds according to their resemblance to known natural products [81].
- **Success Rate Calculation**: Track the proportions of each compound class through clinical phases to determine phase transition probabilities and overall approval rates.
- **Statistical Analysis**: Use appropriate statistical methods to determine the significance of differences in success rates between compound classes.

Key Considerations: The classification system must account for the fact that unmodified natural products often cannot be patented due to legal restrictions on patenting "products of nature," which influences the types of NP-based compounds advanced into clinical development [81].

Diversity-Enhanced Extracts and Chemical Engineering

Objective: To enhance the structural diversity of natural product libraries through chemical modification of crude extracts, generating novel scaffolds with potential bioactivity.

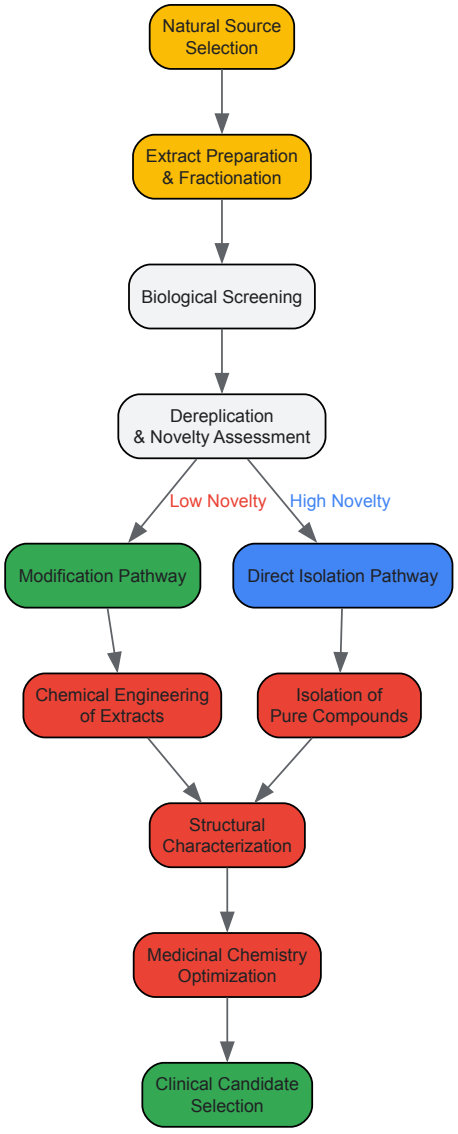
Methodology:

- **Extract Preparation**: Prepare standardized extracts from natural sources (e.g., plants, marine organisms, microorganisms) using solvents of varying polarity [3].
- **Chemical Modification**: Subject extracts to reaction conditions that remodel molecular scaffolds (e.g., acid-mediated rearrangements, ring distortion reactions) [3].
    - **Example Protocol**: Treat dichloromethane extract of *Ambrosia tenuifolia* (enriched in sesquiterpene lactones) with p-toluenesulfonic acid in toluene under reflux conditions for 1 hour to induce skeletal rearrangements [3].
- **Bioactivity-Guided Fractionation**: Fractionate chemically modified extracts using chromatographic techniques guided by biological assays relevant to target diseases [3].
- **Compound Characterization**: Isulate pure compounds from active fractions and determine structures using NMR, MS, and other spectroscopic techniques.
- **Chemical Space Analysis**: Calculate molecular descriptors and fingerprints to evaluate structural diversity and position within chemical space compared to unmodified natural products and synthetic libraries [3] [83].

Research Workflow and Signaling Pathways

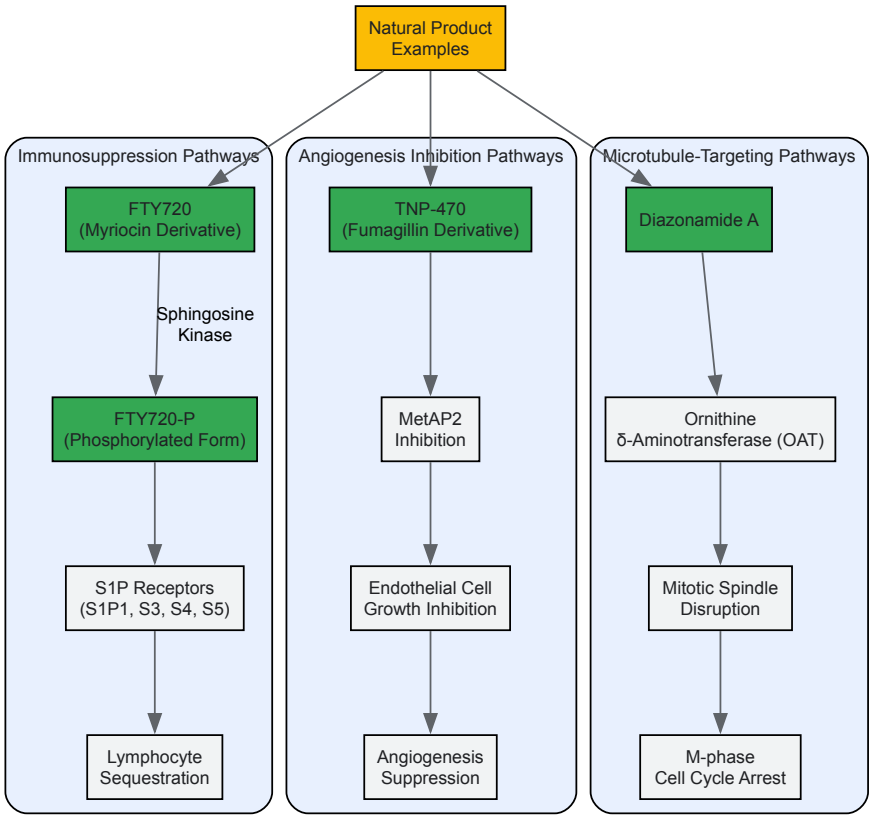
Natural Product Drug Discovery Workflow

The following diagram illustrates the integrated workflow for natural product-based drug discovery, highlighting key decision points and strategic approaches:



Target Pathways of Selected Natural Products

Natural products frequently modulate complex signaling pathways and cellular processes. The following diagram illustrates the molecular targets and pathways of selected clinically significant natural products:



The Scientist's Toolkit: Essential Research Reagents and Materials

Table 3: Key Research Reagents for Natural Product Drug Discovery

Reagent/Material	Function/Application	Key Characteristics
SureChEMBL Database [81]	Patent compound data mining	Contains over 1 million patent applications with NP structures; enables analysis of NP representation in patents
Natural Products Atlas [8]	Microbial NP database	36,454 microbial natural products; enables chemical diversity analysis and novelty assessment
ChEMBL & PubChem [81] [6]	Bioactivity databases	Annotated compound bioactivity data; enables assessment of NP-likeness and biological potential
p-Toluenesulfonic Acid [3]	Chemical remodeling of extracts	Catalyst for scaffold diversification reactions; enables generation of novel chemotypes from natural extracts
LC-HRMS-NMR Platforms [14]	Structural characterization	Hyphenated analytical techniques for rapid dereplication and structure elucidation
Sephadex LH-20 & Silica Gel [3]	Chromatographic separation	Stationary phases for bioactivity-guided fractionation of complex natural extracts

The quantitative evidence presented unequivocally demonstrates that natural products and their derivatives continue to play a disproportionately successful role in drug discovery, constituting approximately 45% of approved small molecule drugs despite representing a minority of compounds in early-stage development. Their increasing success rates through clinical trial phases—from approximately 35% in phase I to 45% in phase III—highlight their superior **drug-like properties** and **reduced toxicity profiles** compared to purely synthetic compounds [81].

The structural diversity inherent to natural products positions them strategically within the **biologically relevant chemical space**, enabling them to address challenging therapeutic targets that have proven difficult with synthetic compounds alone [46] [6]. As drug discovery faces new challenges, including antimicrobial resistance and difficult-to-treat cancers, natural products offer privileged scaffolds that have been evolutionarily optimized for biological interactions [14] [8].

Future natural product research will be increasingly guided by chemical space concepts, with strategic exploration of underexplored regions including macrocycles, metal-containing NPs, and complex marine natural products [84] [6]. The integration of **chemical engineering approaches** with traditional natural product chemistry, complemented by advanced analytical and computational methods, promises to further enhance our ability to

mine nature's chemical diversity for the next generation of therapeutic agents [3] [14]. As the field evolves, the quantitative success rates of natural products provide a compelling rationale for their continued prioritization in drug discovery campaigns aimed at addressing unmet medical needs.

The concept of **chemical space**—a multidimensional domain where molecular properties define coordinates and relationships—serves as a fundamental framework for modern drug discovery [6]. Within this vast universe, the **biologically relevant chemical space (BioReCS)** comprises molecules with documented biological activity, both beneficial and detrimental [6]. Drug discovery efforts rely on screening collections to navigate this space, with these libraries broadly categorized as **synthetic libraries** (designed and synthesized compounds) and **natural product (NP)** libraries (compounds derived from natural sources) [85]. While synthetic libraries, including fragment-like (MW < 250 Da) and lead-like (250 ≤ MW < 350 Da) categories, offer advantages in purity and synthetic tractability, they often explore limited regions of chemical space [86] [85]. In contrast, natural products provide invaluable structural diversity that has been evolutionarily optimized for biological interactions, making them essential for comprehensive chemical space coverage in screening campaigns [87] [7].

Fundamental Differences Between Natural Products and Synthetic Compounds

Structural and Physicochemical Properties

Natural products and synthetic compounds occupy distinct yet complementary regions of chemical space, with significant differences in their structural characteristics and property distributions. The table below summarizes key comparative features:

Table 1: Structural and Physicochemical Properties of Natural Products vs. Synthetic Compounds

Property	Natural Products	Synthetic Libraries	Biological Implications
Structural Complexity	Higher molecular complexity, diverse stereocenters [7]	Generally lower complexity, fewer chiral centers	NPs better suited for targeting complex protein interfaces
Scaffold Diversity	Extensive structurally unique scaffolds [87]	Limited to known synthetic frameworks	NPs access novel biological mechanisms
Molecular Rigidity	Balanced sp2:sp3 hybridization ratios [85]	Often more planar structures	Improved binding specificity and ADMET profiles for NPs
Chemical Space Coverage	Explore underexplored BioReCS regions [6]	Cover heavily explored synthetic subspaces	NPs provide access to novel bioactive chemotypes
Drug-likeness	Often beyond Rule of 5 (bRo5) [6] [7]	Typically compliant with Rule of 5 [85]	NPs suitable for challenging targets but may have poor oral bioavailability

Target Engagement and Bioactivity Profiles

The structural differences between natural products and synthetic compounds translate to distinct bioactivity profiles. Natural products exhibit **privileged scaffolds** that have been evolutionarily optimized to interact with biological macromolecules [7]. Approximately **30% of FDA-approved drugs** from 1981 to 2019 originated from natural products or their derivatives, particularly in anti-infectives (e.g., penicillin) and anti-tumors (e.g., paclitaxel) [7]. This success rate highlights their exceptional biological relevance. Synthetic compounds, while often designed with favorable **ADMET (absorption, distribution, metabolism, excretion, and toxicity)** properties in mind, frequently lack the structural sophistication needed to engage complex biological targets effectively [85]. The presence of **stereochemical complexity** in natural products enables them to bind selectively to chiral biological macromolecules, a feature often missing in synthetic libraries [7].

Quantitative Analysis of Chemical Space Coverage

Library Size and Diversity Metrics

The quantitative assessment of chemical space coverage reveals significant differences between natural product and synthetic libraries. Fragment-based screening represents a particularly efficient approach to exploring chemical space, as the estimated number of fragment-like compounds is approximately **10^11** molecules, compared to **10^23 to 10^60** for drug-like molecules [86]. This makes comprehensive sampling more feasible with fragment libraries. Advances in synthetic chemistry have enabled the creation of ultralarge compound collections, with **>30 billion compounds** now available in make-on-demand catalogs [86]. However, even these massive libraries represent only a tiny fraction of possible chemical space.

Table 2: Library Size and Diversity Comparisons

Library Type	Representative Size	Structural Diversity	Hit Rate for Novel Targets	Notable Examples
Natural Product Libraries	Hundreds to thousands of physical compounds [86]	High scaffold diversity [87]	Historically high for antibiotics, anticancer [7]	Penicillin, paclitaxel, vancomycin [7]
Fragment Libraries	14 million compounds screened virtually [86]	Moderate (limited by size)	Effective for difficult targets [86]	OGG1 inhibitors [86]
Lead-like Libraries	235 million compounds [86]	Moderate to high	Varies by target class	DOCK3.7 screened libraries [86]
Computational Libraries	GDB-17 (160 billion molecules) [85]	Theoretical maximum	Limited by synthetic accessibility	CHIPMUNK library [85]

Performance in Screening Campaigns

Comparative studies demonstrate that natural products and synthetic libraries exhibit different performance characteristics in screening campaigns. Virtual fragment screening of **14 million fragments** against 8-oxoguanine DNA glycosylase (OGG1) identified four binding compounds, with X-ray crystallography confirming the predicted binding modes [86]. This success rate of approximately **14%** (4 hits from 29 tested compounds) demonstrates the effectiveness of combining vast chemical libraries with structure-based virtual screening. In contrast, high-throughput screening (HTS) of traditional synthetic libraries often yields **lower hit rates** and weakly active compounds that require extensive optimization [86]. Natural product libraries typically show **higher hit rates** for novel targets but face challenges in follow-up optimization due to their complex structures [7].

Methodologies for Integrating NPs and Synthetic Libraries

Experimental Screening Protocols

Virtual Fragment Screening Protocol

The virtual screening methodology applied to OGG1 inhibitors provides a robust protocol for integrating natural products and synthetic compounds in screening campaigns [86]:

- **Target Preparation**: Obtain crystal structure of target protein (e.g., OGG1) in complex with a small molecule inhibitor. Prepare the structure by adding hydrogen atoms, optimizing side-chain orientations, and defining binding site parameters.
- **Library Preparation**: Curate fragment-like (MW < 250 Da) and lead-like (250 ≤ MW < 350 Da) libraries. For the OGG1 study, 14 million fragment-like and 235 million lead-like compounds were used [86].
- **Molecular Docking**: Perform docking calculations using programs such as DOCK3.7. Evaluate each molecule in multiple conformations and thousands of orientations in the active site. For the OGG1 screen, **13 trillion fragment complexes** and **149 trillion lead-like complexes** were evaluated [86].
- **Hit Selection**: Cluster top-ranked compounds (e.g., 10,000 for fragments, 100,000 for lead-like) by topological similarity. Select diverse candidates through visual inspection of predicted complexes, focusing on complementarity to the binding site.
- **Experimental Validation**: Synthesize selected compounds (4-5 weeks for make-on-demand compounds). Test binding using thermal shift assays (e.g., differential scanning fluorimetry at 495 μM for fragments, 99 μM for lead-like compounds). Confirm binding modes through X-ray crystallography.

Nano-Drug Delivery System Evaluation Protocol

For natural products with bioavailability challenges, nano-drug delivery systems (NDDS) provide a mechanism to enhance therapeutic potential [87]:

- **Nanocarrier Selection**: Choose appropriate nanocarrier based on natural product properties and target tissue. Common options include liposomes, polymeric nanoparticles, micelles, and inorganic nanoparticles [87].
- **Formulation Optimization**: Incorporate natural products into nanocarriers using methods such as thin film hydration, high-pressure homogenization, nanoprecipitation, or self-assembly [87].
- **In Vitro Characterization**: Evaluate particle size, zeta potential, drug loading capacity, and release kinetics. Assess cellular uptake and cytotoxicity in relevant cell lines.
- **In Vivo Evaluation**: Administer formulations to disease models (e.g., cancer xenografts). For example, in hepatocellular carcinoma studies, galactosylated-chitosan-triptolide-nanoparticles were administered at 6.76-16.90 mg/kg (triptolide equivalent) [87]. Assess biodistribution,

efficacy, and toxicity parameters.

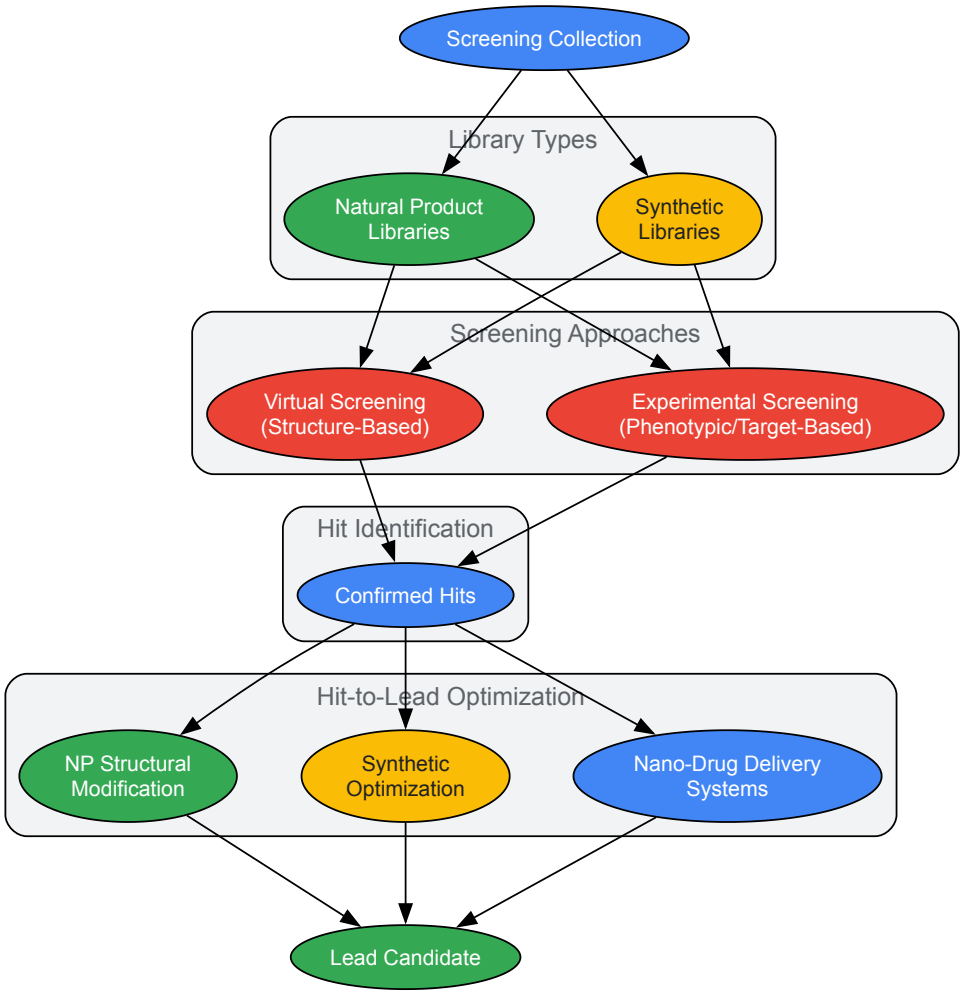


Figure 1: Integrated Screening Workflow Combining Natural Products and Synthetic Libraries. This workflow leverages the complementary strengths of both library types through parallel screening approaches and optimization strategies.

Computational Approaches for Chemical Space Analysis

Molecular Generation Models for NP Optimization

Artificial intelligence and generative models provide powerful tools for bridging the gap between natural products and synthetic libraries [7]:

- **Target Interaction-Driven Models**: Utilize protein-ligand interaction data to guide structural modifications of natural products with known targets.
  - **Fragment Splicing Methods**: Models such as DeepFrag, FREED, and DEVELOP remove ligand fragments from protein-ligand complexes and predict optimal replacements [7].
  - **Molecular Growth Methods**: 3D-MolGNNRL and DiffDec generate molecules directly in 3D target pockets through atom-by-atom or substructure addition [7].
- **Molecular Activity Data-Driven Models**: Apply machine learning to optimize natural products when target information is limited.
  - **Scaffold Hopping Models**: Modify core scaffolds while preserving bioactivity using models such as LinkINVENT and ScaffoldGPT [7].
  - **Property Optimization Models**: Use reinforcement learning to optimize ADMET properties while maintaining desired activity [7].

Quantitative Nanostructure-Activity Relationship (QNAR) Modeling

For nano-drug delivery systems incorporating natural products, QNAR modeling enables prediction of biological activities based on nanostructural characteristics [88]:

- **Virtual Nanoparticle Construction**: Create virtual gold nanoparticles (vGNPs) by inputting structural parameters (particle size, surface ligand structure, ligand density).

- **Descriptor Calculation**: Calculate 86 nanodescriptors representing surface area, potential energy, and other physicochemical properties from optimized vGNP structures.
- **Model Development**: Build predictive QNAR models that quantitatively relate nanostructures to complex bioactivities (e.g., cellular uptake).
- **Nanoparticle Design**: Screen external vGNP libraries using QNAR models to prioritize nanoparticles with desired bioactivities before synthesis.

The Scientist's Toolkit: Essential Research Reagents and Materials

Table 3: Essential Research Reagents and Materials for Chemical Space Exploration

Category	Specific Items	Function/Application	Key Considerations
Natural Product Libraries	Pre-fractionated natural product extracts; Pure NP compounds [86]	Provide diverse scaffolds for screening; Source of novel bioactivities	Standardize storage conditions; Validate authenticity and purity
Fragment Libraries	14 million fragment-like compounds [86]	Efficient exploration of chemical space; Weak binders for difficult targets	MW < 250 Da; comply with "rule of 3" for fragments
Lead-like Libraries	235 million lead-like compounds [86]	Intermediate optimization; Balance of properties	250 ≤ MW < 350 Da; favorable drug-like properties
Nanocarrier Systems	Liposomes; Polymeric nanoparticles; Micelles; Inorganic nanoparticles [87]	Improve NP bioavailability; Enable targeted delivery	Biocompatibility; Drug loading capacity; Release kinetics
Characterization Tools	Thermal shift assays; X-ray crystallography; DOCK3.7 software [86]	Validate binding; Determine structures; Virtual screening	High sensitivity for weak binders; Atomic resolution
Generative AI Tools	DeepFrag; FREED; 3D-MolGNNRL [7]	NP structural modification; De novo design	Integration with target structural data; Synthetic accessibility

Natural products and synthetic libraries occupy complementary regions of chemical space, with each offering distinct advantages for drug discovery. **Natural products** provide unparalleled structural diversity, evolutionary optimization for biological interactions, and access to underexplored regions of biologically relevant chemical space. **Synthetic libraries** offer synthetic tractability, favorable drug-like properties, and the ability to generate massive numbers of compounds for screening. The integration of both approaches through **virtual screening**, **fragment-based methods**, and **AI-driven design** represents the most effective strategy for comprehensive chemical space coverage. Furthermore, **nano-drug delivery systems** address the inherent bioavailability challenges of natural products, enhancing their therapeutic potential. As drug discovery faces increasingly challenging targets, the synergistic combination of natural products and synthetic libraries—leveraging the strengths of each approach—will be essential for identifying novel therapeutic agents.

The exploration of chemical space in natural products research has consistently revealed macrocyclic compounds as structurally unique and biologically privileged scaffolds. These molecules, typically defined as cyclic structures containing 12 or more atoms, occupy a crucial region of chemical diversity that bridges the gap between traditional small molecules and larger biologics [89]. Their structurally constrained three-dimensional configurations facilitate high-affinity and selective interactions with challenging biological targets, notably protein-protein interfaces (PPIs) and other binding sites traditionally considered "undruggable" by conventional small molecules [89] [90]. Natural products have long served as an invaluable source of bioactive macrocycles, offering an array of complex architectures that have evolved over millions of years to interact with biological systems [89]. This review examines key case studies where macrocyclic natural products (NPs) and their analogues have been successfully deployed against challenging therapeutic targets, underpinning their status as privileged structures in modern drug discovery.

Macrocyclic NPs as Therapeutics: Overcoming Traditional Drug Discovery Limitations

Macrocycles exhibit unique properties that enable them to address fundamental challenges in drug development. Their larger ring systems create a higher degree of flexibility compared to the relatively rigid five-to-seven-membered rings common in traditional small molecules, allowing them to adopt multiple conformations [90]. This conformational adaptability, combined with their larger surface area, enables extensive interactions with biological targets, resulting in higher binding affinity and selectivity [90]. A significant advantage of macrocycles is their capacity to target shallow, solvent-exposed protein grooves and disrupt PPIs—a task at which both small molecules and antibodies often fail [89] [90]. Furthermore, despite frequently violating the Rule of Five (R05) guidelines, a substantial proportion (close to 40%) of macrocyclic drugs demonstrate oral bioavailability, challenging historical presumptions about their cell permeability [90].

Table 1: Approved Macrocyclic Drugs and Their Therapeutic Applications

Macrocyclic Drug	Therapeutic Area	Biological Target	Key Feature
Grazoprevir [90]	Hepatitis C	Viral Protease NS3/4A	Result of macrocyclisation of acyclic precursors
Vaniprevir [90]	Hepatitis C	Viral Protease NS3/4A	Result of macrocyclisation of acyclic precursors
MK-0616 [90]	Atherosclerotic Cardiovascular Disease	PCSK9 inhibitor	Utilizes ring bridging strategy to enhance permeability
Abraxane (nab-paclitaxel) [91]	Cancer	Tubulin	Albumin-bound nanoparticle formulation of natural product paclitaxel

Case Studies: Targeting The "Undruggable"

Case Study 1: Targeting Protein-Protein Interactions in Oncology

Background: Protein-protein interactions are fundamental to many cellular processes and are increasingly recognized as valuable yet challenging therapeutic targets due to their large, flat, and often featureless interfaces. The programmed cell death protein 1 and its ligand (PD-1/PD-L1) pathway is a critical immune checkpoint, and its disruption represents a powerful oncological strategy [90]. While antibodies have been successful in targeting this axis, macrocyclic peptides have emerged as effective inhibitors, offering potential advantages in tissue penetration and oral administration.

Macrocyclic Solution: Platforms such as PeptiDream's drug discovery system have facilitated the synthesis and screening of macrocyclic peptide libraries, leading to the identification of a potent PD-L1 macrocycle [90]. The constrained, yet flexible, structure of the macrocycle enables it to effectively bind to the extensive PD-L1 surface, disrupting its interaction with PD-1 and restoring anti-tumor immune activity.

Experimental Protocol: Macrocyclic Peptide Library Screening for PPI Inhibition

- **Library Synthesis:** Generate a diverse library of macrocyclic peptides using a build/couple/pair (B/C/P) combinatorial approach or similar methodology to ensure extensive scaffold diversity [89].
- **In Vitro Screening:** Screen the library against the purified target protein (e.g., PD-L1) using a binding assay, such as surface plasmon resonance (SPR) or an enzyme-linked immunosorbent assay (ELISA)-based binding interference assay.
- **Functional Cellular Assay:** Validate hits in a cell-based reporter assay that measures the inhibition of the specific PPI (e.g., PD-1/PD-L1), confirming functional disruption in a more physiologically relevant context.
- **Hit Optimization:** For confirmed hit compounds, perform structure-activity relationship (SAR) studies. This often involves synthesizing analogue libraries with systematic variations to optimize binding affinity, selectivity, and metabolic stability. Key strategies include N-methylation and incorporation of non-natural amino acids to improve permeability and stability [90].
- **Biophysical Validation:** Use Nuclear Magnetic Resonance (NMR) spectroscopy to obtain experimental restraints on the macrocycle's solution conformation. These restraints are then used in computational models to generate a tractable ensemble of low-energy conformations that represent the bioactive structure when bound to the target [90].

Case Study 2: Targeting Shallow Viral Protease Grooves

Background: The Hepatitis C virus (HCV) protease NS3/4A features a shallow, solvent-exposed substrate-binding groove, making it a difficult target for conventional small molecules [89]. For years, therapeutic options for HCV were limited, creating a significant unmet medical need.

Macrocyclic Solution: Macrocyclization of acyclic precursor peptides proved to be a pivotal strategy in developing effective HCV protease inhibitors like grazoprevir and vaniprevir [90]. The macrocyclic structure pre-organizes the molecule, reducing the entropic penalty upon binding and allowing the inhibitor to make extensive contacts across the flat protein surface. This case exemplifies how macrocycles dominate the inhibitor space for certain challenging viral targets.

Experimental Protocol: Structure-Based Design of Macrocyclic Protease Inhibitors

- **Structural Analysis:** Obtain a high-resolution crystal structure of the target protease (e.g., NS3/4A) in complex with a natural substrate or a weak inhibitor to identify key binding residues and the topology of the active site.
- **Peptide Precursor Design:** Design linear peptide sequences based on the native cleavage site, incorporating non-natural amino acids to enhance binding and stability.
- **Macrocyclization Strategy:** Employ ring-closing metathesis (RCM) or late-stage macrocyclization via C–H activation to form the macrocyclic ring [89]. The cyclization bridge is strategically designed to lock the peptide into a bioactive conformation without disrupting critical interactions.
- **Conformational Analysis:** Utilize computational chemistry tools to model the macrocycle's conformational ensemble and predict the strain energy of the bound state. Advanced methods can fit an ensemble of conformers to electron density from X-ray crystallography, providing a

more accurate model of the bound state and its strain [90].

- **Potency and Selectivity Profiling:** Test inhibitors in a biochemical assay using recombinant NS3/4A protease and a fluorescent substrate. Counter-screen against related human proteases (e.g., elastase) to ensure selectivity and minimize off-target effects.

Case Study 3: Intracellular Target Engagement with Oral Bioavailability

Background: Proprotein convertase subtilisin/kexin type 9 (PCSK9) is a validated target for lowering LDL cholesterol, but developing small, orally bioavailable molecules to inhibit its PPI with the LDL receptor has been exceptionally challenging.

Macrocyclic Solution: MK-0616, an oral PCSK9 inhibitor, demonstrates how modern macrocycle design can overcome multiple drug development hurdles. This macrocycle was engineered using a ring bridging strategy to reduce conformational flexibility and shield polar groups, thereby enhancing metabolic stability and cell permeability to achieve oral bioavailability [90]. Its progression to Phase III trials underscores the therapeutic potential of well-designed macrocycles for intracellular targets.

Experimental Protocol: Engineering Orally Bioavailable Macrocycles

- **Permeability-First Design:** Start with a macrocyclic scaffold known for membrane permeability, often informed by natural products. Intentionally design in features like internal hydrogen bonds and N-methylation to shield polarity [90].
- **Strain Engineering:** Use computational models to design macrocyclic variants with lower ring strain in the protein-bound conformation. Lower strain often correlates with higher binding affinity [90].
- **Ring Bridging/Stapling:** Introduce covalent bridges or "staples" across the macrocyclic ring to rigidify its three-dimensional structure. This pre-organization improves affinity and can dramatically enhance oral exposure by reducing the polarity and molecular flexibility [90].
- **In Vitro ADME Assessment:** Perform parallel artificial membrane permeability assay (PAMPA) or Caco-2 assays to predict intestinal absorption. Assess metabolic stability in liver microsome or hepatocyte assays.
- **In Vivo PK/PD Studies:** Administer the lead macrocyclic compound orally to animal models to confirm bioavailability and measure the pharmacodynamic response (e.g., reduction in plasma PCSK9 or LDL cholesterol levels).

Visualizing Macrocyclic Drug Discovery Workflows

The following diagrams illustrate the logical relationships and key experimental pathways in macrocyclic drug discovery.

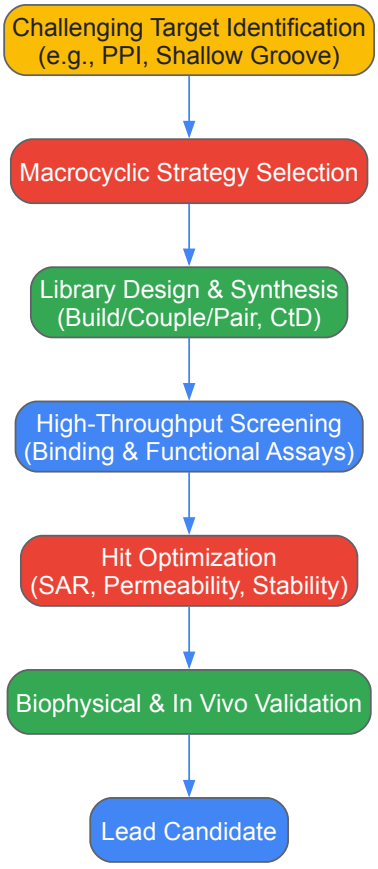


Diagram 1: Macrocyclic drug discovery workflow from target to candidate.

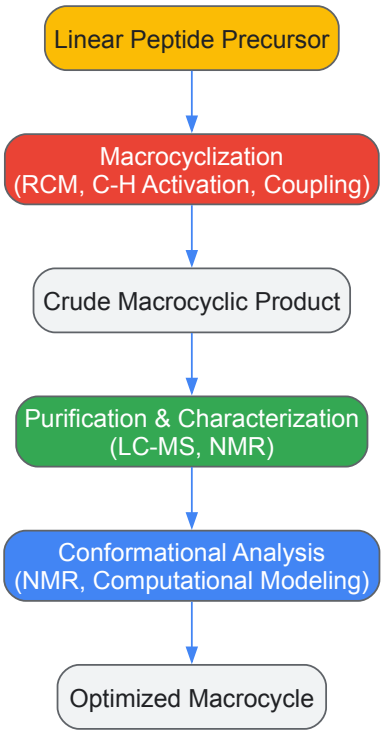


Diagram 2: Key steps in macrocycle synthesis and analysis.

The Scientist's Toolkit: Essential Reagents and Technologies

Successful discovery and development of macrocyclic therapeutics rely on a suite of specialized reagents, technologies, and computational tools.

Table 2: Essential Research Reagent Solutions for Macrocycle Development

Reagent / Technology	Function	Application in Macrocycle Research
DNA-Encoded Libraries (DELs) [89]	Ultra-high-throughput screening platform	Screening vast chemical space of macrocyclic structures against purified protein targets.
Non-Natural Amino Acids [90]	Backbone and side-chain modification	Enhancing binding affinity, metabolic stability, and permeability by replacing natural residues.
N-Methylated Amino Acids [90]	Peptide backbone modification	Improving cell permeability by reducing hydrogen bonding capacity and molecular flexibility.
Ring-Closing Metathesis (RCM) Catalyst [89]	Olefin metathesis for ring formation	Efficient synthesis of large macrocyclic rings from linear diene precursors.
Stabilized Protein Target	High-quality binding partner	For screening and biophysical assays (SPR, X-ray crystallography).
Biophysical Assay Kits (SPR, NMR) [90]	Label-free binding affinity and kinetics measurement	Characterizing macrocycle-target interactions and determining solution structures.

Macrocyclic natural products and their synthetic analogues represent a powerful class of therapeutic agents that uniquely address the challenges of targeting intricate biological interfaces. As evidenced by the case studies against PPIs, viral proteases, and intracellular targets like PCSK9, their constrained three-dimensionality, conformational adaptability, and capacity for engineering make them privileged structures for drug discovery. The continued synergy of innovative synthetic methodologies—such as modular biomimetic assembly and DNA-encoded libraries—with advanced computational modeling and biophysical validation techniques is decisively expanding the accessible macrocyclic chemical space. This cross-disciplinary approach, leveraging the structural diversity inherent to natural products, is poised to deliver a new generation of medicines for previously intractable diseases.

Natural products (NPs) are chemical compounds produced by living organisms that have been evolutionarily selected and validated for optimal interactions with biological macromolecules [46]. They represent a rich resource of bioactive compounds with immense chemical diversity, encoding areas of chemical space explored by nature through billions of years of evolution [92] [24]. This embedded recognition of protein binding sites, captured during their biosynthesis, makes NPs particularly valuable starting points for drug discovery [92]. However, the structural complexity of many NPs presents significant challenges for their direct development as drug candidates. Fragment-based drug discovery (FBDD) has emerged as a

powerful approach to address these challenges by deconstructing complex natural products into simpler fragment-sized molecules with low molecular weight, typically between 100-300 g mol^{-1} [92]. This reductionist approach allows researchers to sample a much greater proportion of chemical space compared to traditional high-throughput screening (HTS) of larger molecules, while providing simpler starting points for subsequent chemical optimization [92].

The chemical space covered by natural products is notably distinct from that explored by traditional combinatorial chemistry libraries. NPs possess broader diversity in chemical space, enriched with biosynthetic intermediates and endogenous metabolites that have undergone long selection processes [92] [46]. This diversity is particularly evident in their three-dimensional structural properties and increased sp^3 carbon fraction (Fsp^3), which may provide better starting points for drug discovery compared to the predominantly flat architectures found in many synthetic libraries [92]. The exploration of this privileged chemical space through fragment-based approaches represents a strategic mind-shift in medicinal chemistry, acknowledging that escaping "flat-land" is crucial for increasing the chances of clinical success in drug development, particularly for challenging targets such as those involved in parasitic diseases [92].

Methodological Frameworks for Natural Product Fragment Library Design

Strategic Approaches to Library Construction

Several sophisticated methodologies have been developed to create fragment libraries derived from or inspired by natural products, each offering distinct advantages for exploring novel chemical space:

Table 1: Approaches to Natural Product Fragment Library Design

Approach	Methodology	Key Advantages	Example Applications
Chemical Disassembly of Larger NPs	In silico guided cleavage of larger natural products into fragment-sized components [92]	Retains natural product-like 3D shape and complexity; Generates novel fragments from known bioactive compounds	Generation of 9,000 fragment-like compounds from 17,000 starting natural products; Fragments derived from FK506, sanglifehrin A, and cytochalasin E [92]
Chemical Modification of Smaller NPs	Systematic modification of smaller natural products to remove reactive sites and introduce novel chiral sp^3 centers [92]	Maintains natural product character while improving drug-like properties; Enhances synthetic tractability	Creation of fragments with improved stability and novel chiral centers while preserving natural product-like features [92]
Pseudo Natural Product Design	Synthetic combination of biosynthetically unrelated natural product fragments to create novel compound classes [92]	Generates truly novel chemotypes beyond natural product space; Enables target-agnostic discovery	"Indotropanes" (combining indole and tropane scaffolds); "Chromopynones" (combining chromane and tetrahydropyrimidinone fragments) [92]
Computational Generation	Recurrent neural networks trained on known natural product structures to generate novel natural product-like compounds [24]	Massive expansion of accessible chemical space (67 million compounds); Explores novel regions of natural product space	Generation of 67,064,204 valid natural product-like SMILES with NP Score distribution similar to known natural products [24]

Quantitative Foundations of Natural Product Fragment Libraries

The foundation for establishing fragment screening libraries from natural products is supported by extensive analysis of natural product databases. According to the Dictionary of Natural Products, there are 7,365 non-flat fragment-sized natural products rich in sp^3 centers, identified through calculation of carbon bond saturation ($Fsp3^* > 0.45$) [92]. These fragment-sized natural products cover approximately 66% of the pharmacological features found in their larger counterparts, demonstrating their significant coverage of bioactive chemical space despite their reduced size and complexity [92]. Recent database expansions have further enhanced these resources, with NPASS now containing 94,413 natural products, 32,561 organism sources (including 444 co-culture combinations and 427 engineered species), and 958,866 activity records against 7,753 targets [93]. This represents a 205.3% increase in total natural products and a 192.8% increase in organism-NP pairs compared to previous versions [93].

Table 2: Quantitative Analysis of Natural Product Databases for Fragment-Based Discovery

Database Metric	NPASS-2018	NPASS-2023	Change
Natural Products with Activity Values	30,926	43,285	+40.0%
Total Natural Products	30,926	94,413	+205.3%
Natural Organisms	25,041	31,690	+26.6%

Database Metric	NPASS-2018	NPASS-2023	Change
Co-culture Organisms	Not available	444 combinations	New
Engineered Organisms	Not available	427 species	New
Activity Records	446,552	958,866	+114.7%
Targets	5,863	7,753	+32.2%
Composition/Concentration Records	Not available	95,004	New

Experimental Workflows and Technical Implementation

Fragment Screening Methodologies

The screening stage of FBDD requires specialized biophysical techniques significantly more sensitive than those utilized in HTS, as fragments typically exhibit weak binding affinities in the 0.1-10 mM range [92]. A bibliographic analysis of 3,642 publications on FBDD between 1953 and 2016 revealed the following distribution of techniques, ordered from highest to lowest usage: X-ray crystallography, surface plasmon resonance, nuclear magnetic resonance, thermal shift assay, isothermal titration calorimetry, and mass spectrometry [92]. Each technique offers unique advantages for detecting the weak interactions characteristic of fragment binding.

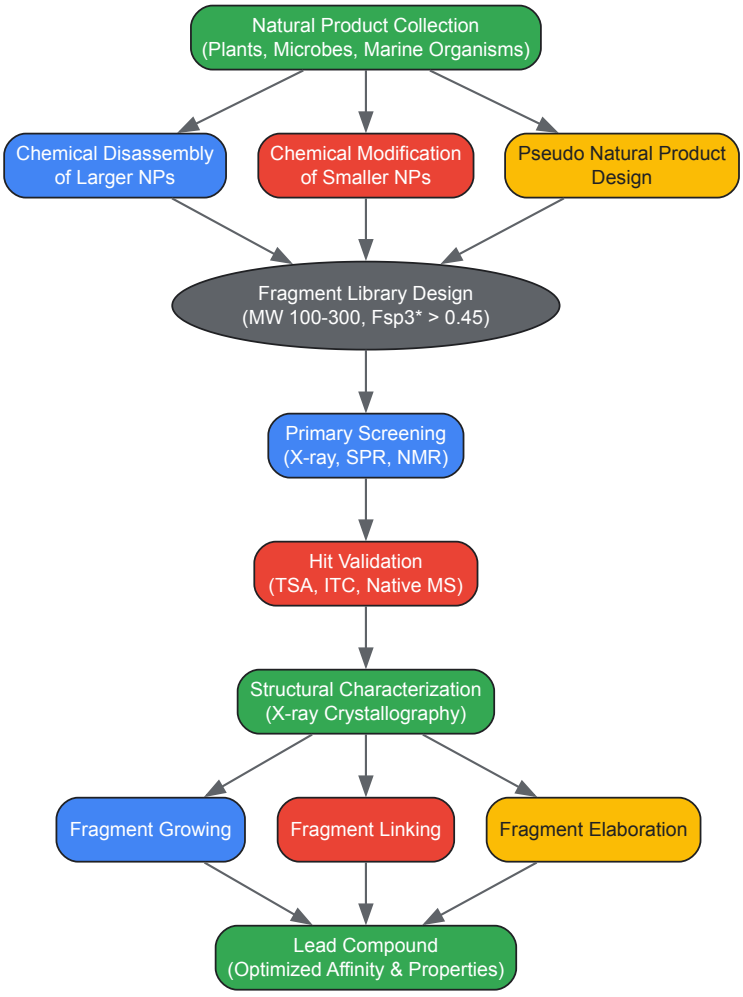


Diagram 1: NP Fragment-Based Discovery Workflow

Successful implementation of fragment-based approaches using natural products requires access to specialized databases, software tools, and experimental resources:

Table 3: Essential Research Resources for NP Fragment-Based Discovery

Resource Category	Specific Tools/Resources	Function and Application
Natural Product Databases	NPASS [93], COCONUT [24], Dictionary of Natural Products [92]	Source structures, activity data, and species information for natural products; NPASS contains 94,413 NPs with 958,866 activity records [93]
Cheminformatics Tools	RDKit [24], NP Score [24], NPClassifier [24]	Calculate molecular descriptors, natural product-likeness scores, and classify NPs by biosynthetic pathway; NP Score uses Bayesian measure of similarity to NP structural space [24]
Structure Generation	SMILES-based LSTM [24], Chemical Checker [93]	Generate novel natural product-like compounds; 67 million NP-like molecules generated via molecular language processing [24]
Biophysical Screening	X-ray Crystallography [92], Surface Plasmon Resonance [92], Native Mass Spectrometry [92]	Detect weak fragment-protein interactions (0.1-10 mM range); X-ray crystallography is the most frequently used technique [92]
Computational Prediction	SPiDER software [92], ADMETlab 2.0 [93]	Predict targets of fragment-like natural products and calculate drug-likeness/ADMET properties; SPiDER successfully predicted targets for sparteine [92]

Case Studies and Applications in Drug Discovery

Successful Applications Against Challenging Targets

Fragment-based approaches using natural products have demonstrated particular utility for addressing classically challenging targets in drug discovery, including protein-protein interactions, nucleic acid complexes, and antibacterial targets [46]. The following case studies illustrate the successful implementation of these strategies:

Case Study 1: p38α MAP Kinase Inhibitors from Renieramycin-Derived Fragments The deconstruction of the natural product renieramycin using a scaffold tree approach generated fragments that were identified as weak inhibitors of p38α MAP kinase with an IC_{50} of 1.3 mM [92]. Subsequent synthetic elaboration and co-crystal structures with the protein revealed an allosteric pocket of p38α MAP kinase, leading to the development of a novel class of type III inhibitors [92]. This example demonstrates how natural product fragments can provide access to novel binding modes and mechanisms of inhibition.

Case Study 2: Myokinasib from Indotropane Pseudo Natural Products The combination of biosynthetically unrelated indole and tropane natural product fragments generated "indotropanes," which were screened using cell-based assays [92]. This approach identified myokinasib as the first selective, isoform-specific inhibitor of myosin light chain kinase 1 (MLCK1), a target that had proven difficult to address with conventional compound libraries [92]. The indotropanes were found to occupy an area of natural product space not accessible via known biosynthetic pathways, highlighting the power of pseudo natural product approaches to explore truly novel chemical space [92].

Case Study 3: Glucose Uptake Inhibitors from Chromopynones The combination of chromane and tetrahydropyrimidinone natural product fragments led to the development of "chromopynones," which represent a novel, structurally unprecedented glucose uptake inhibitor chemotype [92]. These compounds selectively target glucose transporters GLUT-1 and GLUT-3, enabling modulation of tumor metabolism through a previously unexplored mechanism [92]. This case study demonstrates how natural product fragment combination can yield new therapeutic strategies for cancer treatment.

Computational Prediction and Validation

Target prediction of fragment-like natural products with innovative scaffolds provides valuable starting points for chemical biology and medicinal chemistry programs. The SPiDER software successfully predicted targets for 23,340 (36%) of low molecular weight natural products from the Dictionary of Natural Products, compared to 31,556 (22%) of larger natural products [92]. In a prospective experiment using sparteine, SPiDER correctly predicted known targets (muscarinic and nicotinic receptors) among the top three predictions, while also identifying the kappa opioid receptor as a novel target [92]. Subsequent binding and functional assays confirmed this prediction and revealed sparteine as a ligand-efficient fragment (ligand efficiency = 0.30) suitable for further development [92].

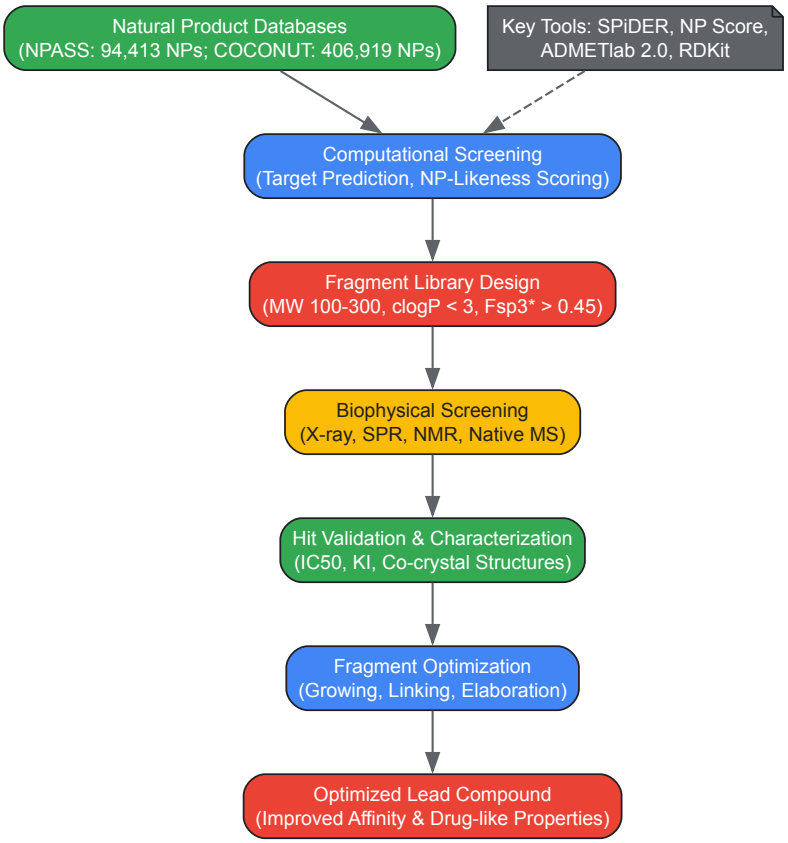


Diagram 2: Computational Screening & Validation Pipeline

Future Directions and Concluding Perspectives

The field of fragment-based approaches using natural products continues to evolve with several emerging trends shaping its future trajectory. The integration of artificial intelligence and deep learning methodologies is dramatically expanding accessible chemical space, as demonstrated by the generation of 67 million natural product-like compounds through molecular language processing [24]. This represents a 165-fold expansion over the approximately 400,000 known natural products and significantly enhances the probability of discovering novel bioactive scaffolds [24]. Furthermore, the exploration of underexplored biological sources, including co-cultured microbes (444 combinations in NPASS) and engineered microorganisms (427 species in NPASS), provides access to previously inaccessible chemical diversity [93].

The ongoing development of more sophisticated library design strategies that explicitly incorporate three-dimensional structural complexity and sp^3 richness addresses historical limitations in exploring chemical space [92]. As these methodologies mature, combined with advances in structural biology and biophysical screening techniques, fragment-based approaches using natural products are poised to make increasingly significant contributions to drug discovery, particularly for challenging target classes that have proven intractable to conventional screening approaches. The strategic deconstruction of natural products provides a powerful framework for navigating the complex landscape of chemical space while maintaining the favorable biological properties inherent to evolutionarily optimized natural scaffolds.

Natural products (NPs) have long served as a cornerstone in drug discovery, with over 50% of approved small-molecule drugs originating directly or being inspired by natural scaffolds [94]. These molecules, refined through millions of years of evolution, occupy chemical spaces far beyond the reach of synthetic libraries, exhibiting superior molecular complexity, higher proportions of sp^3-hybridized carbon atoms, increased oxygenation, and lower lipophilicity compared to synthetic compounds [19]. Despite their proven therapeutic value, only a fraction of Earth's biodiversity has been systematically explored for its pharmacological potential [94]. This whitepaper examines future directions for expanding into untapped biological sources and chemical domains within the broader context of chemical space and structural diversity in natural products research, providing technical guidance for researchers and drug development professionals seeking to overcome current limitations in the field.

Current Landscape and Taxonomic Gaps

Table 1: Distribution of Natural Products Across Biological Sources and Their Characteristics

Biological Source	Representation in NPs	Promising Reservoirs	Notable Characteristics	Research Challenges
Plants	67% of NPBS Atlas entries [94]	Medicinal plants with Traditional Chinese Medicine applications [94]	Strong association with traditional medicine; Extensive ethnopharmacological knowledge base [94] [19]	Sustainable sourcing; Over-harvesting concerns; Identification of active constituents [19]
Marine Organisms	77% of animal-derived NPs [94]	Sponges (Porifera); Marine fungi; Associated microbes [94] [95]	Evolutionary innovation in chemical defense; Unique structural features [94] [95]	Sample collection; Cultivation difficulties; Structural complexity [95]
Fungi	Significant overlap with plant metabolites (3,520 shared NPs) [94]	Endophytic fungi; Marine fungi [94]	Superior Quantitative Estimate of Drug-likeness (QED) profiles (50% > QED 0.5) [94]	Laboratory cultivation; Genetic accessibility [94]
Bacteria	17% annotated with bioactivities [94]	Underexplored genera; Marine bacteria [94] [19]	Exceptional scaffold diversity; Rare macrocyclic and hybrid architectures [94]	Silent biosynthetic gene cluster activation [19]

The systematic analysis of natural product origins within resources like NPBS Atlas reveals distinct taxonomic patterns with implications for future exploration [94]. Plants currently dominate as NP sources (67% of entries), reflecting their historical prominence in ethnopharmacology and combinatorial diversity of secondary metabolism [94]. Cross-taxon analyses have identified significant metabolite overlap between plants and fungi (3,520 shared natural products), suggesting convergent biosynthetic strategies or ecological interactions like endophytic symbiosis as promising research directions [94].

Marine ecosystems represent particularly promising frontiers, contributing disproportionately to animal-derived natural products (77%) with sponges (Porifera) exhibiting exceptional chemical innovation [94]. Marine environments continue to yield structurally unique compounds from both macroorganisms and their associated microorganisms, though considerable challenges remain in sustainable sourcing and cultivation [95].

Sustainable Exploration Strategies

Future exploration must prioritize sustainable practices to mitigate environmental impact and ensure long-term resource availability:

- **Advanced cultivation techniques**: Implementation of optimized cultivation, agroforestry, and microbial fermentation offers scalable alternatives to wild harvesting [19].
- **Waste valorization**: Utilization of agricultural and food processing byproducts represents an eco-friendly sourcing strategy that aligns with circular economy principles [19].
- **Genome mining**: Computational identification of biosynthetic gene clusters in sequenced genomes enables targeted discovery without extensive biological sampling [19].
- **Microbial fermentation**: Engineering microbial hosts for NP production provides sustainable, scalable manufacturing while preserving sensitive ecosystems [19].

Expanding Chemical Domains

Chemical Engineering of Natural Product Scaffolds

Table 2: Chemical Diversification Strategies for Natural Products

Strategy	Key Methodologies	Structural Outcomes	Applications
Diversity-Enhanced Extracts	Acid/base treatment of crude extracts; Scaffold remodeling [3]	New molecular skeletons; Ring distortions; Unnatural arrangements [3]	Rapid generation of structural diversity; Anti-glioblastoma agents [3]
Pseudo-Natural Products (PNPs)	Biology-oriented synthesis (BIOS); Fragment-based design [96]	Non-biogenic fusions of NP-derived fragments; Novel scaffold arrangements [96]	Exploration of biological relevance; Addressing challenging targets [96]
Semi-synthetic Modifications	Targeted functional group manipulation; Biocatalysis [3] [19]	Analogues with improved properties; Enhanced selectivity [3]	Optimization of pharmacokinetics; Reduction of toxicity [3]
Biosynthetic Engineering	Pathway engineering; Combinatorial biosynthesis [19]	New-to-nature compounds; Non-natural analogues [19]	Access to cryptic metabolites; Structural diversification [19]

Chemical diversification represents a powerful approach to expand the structural diversity of natural products beyond what is readily accessible from nature:

- **Diversity-enhanced extracts**: This innovative approach involves applying chemical reactions that remodel molecular scaffolds directly on extracts of natural resources [3]. For example, treatment of Ambrosia tenuifolia dichloromethane extract with p-toluenesulfonic acid in toluene under reflux conditions successfully generated compounds with unprecedented skeletons through ring distortions including expansions, contractions, fusions, cleavages, and rearrangements [3].
- **Pseudo-natural products (PNPs)**: PNPs combine natural product fragments in novel arrangements not accessible through biosynthetic pathways, creating non-biogenic fusions of NP-derived fragments [96]. This strategy effectively merges the biological relevance of natural product scaffolds with synthetic innovation to explore new chemical space [96].
- **Biosynthetic pathway engineering**: Advances in synthetic biology enable the manipulation of biosynthetic pathways to produce novel analogues, with tools like CRISPR-Cas systems facilitating precise engineering of NP biosynthetic gene clusters [19].

Experimental Protocol: Chemical Engineering of Natural Extracts

The following detailed methodology for chemical engineering of natural extracts is adapted from published work on Ambrosia tenuifolia [3]:

Step 1: Extract Preparation

- Obtain global ethanolic extract of source material (e.g., Ambrosia tenuifolia) using standard maceration or Soxhlet extraction.
- Partition the global extract with solvents of increasing polarity (hexane, CH_2Cl_2, EtOAc) to obtain subextracts enriched in target metabolite classes.
- Characterize initial composition of active subextracts using analytical techniques (^{1}H-NMR, LC-MS). For Ambrosia tenuifolia, the dichloromethane (ATD) extract contained psilostachyins A (21.8%), B (47.3%), and C (19.5%) as major sesquiterpene lactones.

Step 2: Chemical Modification

- Dissolve the selected subextract (e.g., ATD) in anhydrous toluene at concentration of 10-50 mg/mL.
- Add p-toluenesulfonic acid (p-TsOH) as catalyst (10-20 mol% relative to estimated NP concentration).
- Reflux the reaction mixture for 1 hour under inert atmosphere with monitoring by TLC or LC-MS.
- Quench the reaction with saturated $NaHCO_3$ solution, extract with ethyl acetate, and dry over anhydrous Na_2SO_4.

Step 3: Bioactivity-Guided Fractionation

- Subject the chemically modified extract to chromatographic separation (e.g., vacuum liquid chromatography, flash chromatography) using gradient solvent systems.
- Screen all fractions for target bioactivity (e.g., anti-glioblastoma activity against T98G cell cultures).
- Further purify active fractions through repeated column chromatography and preparative HPLC until pure compounds are obtained.
- Characterize pure compounds using spectroscopic techniques (NMR, MS, IR) and evaluate their bioactivity.

This protocol successfully generated natural product derivatives containing new molecular skeletons with demonstrated anti-glioblastoma activity in T98G cell cultures [3].

Technological Enablers for Exploration

Analytical and Computational Tools

Table 3: Research Reagent Solutions for Natural Product Exploration

Reagent/Category	Specific Examples	Function/Application	Technical Specifications
AI/Bioinformatics Platforms	DeepBGC; AntiSMASH; NPClassifier [94] [19]	Biosynthetic gene cluster prediction; Chemical classification [94] [19]	Python-based; Integration with genomic databases [19]
Analytical Instruments	UPLC-Q-TOF-MS; High-resolution MS [96] [19]	Metabolite profiling; Dereplication; Structural characterization [96] [19]	High mass accuracy (<5 ppm); MS/MS fragmentation capability [19]
Taxonomic Databases	Catalogue of Life (CoL); WoRMS [94]	Organism identification; Taxonomic classification [94]	API access; Regular updates [94]
Cheminformatics Tools	RDKit; Bingo cartridge [94]	Chemical structure standardization; Database searching [94]	SMILES/InChIKey generation; QED/SA Score calculation [94]

Reagent/Category	Specific Examples	Function/Application	Technical Specifications
Chemical Biology Reagents	CETSA reagents [97]	Target engagement validation in intact cells [97]	Cellular context maintenance; Quantitative binding data [97]

Advanced technologies are revolutionizing natural product discovery by addressing traditional bottlenecks:

- **Artificial intelligence and machine learning**: AI has evolved from a disruptive concept to a foundational capability, with models now routinely informing target prediction, compound prioritization, pharmacokinetic property estimation, and virtual screening strategies [97]. Integration of pharmacophoric features with protein-ligand interaction data can boost hit enrichment rates by more than 50-fold compared to traditional methods [97].
- **High-throughput screening and automation**: Implementation of robotic screening systems enables evaluation of thousands of extracts or compounds against multiple targets simultaneously, dramatically accelerating the discovery process [19].
- **Omics technologies integration**: Strategic combination of genomics, transcriptomics, proteomics, and metabolomics provides comprehensive insights into biosynthetic potential and compound functionality [19].

Workflow Visualization

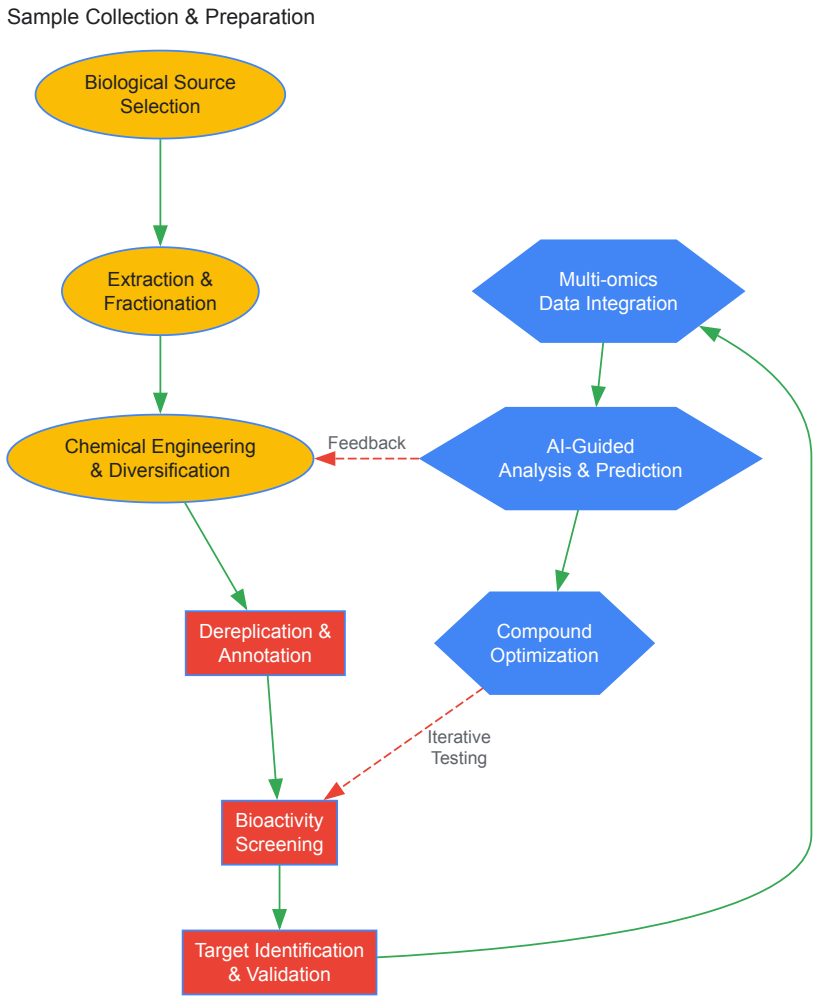


Diagram 1: Integrated Natural Product Discovery Workflow. This diagram illustrates the multidisciplinary approach required for modern natural product discovery, incorporating chemical engineering, multi-omics data integration, and AI-guided analysis in an iterative feedback loop.

The future of natural product research lies in strategic integration of innovative approaches spanning biological exploration, chemical diversification, and technological advancement. By focusing on underexplored biological sources, implementing sustainable sourcing practices, applying sophisticated chemical engineering strategies, and leveraging cutting-edge analytical and computational tools, researchers can significantly expand access to novel chemical space. These approaches will enable the discovery of unprecedented molecular scaffolds with potential to address pressing therapeutic challenges, particularly in areas of unmet medical need such as antimicrobial resistance and complex chronic diseases. Success in this

endeavor will require multidisciplinary collaboration across traditional scientific boundaries, combining expertise in natural product chemistry, synthetic biology, data science, and pharmacology to fully realize the potential of nature's chemical ingenuity.

Conclusion

The systematic exploration of natural product chemical space represents a powerful paradigm for modern drug discovery. Through the integration of foundational concepts, advanced cheminformatics methodologies, targeted troubleshooting approaches, and rigorous validation, researchers can effectively navigate this complex landscape. Natural products continue to demonstrate their immense value by providing unique structural scaffolds that access biological target space often inaccessible to synthetic compounds. The future of NP-based drug discovery lies in the continued development of integrated workflows that combine computational prediction with experimental validation, the expansion into under-explored biological sources and geographical regions, and the application of artificial intelligence to unlock patterns within NP chemical space. As these approaches mature, they promise to accelerate the identification of novel therapeutic agents for complex and untreated diseases, ensuring natural products remain a vital component of the drug discovery arsenal for years to come.

References

- 1. Growth vs. Diversity: A Time-Evolution Analysis of the ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11870478/]
- 2. Frontiers | Exploring chemical space for “druglike” small molecules in the age of AI [https://www.frontiersin.org/journals/molecular-biosciences/articles/10.3389/fmolb.2025.1553667/full]
- 3. Enhancing structural diversity through chemical ... [https://www.nature.com/articles/s41598-024-63639-y]
- 4. Exploring Dimensionality Reduction for Chemical Space ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11733715/]
- 5. Visualizing chemical space networks with RDKit and NetworkX [https://jcheminf.biomedcentral.com/articles/10.1186/s13321-022-00664-x]
- 6. On the biologically relevant chemical space: BioReCS [https://www.frontiersin.org/journals/drug-discovery/articles/10.3389/fddsv.2025.1674289/full]
- 7. Bridging chemical space and biological efficacy: advances ... [https://link.springer.com/article/10.1007/s13659-025-00521-y]
- 8. An Assessment of Chemical Diversity in Microbial Natural ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12464764/]
- 9. Accurate Structure Prediction of Natural Products with ... [https://chemrxiv.org/engage/chemrxiv/article-details/6820654c927d1c2e660f6511]
- 10. Evaluation of the stereochemical quality of predicted RNA ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC8906551/]
- 11. Developing a quantitative structure–property relationships ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12355335/]
- 12. The Chemical and Products Database v4.0, an updated ... [https://www.nature.com/articles/s41597-025-05240-0]
- 13. Time-Dependent Comparison of the Structural Variations ... [https://www.mdpi.com/1422-0067/25/21/11475]
- 14. Natural products in drug discovery: advances and ... [https://www.nature.com/articles/s41573-020-00114-z]
- 15. Fragment libraries from large synthetic compounds and ... [https://chemrxiv.org/engage/chemrxiv/article-details/67a3b0db81d2151a024b2e66]
- 16. SAVI Space—combinatorial encoding of the billion-size ... [https://www.nature.com/articles/s41597-025-05384-z]
- 17. Fragment Libraries from Large and Novel Synthetic ... [https://pubmed.ncbi.nlm.nih.gov/40321506/]
- 18. Bridging chemical space and biological efficacy [https://pubmed.ncbi.nlm.nih.gov/40478370/]
- 19. Revitalizing natural products for sustainable drug Discovery [https://www.sciencedirect.com/science/article/abs/pii/S2352554125001809]
- 20. A new paradigm for drug discovery in the treatment of ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11931805/]
- 21. Building Natural Product Libraries Using Quantitative ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC8547436/]
- 22. Comparative Analysis of Chemical Distribution Models for ... [https://www.mdpi.com/2305-6304/13/6/439]
- 23. The application of chemical similarity measures in an ... [https://www.nature.com/articles/s41598-024-71892-4]
- 24. 67 million natural product-like compound database ... [https://www.nature.com/articles/s41597-023-02207-x]
- 25. Download [https://coconut.naturalproducts.net/download]
- 26. COCONUT 2.0: a comprehensive overhaul and curation of ... [https://pubmed.ncbi.nlm.nih.gov/39588778/]
- 27. npatlas - Database Commons [https://ngdc.cncb.ac.cn/databasecommons/database/id/7849]
- 28. SuperNatural - Database Commons [https://ngdc.cncb.ac.cn/databasecommons/database/id/1597]
- 29. COCONUT - Database Commons [https://ngdc.cncb.ac.cn/databasecommons/database/id/7627]
- 30. NPDBEjeCol: A Natural Products Database from Colombia [https://pmc.ncbi.nlm.nih.gov/articles/PMC11904848/]
- 31. Cheminformatics to Characterize Pharmacologically Active ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC7698493/]
- 32. Effectiveness of molecular fingerprints for exploring the ... [https://jcheminf.biomedcentral.com/articles/10.1186/s13321-024-00830-3]
- 33. Effectiveness of molecular fingerprints for exploring the ... [https://pubmed.ncbi.nlm.nih.gov/38528548]
- 34. Scaffold hopping from natural products to synthetic ... [https://www.nature.com/articles/s42004-018-0043-x]
- 35. From Patterns to Pills: How Informatics Is Shaping Medicinal ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12114992/]
- 36. Privileged Scaffold Analysis of Natural Products with Deep ... [https://pubmed.ncbi.nlm.nih.gov/32406179/]
- 37. Comparative study of the mechanism of natural ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC10555731/]
- 38. Novel Chemical Space Exploration via Natural Products - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC2696019/]

- 39. Structural Diversity Assessment of Natural Products [https://comlibrary.bocsci.com/services/structural-diversity-assessment-of-natural-products.html]
- 40. Natural product-informed exploration of chemical space to ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC8130614/]
- 41. tuned for navigation in biologically relevant chemical space [https://pubmed.ncbi.nlm.nih.gov/17439280/]
- 42. Scaffold Hunter: a comprehensive visual analytics framework ... [https://jcheminf.biomedcentral.com/articles/10.1186/s13321-017-0213-3]
- 43. Comparison of dimensionality reduction techniques for the ... [https://www.sciencedirect.com/science/article/pii/S2949747724000137]
- 44. Connecting chemical and protein sequence space to ... [https://www.nature.com/articles/s41586-025-09519-5]
- 45. Chemical space visual navigation in the era of deep ... [https://www.sciencedirect.com/science/article/pii/S1359644625001059]
- 46. Natural product diversity and its role in chemical biology ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3110584/]
- 47. A Comprehensive Survey of Prospective Structure-Based ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC9781938/]
- 48. Virtual Screening - an overview [https://www.sciencedirect.com/topics/computer-science/virtual-screening]
- 49. Progress of AI-Driven Drug–Target Interaction Prediction and ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12563295/]
- 50. Natural products as an inspiration in the diversity-oriented ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC2496956/]
- 51. Diversity-oriented synthesis as a tool for the discovery of ... [https://www.nature.com/articles/ncomms1081]
- 52. Fully-Functionalized Natural Product Probes to Expand the ... [https://pubmed.ncbi.nlm.nih.gov/40937893/]
- 53. A skeletally diverse library of bioactive natural-product-like ... [https://www.sciencedirect.com/science/article/abs/pii/S2451929424004182]
- 54. natural product-like compound generation with GPT-based ... [https://link.springer.com/article/10.1007/s11227-024-06860-w]
- 55. NaPLeS: a natural products likeness scorer—web application ... [https://jcheminf.biomedcentral.com/articles/10.1186/s13321-019-0378-z]
- 56. Natural product-likeness score and its applications in the drug ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4236212/]
- 57. NaPLeS - Natural Product Likeness Score calculator [https://naples.naturalproducts.net/]
- 58. ScaffoldGVAE: scaffold generation and hopping of drug ... [https://jcheminf.biomedcentral.com/articles/10.1186/s13321-023-00766-0]
- 59. Assigning the Stereochemistry of Natural Products by ... [https://chemrxiv.org/engage/chemrxiv/article-details/689e042d23be8e43d63c85ca]
- 60. A Practical Guide to Chemogenomics Data Curation - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC5657146/]
- 61. Objective, Quantitative, Data-Driven Assessment of Chemical ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC5814752/]
- 62. A chiral LC–MS strategy for stereochemical assignment of ... [https://www.beilstein-journals.org/bjoc/articles/21/171]
- 63. Data Curation Issues in the Chemical Sciences [http://www.niso.org/niso-io/2013/09/data-curation-issues-chemical-sciences]
- 64. Rare Earth Supply Chain Risks & Strategic Solutions [https://discoveryalert.com.au/rare-earth-elements-critical-applications-2025/]
- 65. World Earth Day 2025: Hidden Supply Chain Risks ... [https://www.dynamicdesign.org/post/world-earth-day-2025-hidden-supply-chain-risks-threatening-renewable-energy-growth]
- 66. The Most Destructive Supply Chain Challenges in 2025 ... [https://www.z2data.com/insights/most-destructive-supply-chain-challenges-2025-how-to-navigate]
- 67. Advances in structural modification of natural products with ... [https://www.sciencedirect.com/science/article/abs/pii/S0367326X25004411]
- 68. Genome mining-driven discovery of enzymes catalyzing ... [https://www.nature.com/articles/s41429-025-00881-0]
- 69. Estimation of synthetic accessibility score of drug-like ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3225829/]
- 70. Exploring and expanding the natural chemical space of ... [https://www.nature.com/articles/s41467-025-57145-6]
- 71. Evolutionary Tree in Chemical Space of Natural Products [https://pubmed.ncbi.nlm.nih.gov/40950011/]
- 72. Integrated molecular and ADME-toxicity profiling identifies ... [https://www.nature.com/articles/s41598-025-02858-3]
- 73. In Silico ADME/Tox Comes of Age: Twenty Years Later - PMC [https://pmc.ncbi.nlm.nih.gov/articles/PMC10850432/]
- 74. First multifaceted ADME profile of ACP-105 (CAS: 1048998 ... [https://link.springer.com/article/10.1007/s00204-025-04170-5]
- 75. The graphical representation of ADME-related molecule ... [https://www.sciencedirect.com/science/article/abs/pii/S1359644610008056]
- 76. A beginners guide to ADME Tox [https://www.cellgs.com/blog/a-beginners-guide-to-adme-tox.html]
- 77. Computational Prediction and Validation of an Expert's ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4955571/]
- 78. Exploring chemical space for lead identification by ... [https://www.sciencedirect.com/science/article/pii/S200103702300291X]
- 79. Computational approaches streamlining drug discovery [https://www.nature.com/articles/s41586-023-05905-z]
- 80. Medicinal Plants Conference | Natural Products 2025 [https://naturalproducts.pharmaceuticalconferences.com/2025]
- 81. Natural Products Have Increased Rates of Clinical Trial ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11287737/]
- 82. Natural Products as Sources of New Drugs over the 30 ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3721181/]
- 83. NPDBEjeCol: a natural products database from Colombia [https://chemrxiv.org/engage/chemrxiv/article-details/66b56c7ec9c6a5c07a8d8157]
- 84. Pharmaceuticals from marine sources: past, present and ... [https://pharmaceutical-journal.com/article/research/pharmaceuticals-from-marine-sources-past-present-and-future]
- 85. Exploring chemical space for “druglike” small molecules in ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11955463/]
- 86. Virtual fragment screening for DNA repair inhibitors in vast ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC11836371/]
- 87. Nano-Drug Delivery Systems Based on Natural Products [https://pmc.ncbi.nlm.nih.gov/articles/PMC10802180/]
- 88. Predicting Nano–Bio Interactions by Integrating ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC5772766/]

- 89. Exploring Macrocyclic Chemical Space: Strategies and ... [https://www.mdpi.com/1424-8247/18/5/617]
- 90. Obstacles and innovations of macrocyclic drug development [https://www.drugtargetreview.com/article/155484/obstacles-and-innovations-of-macrocyclic-drug-development/]
- 91. Challenges in Development of Nanoparticle-Based ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3326161/]
- 92. Fragment-based screening with natural products for novel ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC6816479/]
- 93. NPASS database update 2023: quantitative natural product ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC9825494/]
- 94. NPBS Atlas: a comprehensive data resource for exploring the ... [https://jcheminf.biomedcentral.com/articles/10.1186/s13321-025-01116-y]
- 95. Section Structural Studies on Marine Natural Products [https://www.mdpi.com/journal/marinedrugs/sections/marine_natural_product_chemistry]
- 96. Applications of Natural and Pseudo-Natural Products in ... [https://www.mdpi.com/journal/cimb/special_issues/1V6D29J00P]
- 97. Top 5 Drug Discovery Trends 2025 Driving Breakthroughs [https://www.pelagobio.com/cetsa-drug-discovery-resources/blog/drug-discovery-trends-2025/]